

CLD300

SAP Cloud Portal Administration and Operations

PARTICIPANT HANDBOOK
INSTRUCTOR-LED TRAINING

Course Version: 17
Course Duration: 3 Day(s)
e-book Duration: 7 Hours 40 Minutes
Material Number: 50146841

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Typographic Conventions

American English is the standard used in this handbook.

The following typographic conventions are also used.

This information is displayed in the instructor's presentation	
Demonstration	
Procedure	
Warning or Caution	
Hint	
Related or Additional Information	
Facilitated Discussion	
User interface control	Example text
Window title	Example text

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Course Overview

TARGET AUDIENCE

This course is intended for the following audiences:

Developer

Development Consultant

System Administrator

Technology Consultant

UNIT 1

Introduction to SAP Cloud Platform Portal

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UNIT OBJECTIVES

Describe SAP Cloud Platform and cloud platform services

Explain role assignment

Get started with the SAP Cloud Platform Service

Navigate to the Site Directory

Unit 1

Lesson 1

Describing the SAP Cloud Platform Service



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Describe SAP Cloud Platform and cloud platform services

SAP Cloud Platform Services

To experience SAP Cloud Platform Portal and its services, you can register a personal trial account. Access the trial environment at <https://account.hanatrial.ondemand.com>. See the links related to SAP Cloud Platform and SAP Cloud Platform Portal for more detailed information.

The following are important sources of information:



Overview of SAP Cloud Platform at <https://cloudplatform.sap.com>

SAP Portal roadmap in SAP roadmaps at <http://www.sap.com/solution/roadmaps.html>
(SAP Cloud Platform Portal and SAP Enterprise Portal Road Map)

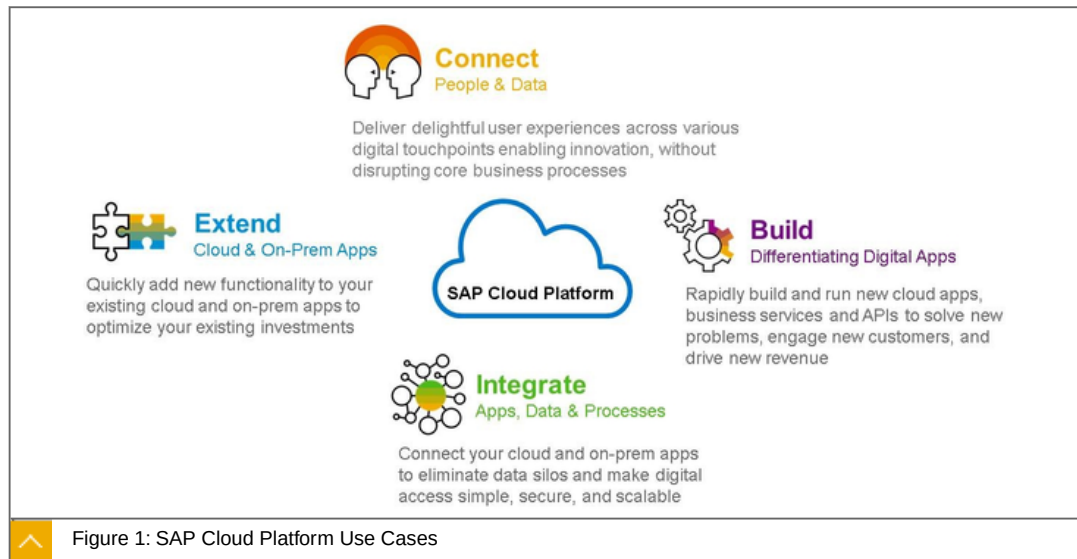
Release notes and documentation at <https://help.hana.ondemand.com>

Portal service general documentation at https://help.sap.com/viewer/p/Portal_Service

Support for SAP Fiori Cloud (SAP Business Suite and SAP S4/HANA) at https://help.sap.com/viewer/p/SAP_FIORI_CLOUD_ALL

SAP community at <https://www.sap.com/community/topic/portal.html>

Use Cases



With comprehensive services and capabilities, SAP Cloud Platform enables customers and partners to quickly build new solutions, extend SAP cloud and on-premise solutions, and support integration with SAP and non-SAP apps. The following are key benefits:

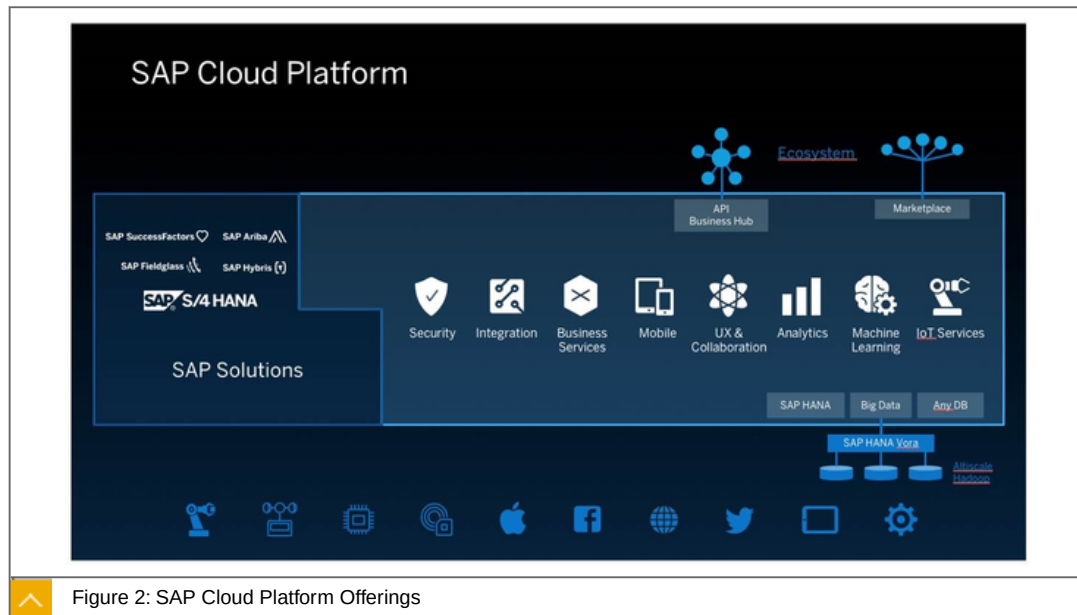
Extend existing applications and cloud or on-premise solutions.

Build next-generation applications with comprehensive services such as mobile, collaboration, big data, and analytic services.

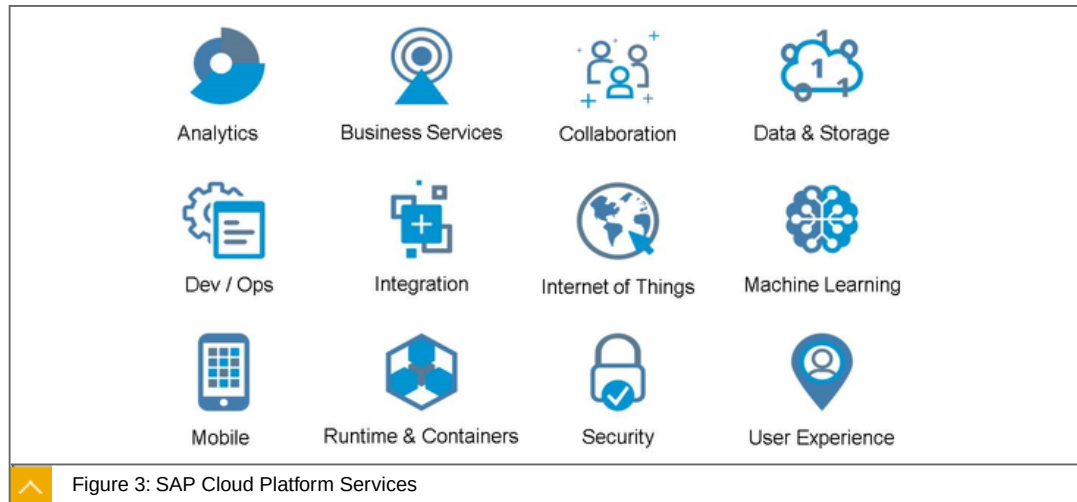
Increase developer productivity with an open, standards-based, easy-to-use cloud development and runtime environment.

Leverage enterprise grade operations, lifecycle management, and administration capabilities.

Offerings



SAP Cloud Platform has evolved into an enterprise, platform-as-a-service (PaaS) with a unique in-memory database and business application services that are designed to meet the needs of companies that want to leverage innovative technologies, such as IoT, machine learning, big data, artificial intelligence, and multi-cloud delivery, all on one platform.



SAP Cloud Platform is capable of connecting people, things, and businesses together under an open platform on which numerous SAP apps, such as SAP Leonardo, are built.

SAP Cloud Platform Portal

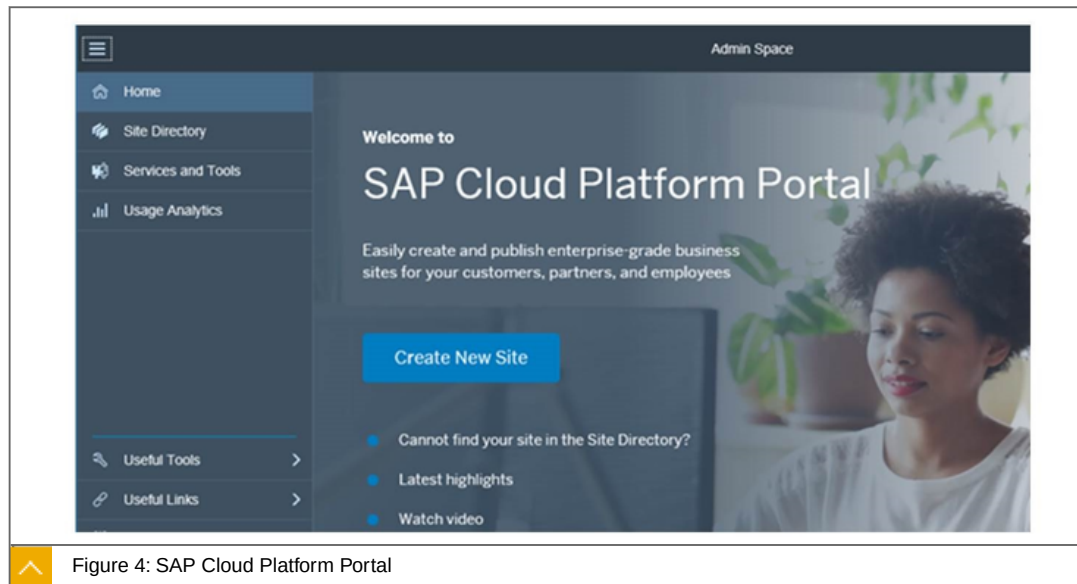


Figure 4: SAP Cloud Platform Portal

The SAP Cloud Platform Portal is part of SAP Cloud Platform. Through the portal, organizations can easily use the tools and services to build enterprise business sites and provide users with a good user experience. The portal supports seamless integration with SAP cloud and on-premise solutions and third-party content. The portal facilitates the following:

- Easy site creation and consumption
- Fast branding and customization
- Smooth user experience through SAP Fiori or business sites and home pages
- Role-based authorization management
- Integration of applications, services, and content to meet every business need
- Integration of SAP and non-SAP solutions
- Open standards support

You can use the SAP Cloud Platform Portal to create the following:

- A self-service launchpad, for example, an employee launchpad that provides a single point of access to role-based applications, processes, and services
- A Line-of-Business (LOB) homepage, for example, a HR site with domain-specific apps, content, information, and news
- An external-facing site, for example, an external support site for registered customers, suppliers, and partners
- A social intranet, for example, for social collaboration and content sharing

The SAP Cloud Platform Portal can also be used in SAP Fiori Cloud solutions. Customers can access SAP Fiori apps via a launchpad site in SAP Fiori Cloud.



LESSON SUMMARY

You should now be able to:

Describe SAP Cloud Platform and cloud platform services

Explaining Role Assignment

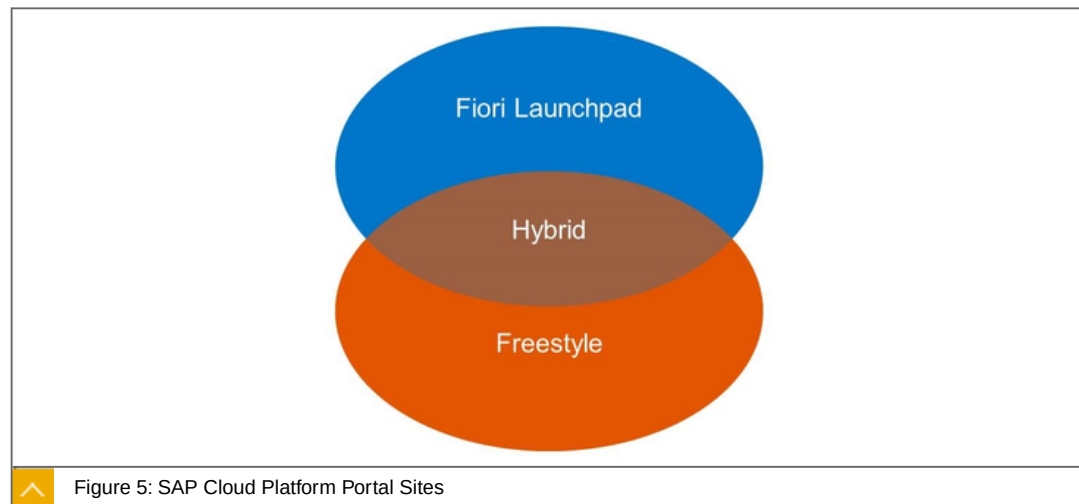


LESSON OBJECTIVES

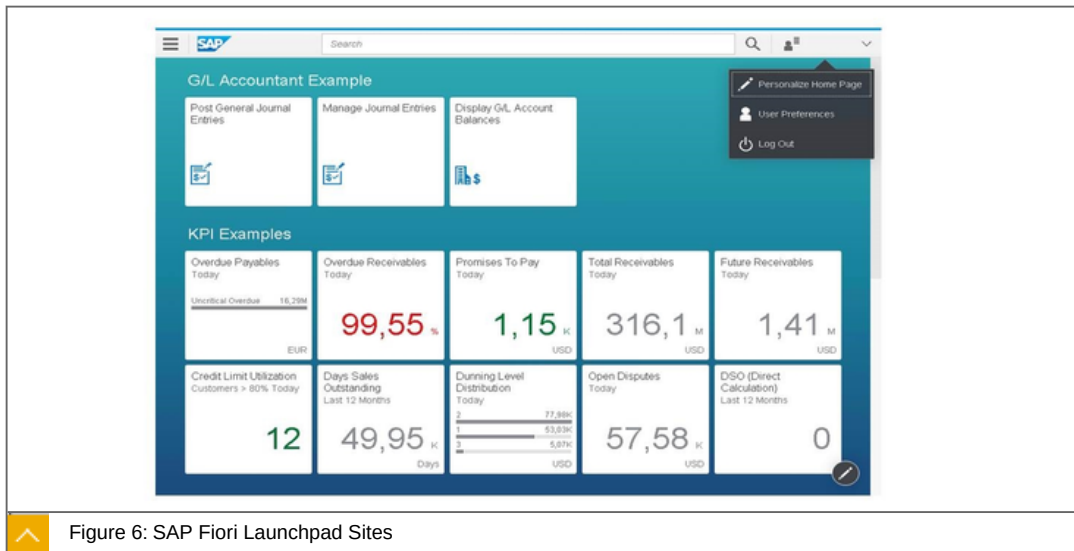
After completing this lesson, you will be able to:

Explain role assignment

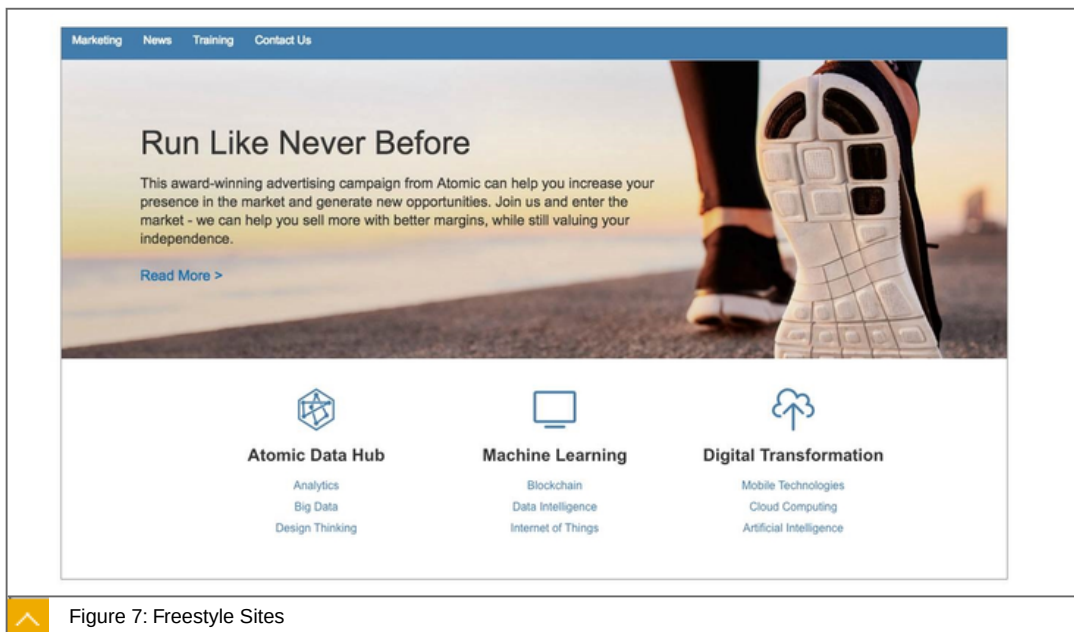
Role Assignment



The SAP Cloud Platform Portal offers two types of sites: freestyle sites and launchpad sites. You can also create a hybrid site by embedding an SAP Fiori launchpad page as an entry point for SAP Fiori apps, together with other pages in a freestyle site. In this way, you can receive the benefits of SAP Fiori launchpad, together with other types of content on other pages of your site.



The SAP Fiori launchpad site is based on the SAP Fiori launchpad site template from SAP. The home page of the site provides end users with tiles and links to SAP Fiori applications hosted in the cloud. The launchpad site is initially created in the Site Directory. Using the SAP Fiori launchpad configuration cockpit, you can then configure the settings and create the content.



Freestyle sites are very versatile and flexible in terms of content and layout. These sites can either be built using SAP templates or templates created by developers using SAP Web IDE options to build freestyle sites. The following are the main options:

Create site using available site template and use as is

Create site using available site template and edit/extend content/page templates, pages, and apps

Create site with pages and apps and embed a SAP Fiori launchpad page with SAP Fiori applications (hybrid sites)

Portal Site Designer is the tool to use for editing/extending the freestyle sites.

Portal Service Roles

The following table shows technical portal service roles:



Table 1: Portal Service Roles: Technical

Role	Description
SAP Cloud Platform Member	Has authorizations for the SAP Cloud Platform cockpit for administration and support tasks. Can configure authorizations, manage applications and services, and provide support
TENANT_ADMIN	Enables administration in the portal (Portal Site Designer and Admin Space) and the SAP Fiori launchpad configuration cockpit
WEB_CONTENT_EDITOR	Anyone assigned to the WEB_CONTENT_EDITOR role for a specific subaccount can create and update the content of the sites in that subaccount. Web content is managed with the Web Content Editor, which enables non-technical users to easily create, edit, and maintain the content of a website.

Technical roles are for subaccount administrators and portal site creators, for example. These are internal predefined roles used for restricting access to design time tools such as the SAP Cloud Platform Cockpit, Portal Site Designer, and the SAP Fiori launchpad configuration cockpit.

The following table shows organizational end user portal service roles:

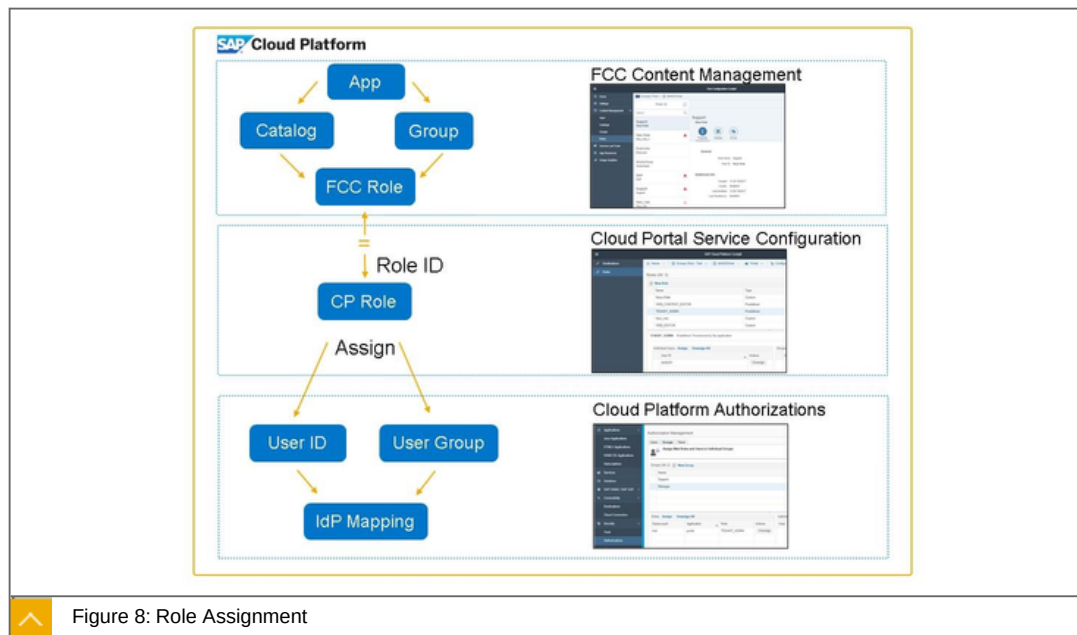


Table 2: Portal Service Roles: Organizational End User

Access Level	Description
Guest	Provides access to the users who are assigned the anonymous role for a site
Role-based	Provides access to users that belong to specific roles
Authenticated	Provides access for users that are logged on

Organizational roles (for specific job functions/tasks) are external custom roles created in the SAP Cloud Platform Cockpit, which are mainly used for restricting access to portal sites, pages, and apps. Examples include accountant, sales person, and manager.

In the general site settings, it is possible to set the access level per site. This is shown in the following figure:



To enable role-based access to apps in a site, roles must be assigned to site objects as follows:

Assign a role to a catalog.

Users assigned with a specific role can access apps in a particular site. This is only possible if you have also assigned the apps to the same catalog as the user's role.

Assign a role to a group (only applicable for a launchpad site).

Assigning roles to a group allows users with this role to view the group in their launchpad. Users can launch apps in the group only if the apps are assigned to the same catalog as the user's role.

Roles can be assigned to individual launchpad users or groups using the SAP Cloud Platform Cockpit.



LESSON SUMMARY

You should now be able to:

Explain role assignment

Unit 1

Lesson 3

Getting Started with the SAP Cloud Platform Service



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Get started with the SAP Cloud Platform Service

SAP Cloud Platform Services Overview

The following are prerequisites for using SAP Cloud Platform Services:



You have an SAP Cloud Platform account.

You have been assigned the administrator role `TENANT_ADMIN`.

The portal service has been enabled in the SAP Cloud Platform Cockpit.

Optional Steps



Subscribe to the SAP Web IDE service.

Create destinations to your content providers (if applicable).

Configure your corporate identity provider in your SAP Cloud Platform subaccount.

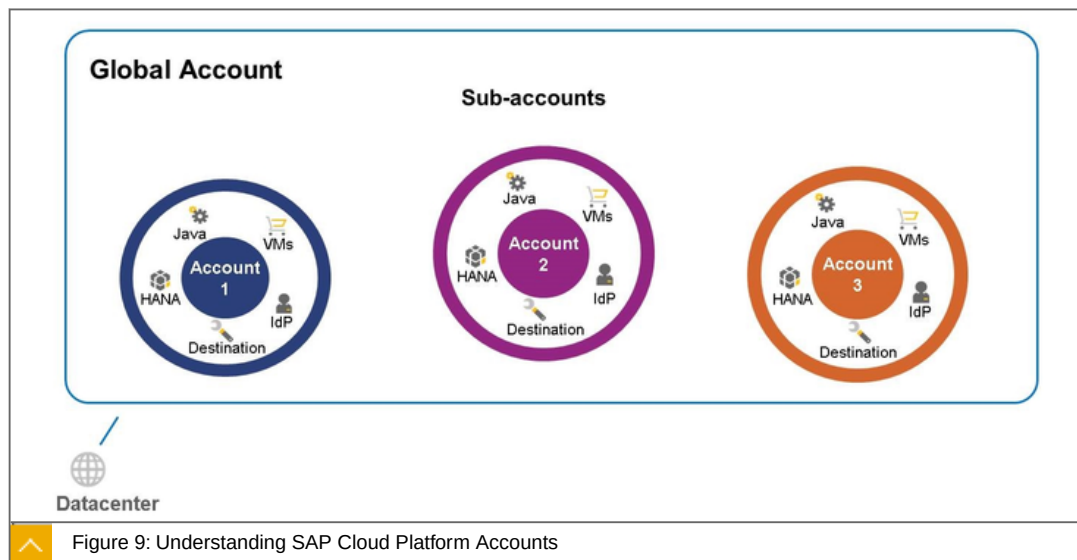


Figure 9: Understanding SAP Cloud Platform Accounts

As shown in the figure, account setup is completed as follows:

1. SAP creates the SAP Cloud Platform global account for every customer.
2. Global account admin creates subaccounts.

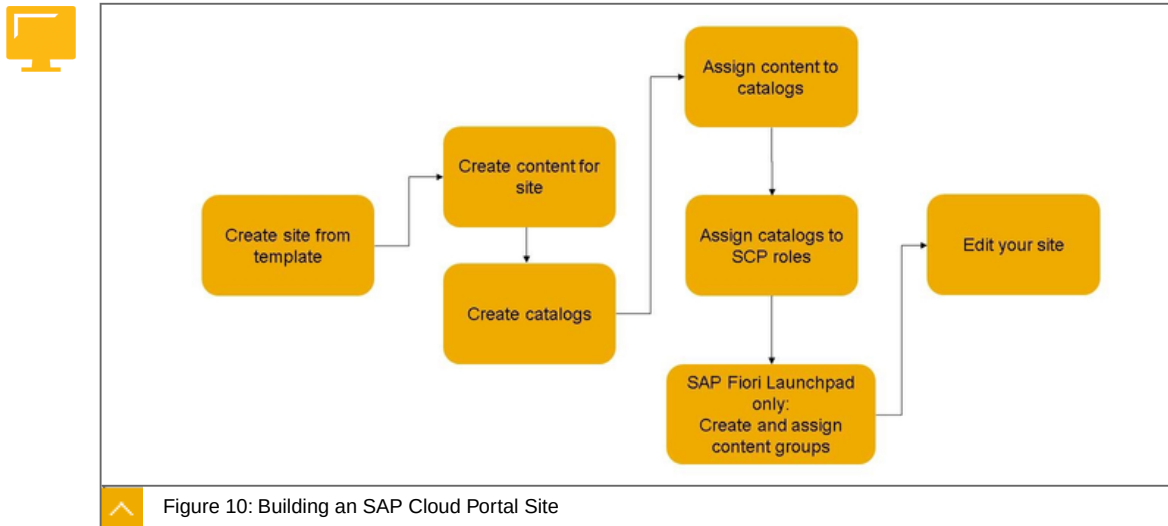
These are created for the following for example:

Development, test, and production

Subsidiaries/Geolocations

Individual business teams

3. Global account admin manages SAP Cloud Platform account users and resources.



The figure shows the recommended steps for building business sites in the portal.



LESSON SUMMARY

You should now be able to:

Get started with the SAP Cloud Platform Service

Unit 1

Lesson 4

Navigating to the Site Directory



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Navigate to the Site Directory

Site Directory

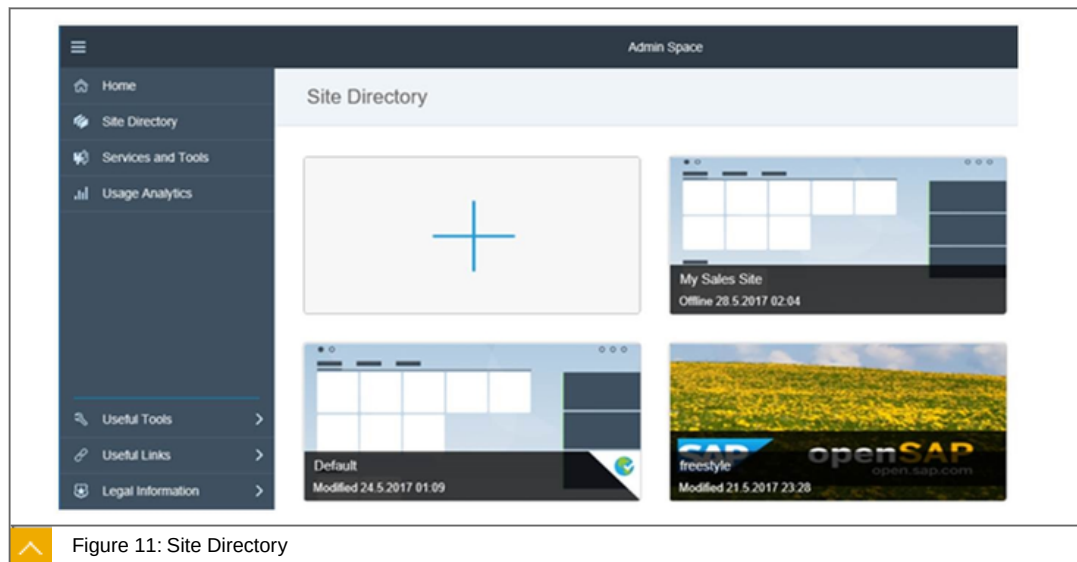


Figure 11: Site Directory

When you connect to the SAP Cloud Platform Portal, the Admin Space is your entry point. The Admin Space is where you manage your subaccount, which includes all your sites, themes, analytics, and assets. In the Site Directory of your Admin Space, you can view, create, edit, duplicate, import, export, delete, and publish your sites. When you edit a site, it takes you into the Site Designer, where you can manage your pages and apps, page templates, menu, content, and settings.

To go to the Site Directory, log on to the SAP Cloud Platform Cockpit using your subaccount. From the navigation panel, you can choose one of the following options:

Through portal and then services

Through subscriptions and then HTML5 applications



LESSON SUMMARY

You should now be able to:

Navigate to the Site Directory

Learning Assessment

1. What are the two basic types of sites you can create with SAP Cloud Platform Portal Service?

Choose the correct answers.

- ☐ **A** Marketing Sites
- ☐ **B** Hybrid Sites
- ☐ **C** Freestyle Sites
- ☐ **D** Fiori Launchpad Sites

2. It is possible to use Fiori Launchpad functionality in a Freestyle site.

Determine whether this statement is true or false.

- ☐ True
- ☐ False

3. An app needs to be assigned to which of the following in order to for an end user to use an application?

Choose the correct answers.

- ☐ **A** Role
- ☐ **B** Catalog
- ☐ **C** Group
- ☐ **D** Identity Provider Mapping

Learning Assessment - Answers

1. What are the two basic types of sites you can create with SAP Cloud Platform Portal Service?

Choose the correct answers.

- ☐ A Marketing Sites
- ☐ B Hybrid Sites
- ☒ C Freestyle Sites
- ☒ D Fiori Launchpad Sites

You are correct! Sites come in two basic types: Freestyle and Fiori Launchpad. The administration cockpit for each differ in their functionality. The Fiori Configuration Cockpit is focused on Fiori use cases. The freestyle Administration Cockpit includes most of the Fiori configuration while also adding administrative controls for Content Management, Site Assets, Menu Configuration and other elements needed in a freestyle site. Refer to the lesson 'Describing the SAP Cloud Platform Service' for more details.

2. It is possible to use Fiori Launchpad functionality in a Freestyle site.

Determine whether this statement is true or false.

- ☒ True
- ☐ False

You are correct! It is possible to integrate a full Fiori Launchpad into a freestyle site template. Refer to the lesson 'Explaining Role Assignment' for more details.

3. An app needs to be assigned to which of the following in order to for an end user to use an application?

Choose the correct answers.

- ☒ **A** Role
- ☒ **B** Catalog
- ☐ **C** Group
- ☐ **D** Identity Provider Mapping

You are correct! Applications need to be assigned to a Role and an Catalog so that a user can access an application. Additionally, you may assign an application to a Group, so that it will show up in the launchpad when the Fiori Launchpad is opened. Refer to the lesson 'Explaining Role Assignment' for more details.

UNIT 2

SAP Fiori Launchpad Site Administration

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UNIT OBJECTIVES

Use the SAP Fiori launchpad configuration cockpit

Describe SAP Fiori launchpad objects

Explore the app types and tile types

Navigate the SAP Fiori launchpad

Create content in the SAP Fiori launchpad configuration cockpit

Consume site content

Configure site settings

Clear app cache

Manage site themes

Unit 2

Lesson 1

Using the SAP Fiori Launchpad Configuration Cockpit



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Use the SAP Fiori launchpad configuration cockpit

SAP Fiori Launchpad Configuration Cockpit

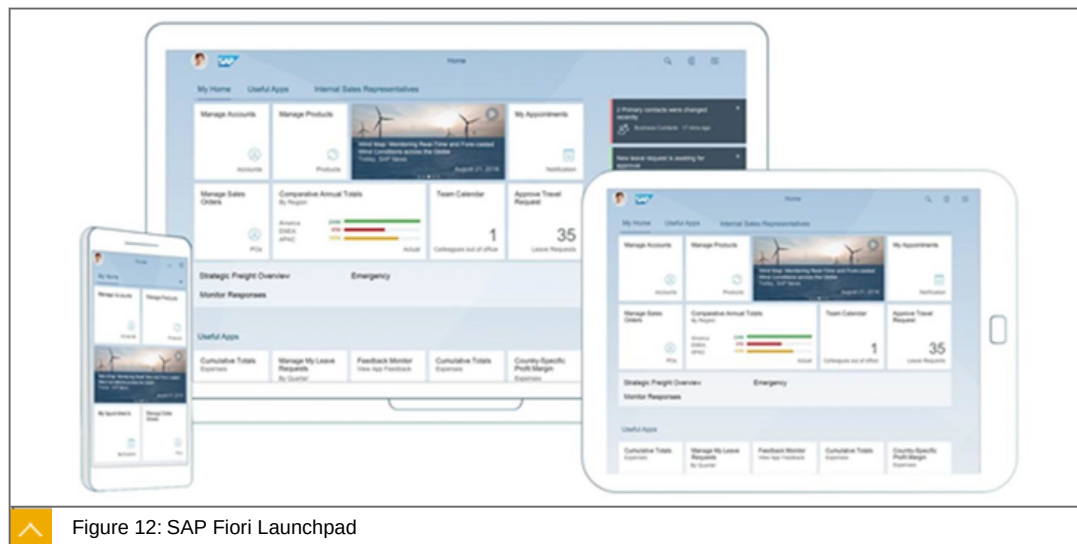


Figure 12: SAP Fiori Launchpad

The SAP Fiori launchpad provides users with a role-based, personalized entry point to SAP Fiori apps across platforms and devices. Depending on the role of the user, specific groups consisting of apps are displayed. Each app is represented by a tile. Tiles can also present summarized or dynamic data to the user and this information can be visualized by number or color or presented as charts.

A SAP Fiori launchpad site is an instance of the cloud portal on the SAP Cloud Platform. The SAP Cloud Platform Portal provides a SAP Fiori launchpad site template by default, and administrators can create new launchpad sites based on this. A SAP Fiori launchpad site is initially created in the portal using the Site Directory and then the launchpad objects are created using the SAP Fiori launchpad configuration cockpit.

The following are typical administrator tasks in a SAP Fiori launchpad project:

- Create and manage sites.

- Manage catalogs, groups, and roles.

Leverage SAP business content (SAP Fiori applications).

Manage themes, translations, and transport portal content across the landscape.

Integrate the apps on SAP Fiori launchpad with on-premise and cloud systems.

SAP Fiori Launchpad Tools

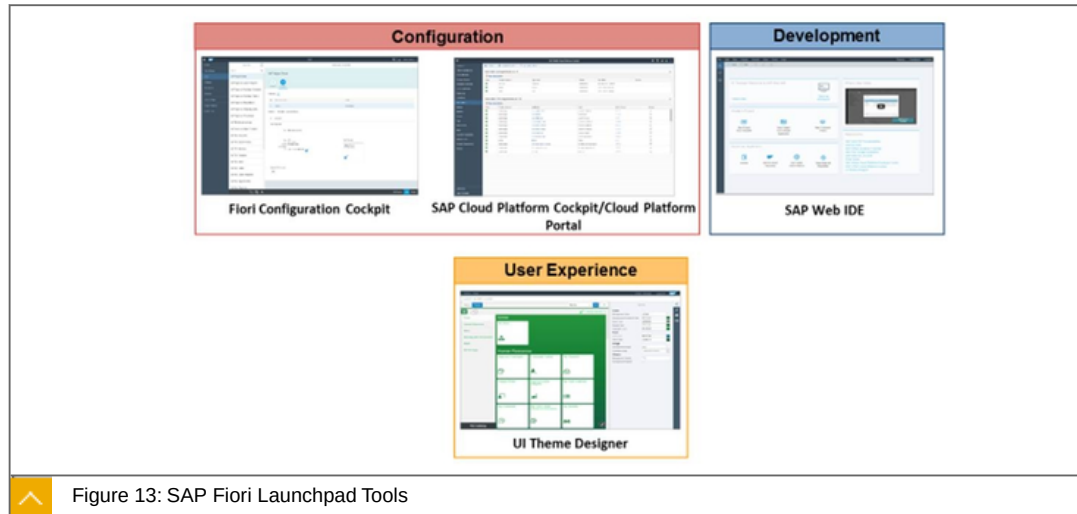


Figure 13: SAP Fiori Launchpad Tools

Several tools are available to support the SAP Fiori launchpad:

SAP Fiori launchpad configuration cockpit

Manage and customize SAP Fiori launchpad content.

SAP Cloud Platform Cockpit/SAP Cloud Platform Portal

Manage accounts, services, sites and more.

SAP Web IDE

Customize, extend, and develop new SAP Fiori apps

UI Theme Designer

Customize styling in the SAP Fiori launchpad site.

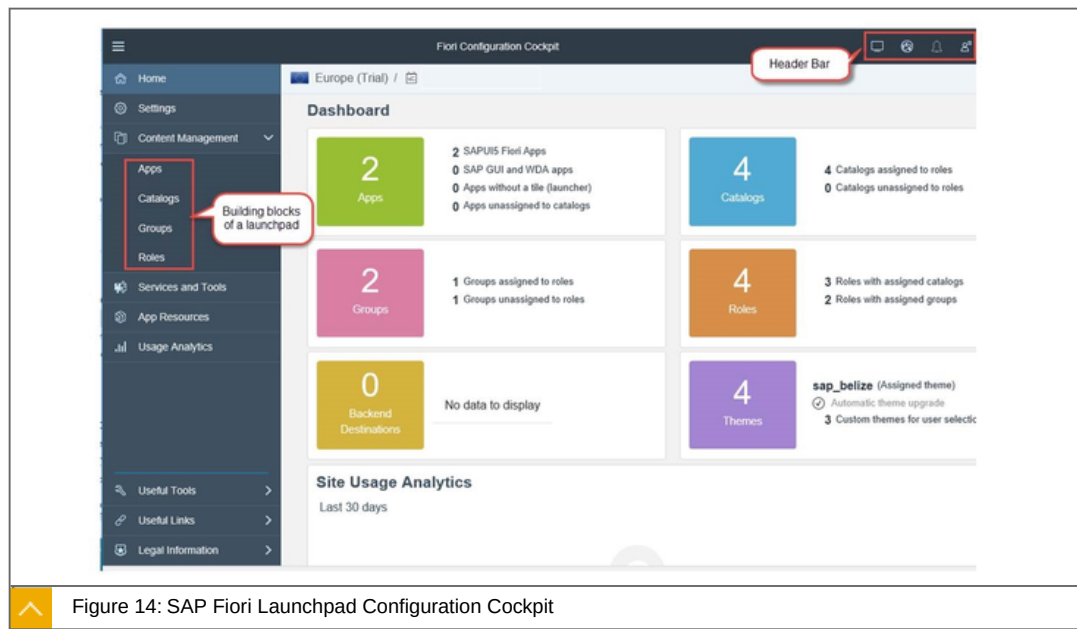


Figure 14: SAP Fiori Launchpad Configuration Cockpit

The SAP Fiori launchpad configuration cockpit is a browser-based tool used by administrators to manage their SAP Fiori launchpad sites and launchpad content such as app tiles, groups, catalogs, and roles. The cockpit also provides various services that you can use such as translation, branding, transport, and usage analytics.

The header bar contains options to preview a site, publish a site, display notifications, and view account information. The navigation menu (on the left-hand side) contains various options, editors, tools, and services.

The following editors are available for creating the SAP Fiori launchpad content:

Configure Apps editor: create and edit app tile configurations

Manage Catalogs editor: create catalogs

Manage Groups editor: create groups

Manage Roles editor: create roles

The following tools and services are available:

Translations: manage translations to support multiple languages for your site

UI Theme Designer: create your own themes to customize the visual appearance of apps

Theme Manager: assign custom themes to apps in your sites

Transport Manager: transport content using export and import functions



LESSON SUMMARY

You should now be able to:

Use the SAP Fiori launchpad configuration cockpit

Unit 2

Lesson 2

Describing SAP Fiori Launchpad Objects



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Describe SAP Fiori launchpad objects

SAP Fiori Launchpad Objects

The following table describes SAP Fiori launchpad objects:



Table 3: SAP Fiori Launchpad Objects

Launchpad Objects	Description
User	Users are assigned to roles. Roles control which apps the users can access.
Role	Roles are created based on SAP Cloud Platform roles. Roles are then linked with catalogs and groups.
Catalog	Catalogs are collections of apps make available for a specific role (or multiple roles). Roles and apps are assigned to a catalog. Users belonging to a role have access to all the apps that are assigned to the same catalog as their role.
App, Tile, and Intents	App: An app represents any SAPUI5 application that users can launch from their SAP Fiori launchpad. Tile: Typically, apps are represented by tiles. A user clicks a tile to launch an app. Intents: Navigation to apps is done via intents. An intent is a unique combination of a semantic object (for example, purchase order) and an action (for example, display) and an optional set of navigation parameters. To summarize, when a user clicks a tile in their SAP Fiori launchpad, the intent (including optional parameters) navigates to the app and opens it.

Launchpad Objects	Description
Group	A group consists of tiles (and links) representing a subset of apps. When you assign a role to a group, all users with this role can view the group in their SAP Fiori launchpad. Users only see those tiles in the group that have been assigned to a catalog with the same role. End users can personalize their launchpad by adding or removing apps from out of the box or self-defined groups.

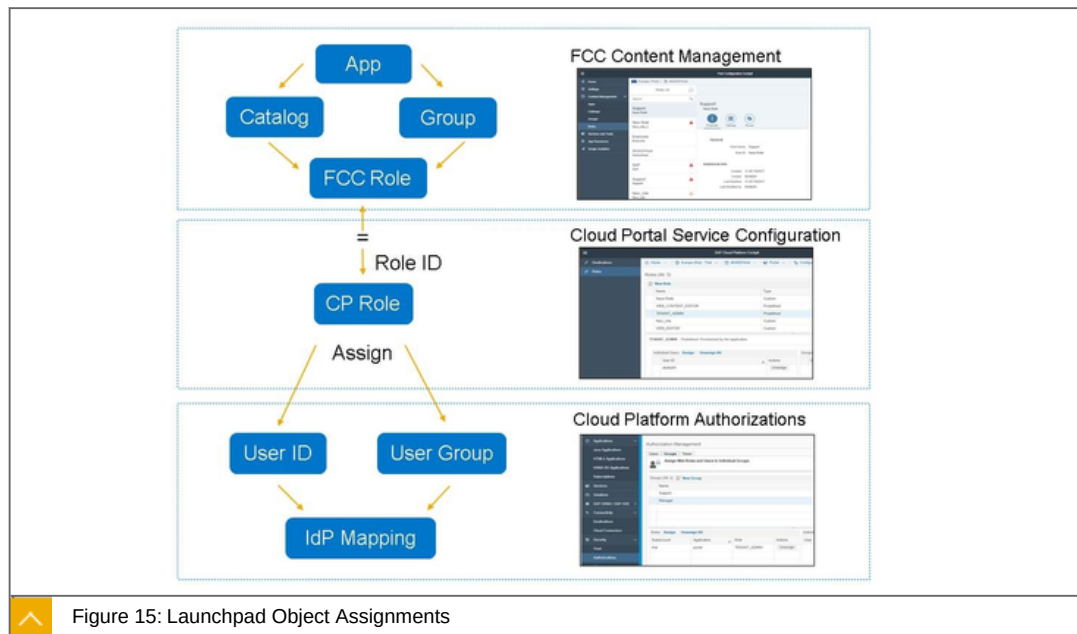


Figure 15: Launchpad Object Assignments

You can perform the following assignment tasks:

Assign a role to a catalog.

Users with a specific role can access apps that are assigned to the same catalog.

Assign apps to a catalog.

Users can access apps in their SAP Fiori launchpad. Note that the user's role must also be assigned to this catalog.

Assign a group to a role.

Users with this role can view the group in their SAP Fiori launchpad. Users only see those app tiles in the group that have been assigned to a catalog with the same role.

Assign a tile and its intent to a group

When a user clicks a tile in the SAP Fiori launchpad, the intent resolves into a URL and navigates to the app.



LESSON SUMMARY

You should now be able to:

Describe SAP Fiori launchpad objects

Unit 2

Lesson 3

Exploring the App Types and Tile Types



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Explore the app types and tile types

App Types and Tile Types

The following table describes app types:

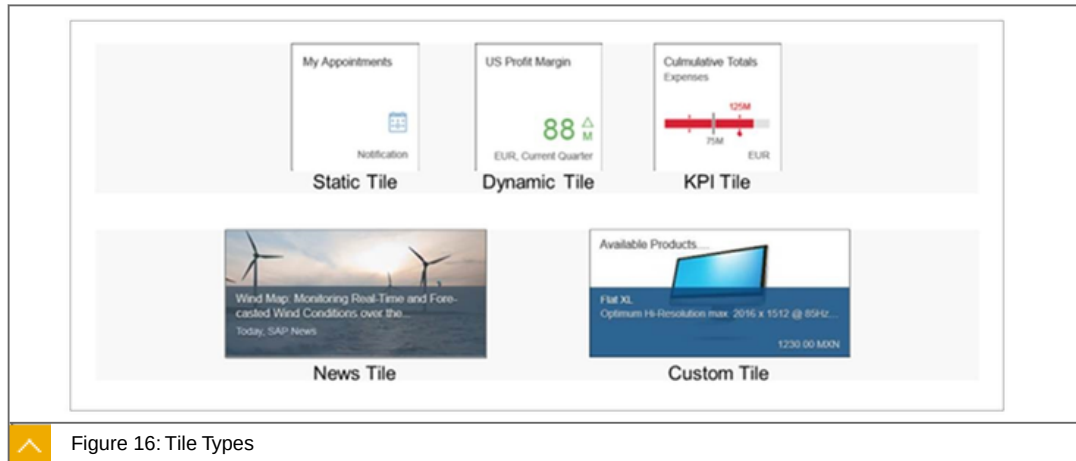


Table 4: App Types

Assignment Type	Description
SAPUI5	Developed using SAPUI5 technology, this type of app can only run in an SAP Fiori launchpad container and is always wrapped with a launchpad header.
URL	Launches a new browser window and navigates to the URL defined in the app configuration
SAPUI5 News	Displays news or other information via an external URL to an RSS feed
Shell Plugin	The Shell Plugin app is executed when the site loads and there is no tile representation. Developers can insert their own code within the shell, headers, or footers or simply run the app in the background for various purposes such as analytics metric collection or the display of a welcome dialog.
SAP GUI and Web Dynpro ABAP	These app types are imported into your SAP Cloud Platform account and they can be found in the Configure Apps editor. These apps are launched via SAP NetWeaver Business Client (SAP NWBC) in a new browser window.
SAP Mobile Documents	An app that provides access to a folder in SAP Mobile Documents

Assignment Type	Description
Dynamic URL	The Dynamic URL is a new app type that is based on destination properties (configured in SAP Cloud Platform Cockpit) and path properties. Depending on the values set for the same destination name on different environments, the same URL can navigate to different apps in these different environments.
Business Server Page	You can enable end users to access their on-premise Business Server Page (BSP) apps from their SAP Fiori launchpad or freestyle sites.

The following figure shows some of the main tile types:



The following table describes tile types:



Table 5: Tile Types

Assignment Type	Description
Static App Launcher	Displays static content such as title, subtitle, and icon
Dynamic App Launcher	Displays data that is updated at regular intervals. The data is retrieved from a back-end system using OData services.
Custom App Launcher	Allows background colors, icons, placement, and so on to be customized. Typical use cases are comparison charts or bullet charts.
News	Displays messages from a news feed. This is only used for the News app type.

Assignment Type	Description
KPI	Used for KPI tiles created in SAP Smart Business. It displays Key Performance Indicators (KPIs) on a tile in real time. This tile type can only be edited in the SAP Smart Business modeler.
No Tile	App is not visible in SAP Fiori launchpad sites. Navigation to the app is executed from a different app or URL using an intent.

**LESSON SUMMARY**

You should now be able to:

Explore the app types and tile types

Unit 2

Lesson 4

Navigating the SAP Fiori Launchpad



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Navigate the SAP Fiori launchpad

Navigation in the SAP Fiori Launchpad

SAP Fiori launchpad supports navigation using intents or URLs. The recommended way of navigating between apps in the launchpad is via intent navigation. An intent is a unique combination of a semantic object (for example, Purchase Order), an action (for example, Display), and an optional set of navigation parameters that are used to define navigation to an application. When a user clicks a tile, the intent and its parameter are resolved into a navigation target that launches the app. To support intent navigation, the following editors are used:



Configure Apps editor: define intent navigation properties (Parameters tab)

Manage Tile Group editor: view intent settings for each tile in a tile group

Multiple intents refer to a case where two (or more) apps are configured with the same intent. When the user clicks on the tiles representing the apps, the assigned role of the user determines which tile is rendered and launched. If the two apps share the same role assignment, the intent of the first app is rendered and launched.

External Content

It is possible to integrate external content into the SAP Fiori launchpad, each type of which is launched in a different way. The following table provides more information:



Table 6: External Content

App Type	How the App is Launched
URL/Dynamic URL	Launches URL in a new browser tab or window (ex-place)
News	Launches as a news tile via an external URL to an RSS feed
SAP GUI for HTML, Web Dynpro ABAP	Can be launched as follows: Ex-place – in a new browser tab/window In-place – within the existing FLP
BSP	Ex-place

App Type	How the App is Launched
SAP Mobile Documents	Ex-pace: Launched via a specific URL that points to a folder in the SAP Mobile Documents folder
SAP Enterprise Portal Apps	Ex-pace: After the user has added the SAP Enterprise Portal to their launchpad, they are launched in a new browser window/tab when the tile is clicked.

**LESSON SUMMARY**

You should now be able to:

Navigate the SAP Fiori launchpad

Unit 2

Lesson 5

Creating Content in the SAP Fiori Launchpad Configuration Cockpit



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Create content in the SAP Fiori launchpad configuration cockpit

Content Creation in the SAP Fiori Launchpad Configuration Cockpit

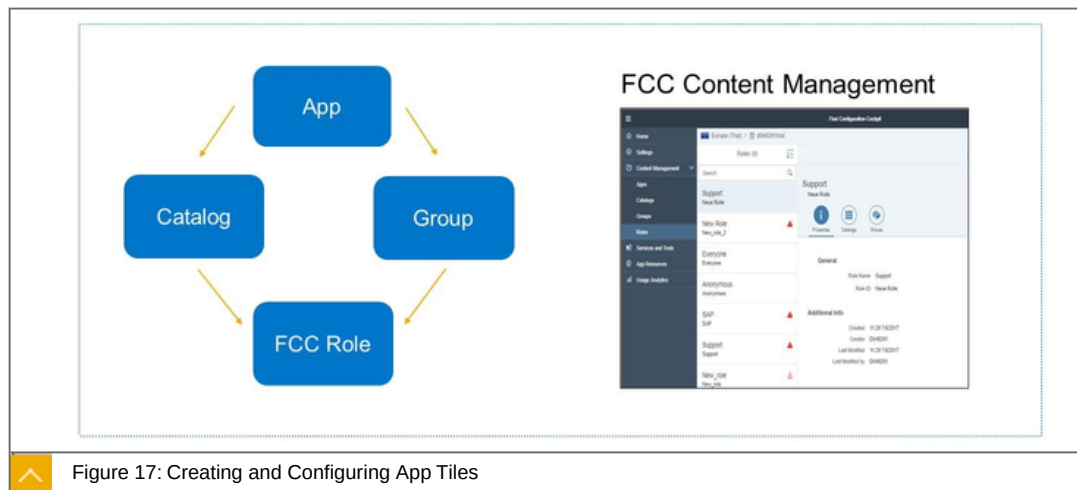


Figure 17: Creating and Configuring App Tiles

Apps, tiles, groups, and catalogs, together with their associated roles, are the building blocks of the SAP Fiori launchpad. The SAP Fiori launchpad configuration cockpit provides dedicated tools and editors to create and maintain these launchpad objects.

An app tile in the SAP Fiori launchpad represents an app. Administrators can configure the app tiles according to business use cases. The steps to create or configure app tiles are as follows:

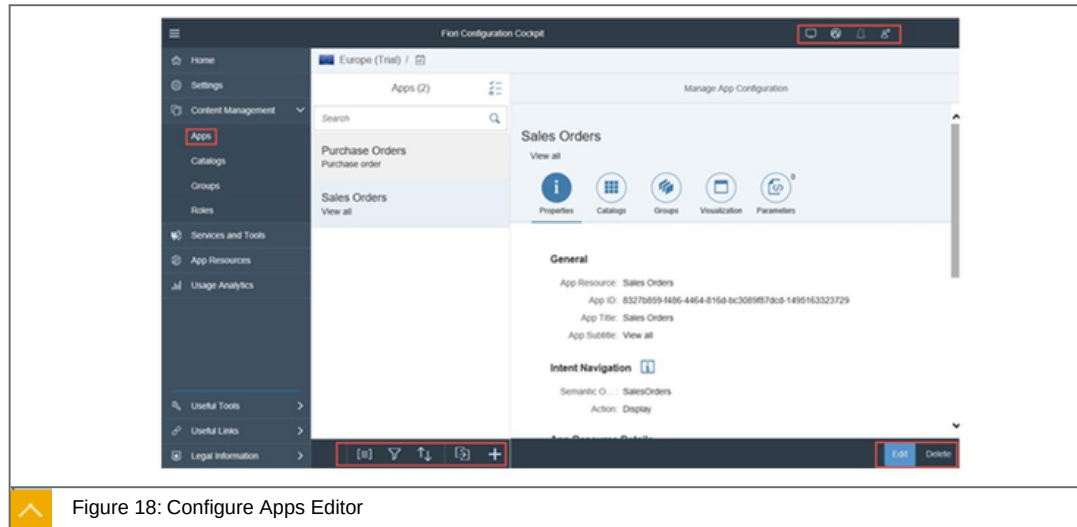
1. Search for an existing app from the following:

- Your SAP Cloud Platform account in the SAP Fiori launchpad configuration cockpit

- Technical catalogs for SAP GUI or Web Dynpro ABAP apps from an ABAP backend system

2. Copy the configuration.
3. Edit the app configuration and properties according to your business scenario using the Configure Apps editor in the SAP Fiori launchpad configuration cockpit.

Configure Apps Editor



Using the Configure Apps editor, you can do the following:

- Create a new app tile.
- Edit an app tile configuration (properties/catalogs/groups/visualization/parameters).
- Delete an app tile configuration.
- Duplicate an app tile configuration.
- Preview an app tile configuration.
- Publish a site.

Creating an App Tile

An app tile can be created using the Configure Apps editor in the SAP Fiori configuration cockpit. An app tile can also be created using SAP Web IDE.



Note:

An application deployed on SAP Cloud Platform can be registered to the SAP Fiori launchpad directly from SAP Web IDE. Once registration is complete, a new app tile is created. Refer to the Web IDE documentation at <https://help.hana.ondemand.com/webide/frameset.htm?b075d767e91f422b8b4be4e8c6fd5ab7.html> for more details.

Maintaining an App Configuration

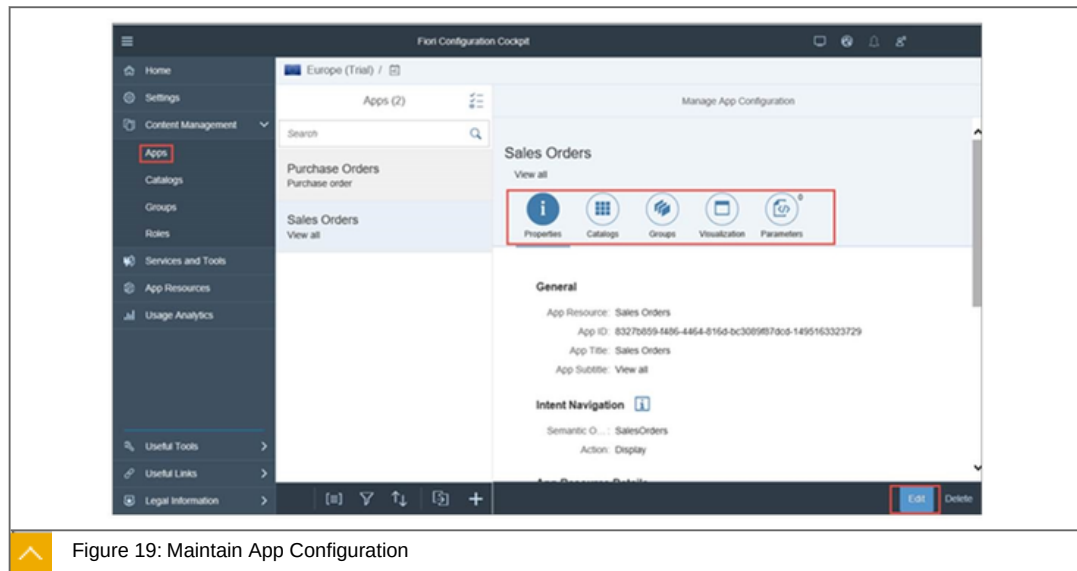


Figure 19: Maintain App Configuration

After an app tile is created, administrators often need to adapt the configuration and change the properties, visualization, parameters, and so on for their app.

Various properties can be maintained for an app as follows:

- App properties

- Visualization properties

- Navigation properties

App Properties

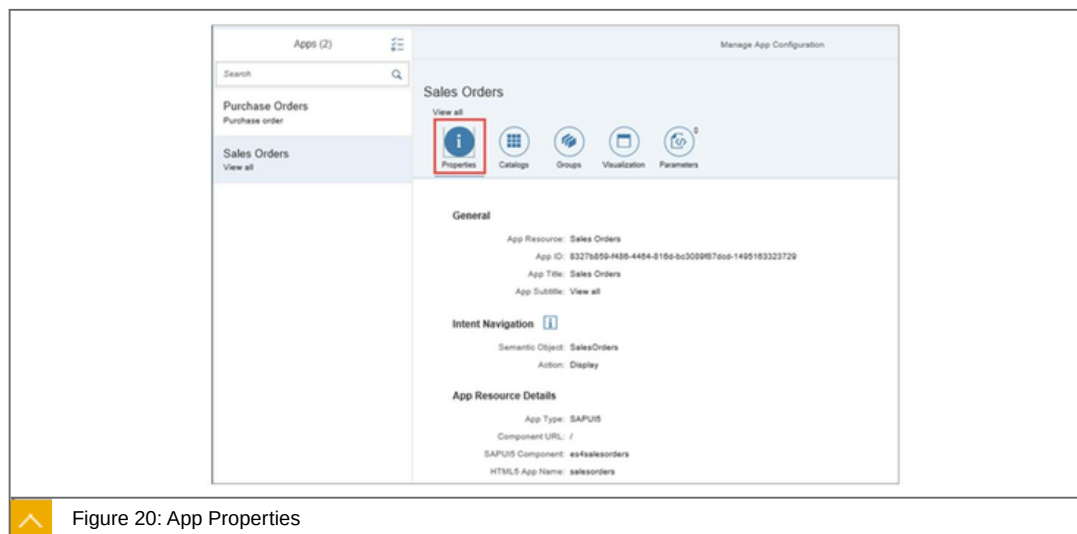


Figure 20: App Properties

App properties include the following:

- General properties such as app resource, title, and subtitle

Intent navigation

Semantic object, for example, sales order

Action, for example, display

App resource details, which include the app type and target properties

App type (SAPUI5/URL/SAPUI5 News/Shell Plugin/SAP GUI and Web Dynpro ABAP/SAP Mobile Documents)

App assignment to catalogs and groups

Visualization Properties

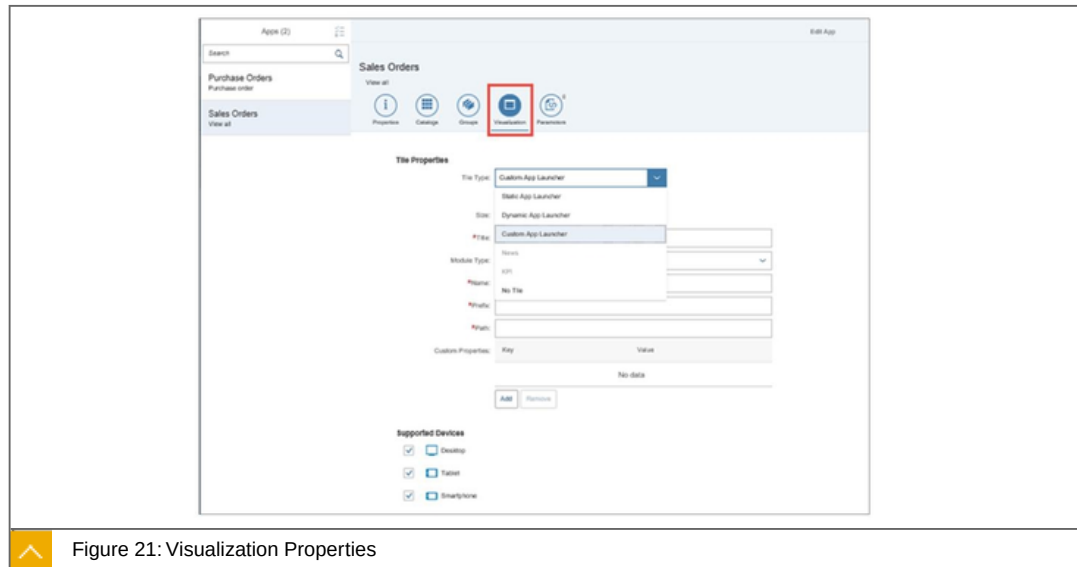


Figure 21: Visualization Properties

Visual properties include tile type, properties, and supported devices. The following tile types are supported:

Static App Launcher

Dynamic App Launcher

Custom App Launcher

News

KPI

No Tile

Navigation Parameters

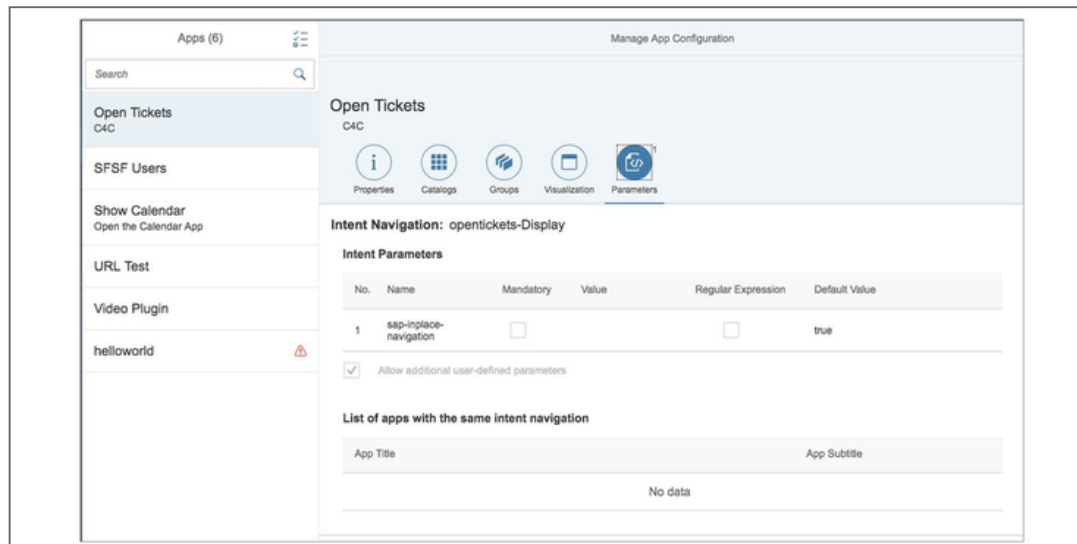


Figure 22: Navigation Parameters

Each intent consists of a combination of a semantic object and an action. You define these on the Properties tab. You can define parameters to pass with the navigation request. By defining parameters, you ensure that users see different views of the same app.

Properties include the following:

- Intent properties

- Parameters to pass with the navigation request

For example, for a Leave Request app, a user chooses a date and clicks a button to create a leave request.

Properties are defined as follows:

- Semantic Object: LeaveRequest

- Action: Create Parameters

- Parameters, as follows:

- Name: startDate
- Value: 10/01/2017

Navigation Parameters with Default Values



Note:

This feature is only available if you are subscribed to SAP Fiori Cloud for S/4HANA Finance content.

User-specified default values can be configured in the navigation parameters by administrators so users do not have to enter the same values every time they navigate to an app.

To use navigation parameters with default values, administrators need to do the following:

1. Import the Default Values shell plugin app using the Transport Manager.
2. In the Configure Apps editor, choose the **Parameters** tab. For each application that has been developed to support user defaults as a navigation parameter, populate the parameter fields using the following structures:

Parameter Name: <parametername>

Parameter Default Value: userdefault.<parametername>

This parameter enables the app to use the Default Values shell plugin app.

Users can then set up the default values in the SAP Fiori launchpad under **User Preferences** **Default Values**.

Configuring an App

Depending on the business scenarios, apps can be configured according to their use case. Various types of app configuration are supported in the SAP Fiori launchpad including the following:

Apps from the SAP Cloud Platform subaccount

SAP GUI for HTML and Web Dynpro ABAP apps

Shell Plugin app

News app

Creating or Configuring Subaccount Apps

You can search for an app in your SAP Cloud Platform Portal subaccount and add the configurations to the Configure Apps editor, which allows you to change the properties according to your use case.

To search for apps deployed in your SAP Cloud Platform Portal subaccount, you can do one of the following:

Use app resources from the SAP Fiori launchpad configuration cockpit. When you select the **Add to site** option, the configuration is copied directly to the Configure Apps editor

Search for an app in the Configure Apps editor. When you select the app, the configuration is copied

Configuring SAP GUI for HTML and Web Dynpro ABAP Apps



The screenshot shows the 'General' configuration page for a new app. Key fields include:

- App Resource:** A button labeled 'Browse for app resources'.
- *App ID:** A text field containing '3348773b-1832-4466-8e90-e90ed63e35cf-1496381183650'.
- *App Title:** A text field containing 'New App'.
- App Subtitle:** An empty text field.
- Intent Navigation:** A section with an information icon.
- *Semantic Object:** A text field containing 'Shell'.
- *Action:** A text field containing 'Plugin'.
- App Resource Details:** A section containing:
 - App Type:** A dropdown menu set to 'SAP GUI'.
 - Transaction Code:** An empty text field.
 - System Alias:** An empty text field.
 - SAP Business Client URI:** An empty text field.

Figure 23: Configuring SAP GUI for HTML and Web Dynpro ABAP Apps

In the SAP Fiori launchpad configuration cockpit, administrators can create a new app or configure an existing SAP GUI for HTML or Web Dynpro ABAP app.

The following are prerequisites:

The backend system is configured with version SAP_UI 751 and onwards, and the backend service `sui_tm_mm_rest` is available and can be accessed.

The HTTP destination for the back-end system is configured in the SAP Cloud Platform Cockpit, where SAP GUI for HTML and Web Dynpro ABAP apps can be selected.

The following destination properties are required:

`sap-client`: Optional property. The client number as defined in your back-end system

Usage: Back end

For Web Dynpro ABAP: Application ID (from transaction SICF), configuration ID, system alias, and SAP Business URI (path to SAP NetWeaver Business Client)

To configure an app from an existing ABAP back-end system:

1. In the Configure Apps editor, select a back-end system and view its technical catalogs.
Technical catalogs contain references to SAP GUI transactions and Web Dynpro ABAP applications that are located in a back-end system.
2. Select an app from the catalog.
All of the properties of the selected app are then automatically populated in the application properties fields.

For more information, refer to the documentation at <https://help.sap.com/viewer/3ca6847da92847d79b27753d690ac5d5/Cloud/en-US/d36ea2f36ae443289d9968c7e1f9b87f.html>.

To create new SAP GUI for HTML or Web Dynpro ABAP apps in the SAP Fiori launchpad:

1. Under Properties in the Configure Apps editor, select the app type SAP GUI or Web Dynpro ABAP.
2. Provide the required properties manually.

For SAP GUI for HTML, the following are required: transaction ID, system alias, and SAP Business URI (path to SAP NetWeaver Business Client).

Configuring a Shell Plugin App



General

App Resource: Browse for app resources

*App ID: 3348773b-1832-4466-8e90-e90ed953e35cf-1496381183650

*App Title: New App

App Subtitle:

App Resource Details

App Type: Shell Plugin

Shell Plugin Type: Custom

Component URL: Location of component.js in HTML5 project

*SAPUI5 Component: SAPUI5 component name without .component suffix

HTML5 App Name: Name of deployed app or app subscription

Figure 24: Configuring a Shell Plugin App

When the SAP Fiori launchpad is launched, all shell plugin apps are initialized while the loading is in progress.

Shell plugin apps are SAPUI5 apps, but they are more flexible and can be extended using any SAP Fiori launchpad API. To consume the SAP Fiori launchpad APIs, go to <https://sapui5.hana.ondemand.com/sdk/#docs/api/symbols/sap.ushell.services.html>.

A prerequisite is that a Shell Plugin app has been developed and deployed to your SAP Cloud Platform account.

The following are examples of SAP Fiori launchpad APIs:

Add buttons to the header of your app or to the actions menu.

Add a popup welcome page.

Create a user feedback dialog box.

To configure a Shell Plugin app:

1. In the Configure Apps editor, select the back-end systems from App Resource, and search for the Shell Plugin app.

The target fields are populated with the values from the app that you selected.

2. In the Configure Apps editor, select Shell Plugin as your App Type and choose the Shell Plugin type.

The following are the Shell Plugin type options:

Custom: a flexible SAPUI5 app that can be extended by using standard out-of-the-box SAPUI5 APIs

User Defaults: a Shell Plugin app that enables users to set default values for the navigation parameters of their apps (only those apps that have been developed to support user defaults)

UI Adaptation: a Shell Plugin app that enables users to adapt the UI of their apps (only those apps that have been developed to support UI adaptation)

For more information, see how UI adaptation at runtime (RTA) allows key users (assigned users) to change the UI of a SAPUI5 app at runtime, for example, by adding, removing, or moving fields and groups. These UI changes apply to all users of this app. See https://help.sap.com/saphelp_uiaddon20/helpdata/en/5c/424437bf794f809087fdce391149f2/frameset.htm.

Configuring a News App

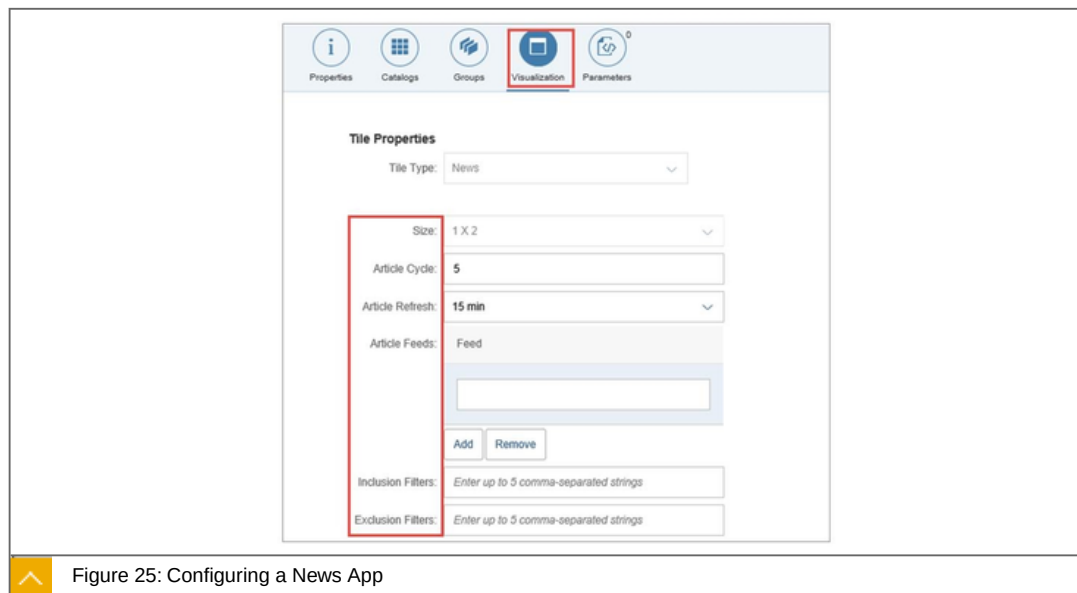


Figure 25: Configuring a News App

With the News app, users can see news headlines in their news tile. When they click the news tile, an external URL navigates to an app that displays an RSS news feed.

To configure a News app:

1. On the Properties tab of the Configure Apps editor, select the app type SAPUI5 News.

This app type is an out-of-the-box news reader application with predefined target properties, so the target properties are automatically populated.

2. On the Visualization tab, configure the following tile properties:

Size

Article cycle

Refresh interval

RSS feeds up to 10

Inclusion and exclusion filters

A news app tile can also be configured to launch a custom app instead of the out-of-the-box news app. A prerequisite is a custom SAPUI5 app that can read RSS feeds from different URLs and display them. Refer to the documentation for more details.

To configure a custom News app, follow the configuration steps above for a News app, replacing the default target values with the values of the custom app in the following fields:

Component URL field: replace the default value with the location of your component.js file in the HTML5 project

SAPUI5 Component Name field: replace the default value with the name of the SAPUI5 component without the .component suffix

HTML5 App Name field: enter the name of the HTML5 app

Configuring a URL App



Figure 26: Configuring a URL App

A URL app is configured by entering a URL in its configuration properties. This URL is opened in a new window or tab.

A Dynamic URL app is based on destination properties (configured in SAP Cloud Platform Cockpit) and path properties. Depending on the values set for the same destination name in different environments, the same URL can navigate to different apps in the different environments.

Note the following:

The URL must include the port number 443.

The destination must have the property „Usage“, which must either be defined as „Backend“ or „Dynamic“.

Configuring a BSP App

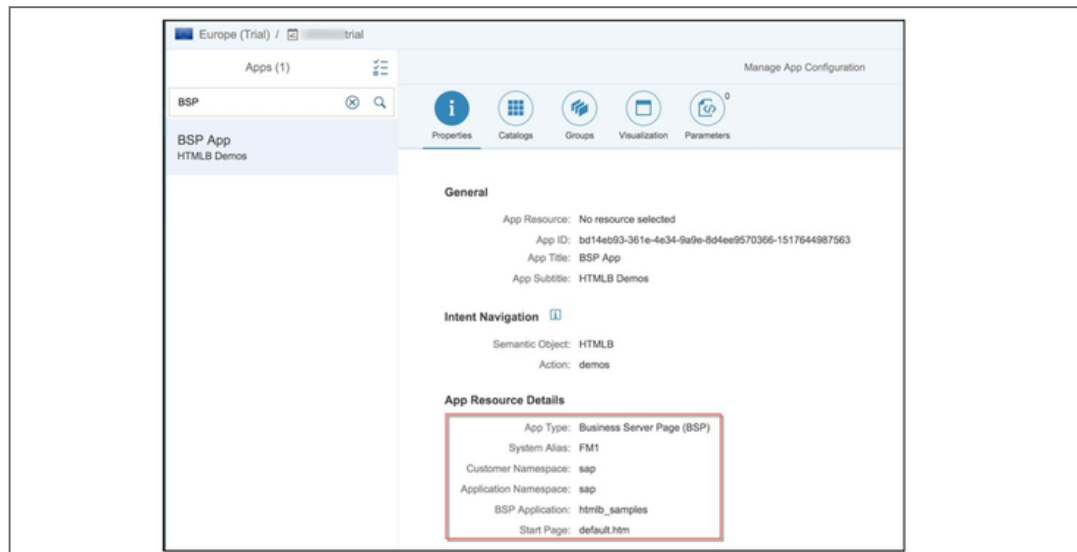


Figure 27: Configuring a BSP App

The following is a prerequisite for configuring a BSP app: destination to the backend, with the properties Usage = Backend and optionally sap-client=<sap client>

Parameters include the following:

system_alias: the destination name of the SAP system that hosts the BSP application

cust_namespace: a unique parameter that appears in the back-end system as a service folder. It is the namespace ID of the BSP application.

bc and bsp: default parameters

app_namespace: a unique parameter used as a prefix for the application name

bsp_app_name: the name of the BSP application (without its namespace)

start_page: the page ID of the BSP application start page, for example, start.html

Creating Catalogs, Roles, and Groups

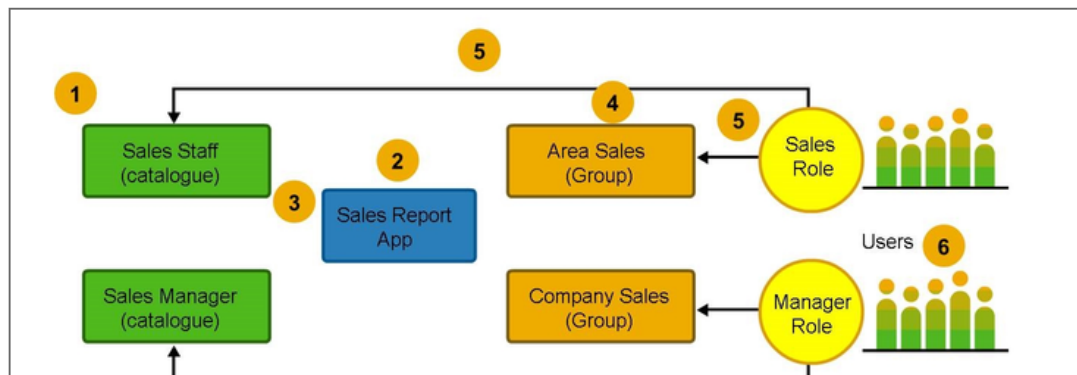
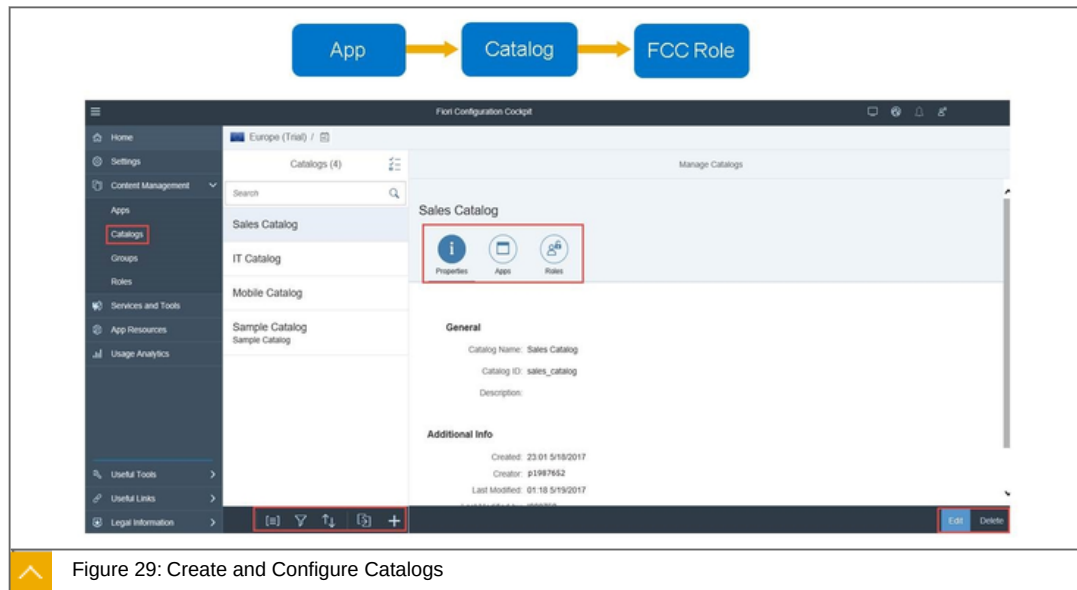


Figure 28: Create Catalogs, Roles, and Groups

Although the steps can be completed in a different order, the following is a typical workflow for creating catalogs, roles, and groups:

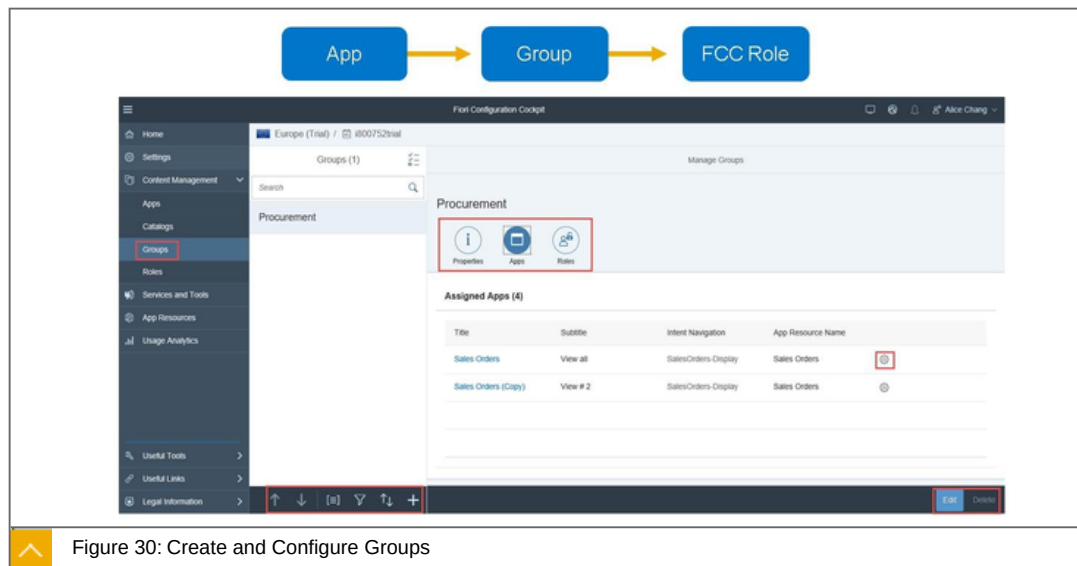
1. Create catalogs
2. Configure apps
3. Assign apps to catalog
4. Create groups
5. Assign roles to catalogs and groups
6. Assign users to roles



User access and permissions to a site, pages, or apps can be managed through catalogs and roles. Catalogs are collections of apps made available for one or more roles. Administrators assign roles and apps to a catalog. Instead of assigning each app to a role, you can bundle them into a catalog and then assign one or more roles to each of the catalogs. With this approach, users who are assigned to a particular role can access the apps that are assigned to the same catalog as their role. Depending on their role, users can browse through a Catalog library from their SAP Fiori launchpad, and choose the apps that they want to display on the entry page of their launchpad.

The Manage Catalogs editor in the SAP Fiori launchpad configuration cockpit is used to create new catalogs or configure existing ones. The Manage Catalogs editor allows you to do the following tasks:

- Create a new catalog.
- Edit a catalog.
- Duplicate a catalog.
- Delete a catalog or delete multiple catalogs.
- View details about the apps assigned to a catalog.



A group is a collection of apps tiles and links representing a subset of apps. Groups are set up to ensure related apps are organized and visible to users in their SAP Fiori launchpad. End users can then personalize their SAP Fiori launchpad by adding apps or removing apps from the out-of-the-box or self-defined groups.

The Manage Groups editor in the SAP Fiori launchpad configuration cockpit is used to create new groups or configure existing ones. The Manage Groups editor allows you to do the following tasks:

- Create a new group.
- Edit a group.
- Duplicate a group.
- Delete a group or delete multiple groups.
- View the navigation target settings.
- Lock a group.
- Reorder the tiles in a group.
- Reorder the group display in the SAP Fiori launchpad.



Note:

You can also create a group on the Groups tab of the Configure Apps editor.

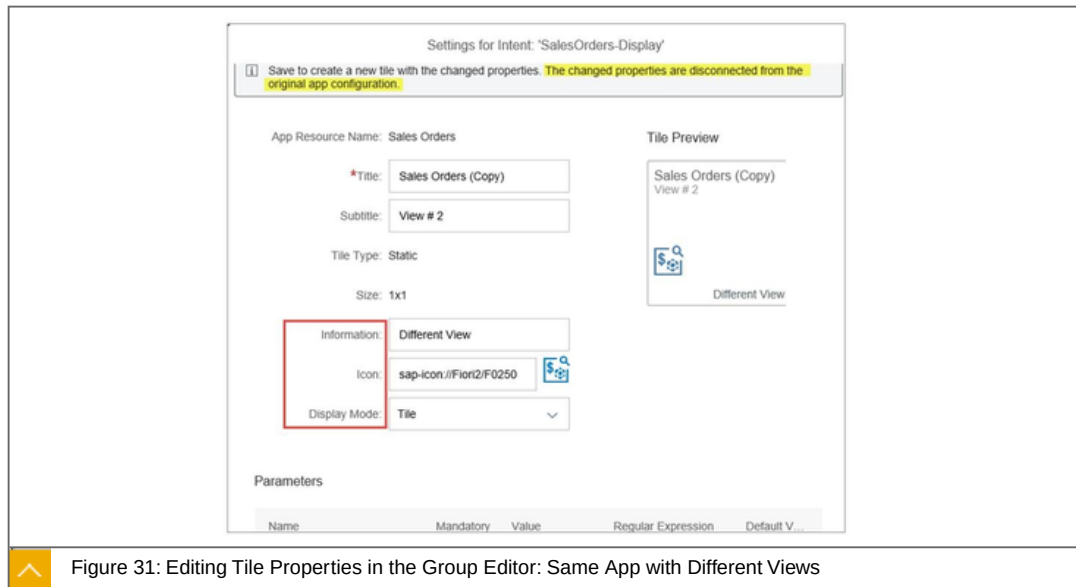


Figure 31: Editing Tile Properties in the Group Editor: Same App with Different Views

The tile properties of an app are usually defined in the Configure Apps editor. However, there might be situations where you want the same app to be represented by different tiles for different users and roles. To support this scenario, you can duplicate a tile in the Manage Groups editor and then edit its properties as follows:

Configure different intent parameters for each tile. The intent remains the same, but because the parameters are different, each tile launches a different view of the same app.

Configure different visual properties for each tile, such as title, subtitle, and description.

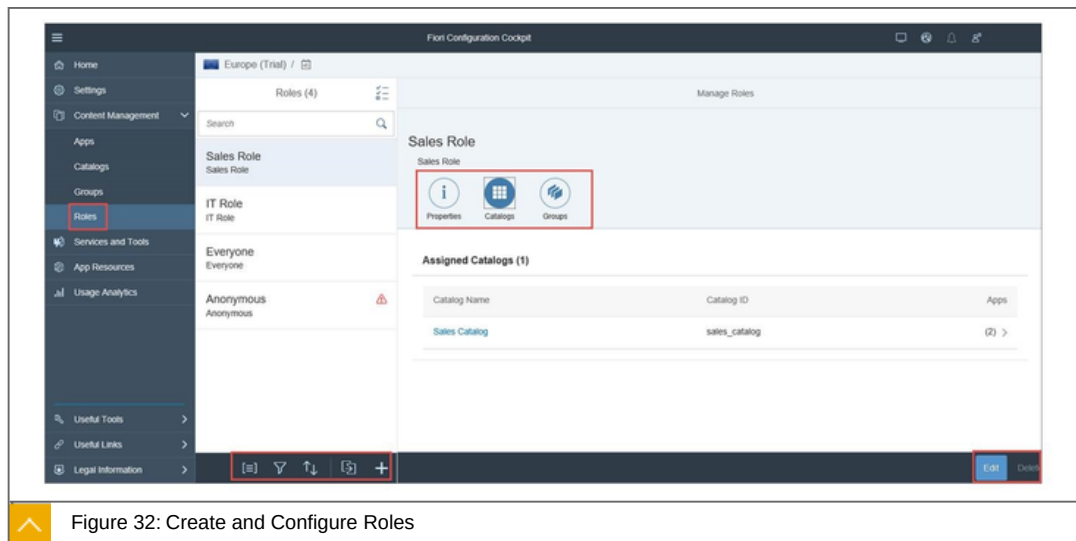


Figure 32: Create and Configure Roles

Access to SAP Fiori launchpad objects is managed through roles. Roles can be created in the SAP Cloud Platform Cockpit under Configure Portal Service, after which they automatically appear in the SAP Fiori launchpad configuration cockpit. Alternatively, you can use the SAP Fiori launchpad configuration cockpit to create and configure roles. New roles created in the Roles editor of the SAP Fiori launchpad configuration cockpit must have a matching role defined in the SAP Cloud Platform account. You can then assign catalogs and groups to a role.

The Roles editor allows you to perform the following tasks:

Create a new role.

Edit a role.

Duplicate a role.

Delete a role or delete multiple roles.

View details about the catalogs and groups assigned to a role.



LESSON SUMMARY

You should now be able to:

Create content in the SAP Fiori launchpad configuration cockpit

Unit 2

Lesson 6

Consuming Site Content



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Consume site content

Consumption of Site Content

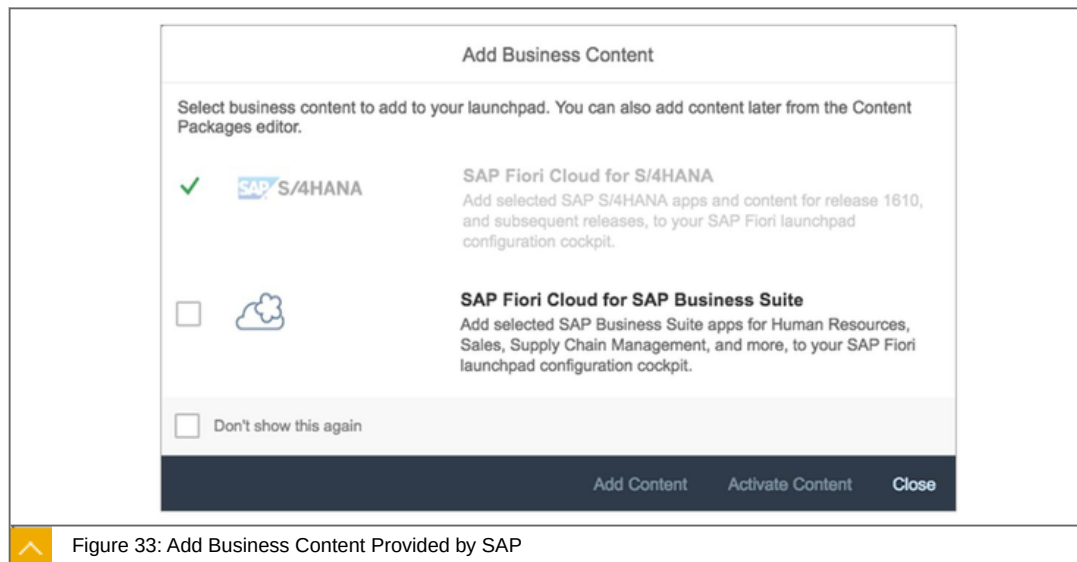


Figure 33: Add Business Content Provided by SAP

Administrators manage business content provided by SAP in the Content Packages editor in the SAP Fiori launchpad configuration cockpit. You can add various types of business content to your site, depending on your license. For example, both SAP S/4HANA content and SAP Business Suite content are included in a single SAP Fiori Cloud license. When you open the SAP Fiori launchpad configuration cockpit for the first time for a specific subaccount, a popup for selecting business content is displayed. The content you select to add to your site is listed in the Content Packages editor.

When SAP updates the business content, the content in your SAP Fiori launchpad configuration cockpit is updated automatically, and you receive a notification that asks you whether you want to publish the site with these updates. The content cannot be modified, except for assigning existing roles to groups or catalogs and unassigning these roles.

SAP roles, catalogs, and apps can be duplicated. The duplicated object is a standalone local object that is no longer connected to the corresponding object in the source subaccount. However, the duplicated object maintains original assignments to content provided by SAP according to the following hierarchy: role, catalog, and app. For example, a duplicated role maintains assignments to catalogs provided by SAP, and a duplicated catalog maintains assignments to apps provided by SAP. In this way, the assigned content provided by SAP

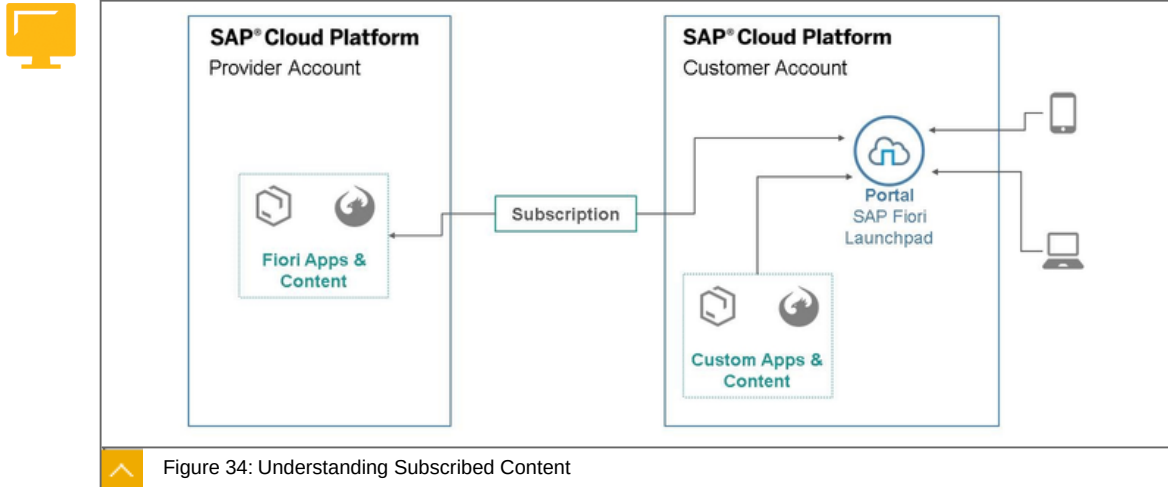
continues to be updated when changes are made by SAP. Note that a duplicated application becomes local and does not receive any updates.

Note the following points:

SAP business content includes translation files.

SAP business content does not include themes.

Subscribed Content



Subscribed content, such as SAP Fiori Cloud for SAP S4/HANA or SAP Business suite, is hosted in a provider account. Therefore, the account owner is responsible for building and running an application for consumption by multiple consumers. Consumers do not deploy applications in their subaccounts, but simply subscribe to the provider application. As a result, a subscription is created in the consumer subaccount. This subscription represents the contract or relation between a subaccount (tenant) and a provider application. Both subscribed and custom apps appear in the SAP Fiori launchpad configuration cockpit and can be added to the SAP Fiori launchpad.



LESSON SUMMARY

You should now be able to:

Consume site content

Unit 2

Lesson 7

Configuring Site Settings



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Configure site settings

Site Settings Configuration

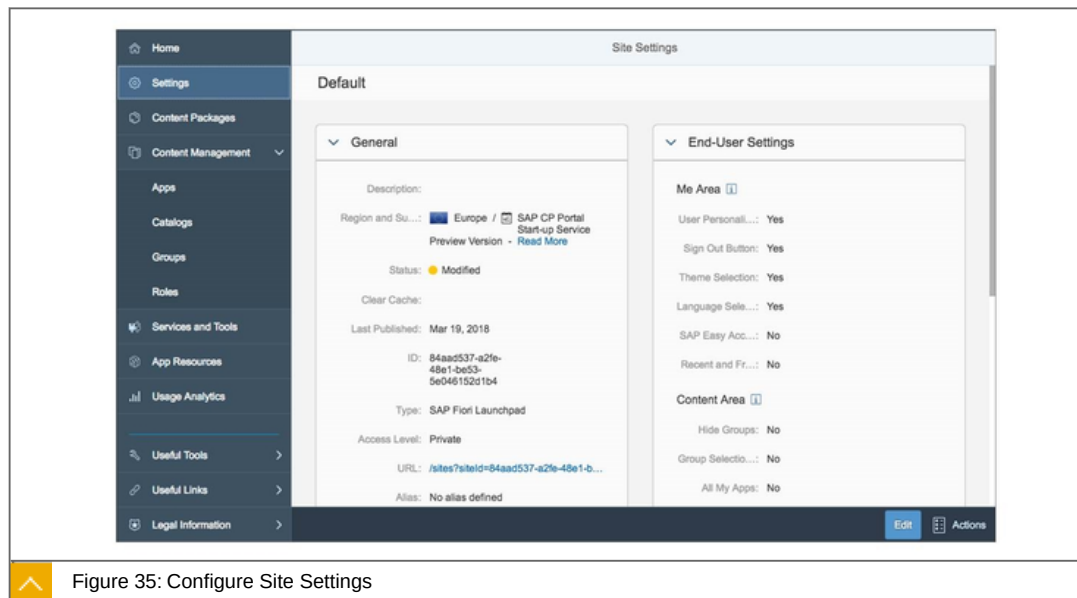


Figure 35: Configure Site Settings

Runtime site settings can be configured using the Site Settings editor in the SAP Fiori launchpad configuration cockpit.

The following properties can be configured:

General settings: set an external alias and determine whether a timeout message should be shown

End user settings: configure what is visible and available in the SAP Fiori launchpad, for example, whether users can personalize the SAP Fiori launchpad and what is available in the Me area

System settings: change the UI5 version used, the access type of the SAP Fiori launchpad, and whether and how classic UI Technologies (WebDynpro and SAP GUI HTML) are shown

After you make changes to the site settings, the site must be published again.

See the documentation relating to site settings at <https://help.sap.com/viewer/3ca6847da92847d79b27753d690ac5d5/Cloud/en-US/cb10beafb67c4ba09c13a3e6a9094b6d.html>.



LESSON SUMMARY

You should now be able to:

- Configure site settings

Unit 2

Lesson 8

Clearing App Cache



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Clear app cache

Clearance of App Cache



The screenshot shows the SAP Fiori Configuration Cockpit interface. The left sidebar contains navigation options: Home, Settings, Content Packages, Content Management, Services and Tools, App Resources (selected), and Usage Analytics. The main area displays 'Business Content: Custom Content Only' and a message 'Last update ended with warnings. Download report' highlighted with a red box. Below this is a table titled 'App Resources (216)' with columns 'App Title', 'HTML5 App Name', and 'Version'. The table lists three apps: 'Display G/L Account Balances' (version 5.0.3), 'Realignment Results - Profitability Analysis' (version 5.0.4), and 'c4ctickets' (version 1.0.0). At the bottom right, a modal dialog is open with the option 'Clear HTML5 Application Cache' and a 'Take Offline' button.

App Title	HTML5 App Name	Version
Display G/L Account Balances	accountbalancesdisplaycnfdef2	5.0.3
Realignment Results - Profitability Analysis	arealignmentresultdisplaye4fb1	5.0.4
c4ctickets	c4ctickets	1.0.0

Figure 36: Clear App Cache

HTML5 static resources are stored in the app cache. If an app is redeployed to the account with different resources, an automatic cache invalidation process is triggered during loading of the SAP Fiori launchpad configuration cockpit. In this case, the app resources are fetched from the repository instead of the cache.

If you want to check the results of the last app replication process, a report is available in the Apps editor and in App Resources.

If you need to manually invalidate the app cache, you can do this using the **Action** button in the Site Settings editor. Choose **Clear HTML5 Application Cache**.



LESSON SUMMARY

You should now be able to:

Clear app cache

Unit 2

Lesson 9

Managing Site Themes



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Manage site themes

Site Theme Management

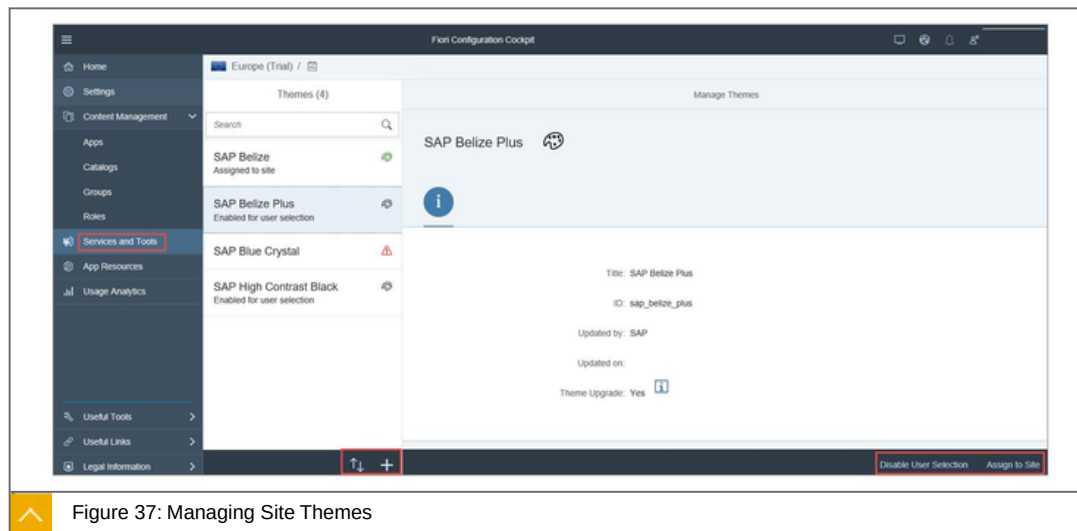


Figure 37: Managing Site Themes

A theme defines the visual appearance of an application, for example, the font size and type, the colors of UI controls, and background images. A default theme provided by SAP is assigned to each SAP Fiori launchpad site and its apps. A custom theme can also be used instead of the default theme. A custom theme is a new theme adapted from an SAP standard to have a different visual appearance. Once the theme is created, the theme needs to be imported and assigned it to the site.



Note:

Whenever the SAPUI5 version changes, the themes have to be upgraded using the upgrade service.

The UI Theme Designer is available under Services and Tools in the SAP Fiori launchpad configuration cockpit. You can use the UI Theme Designer to build themes. Theming parameters such as color values, text sizes, icons, background images, and so on can be configured.

The Theme Manager is also available under Services and Tools in the SAP Fiori launchpad configuration cockpit. You can use the Theme Manager to view, add, assign, enable, or disable

the available themes. All of the available themes are displayed in a list. The icon next to the theme shows whether the theme has already been assigned to a site or enabled for user selection.

To use a custom theme, perform the following steps in the Theme Manager:

1. Add the custom theme.
2. Assign the theme to the SAP Fiori launchpad site and its apps.
3. Enable one or more themes for user selection in runtime.



LESSON SUMMARY

You should now be able to:

Manage site themes

Learning Assessment

1. Catalogs are used to configure which tiles are displayed when opening the Fiori Launchpad.

Determine whether this statement is true or false.

☐ True

☐ False

2. Which of the following UT technologies can be configured using Tiles besides the SAPUI5 applications?

Choose the correct answers.

☐ **A** SAP GUI for HTML

☐ **B** WebDynpro ABAP

☐ **C** WebDynpro Java

☐ **D** Business Server Page

3. Subscribed content, e.g. Fiori Cloud for S4/HANA, is copied into your subaccount.

Determine whether this statement is true or false.

☐ True

☐ False

Learning Assessment - Answers

1. Catalogs are used to configure which tiles are displayed when opening the Fiori Launchpad.

Determine whether this statement is true or false.

☐ True

☒ False

You are correct! Catalogs are collections of applications which are assigned to roles. A user may only use applications which are in the catalogs assigned to their role. Groups are used to configure which tiles are shown in the Fiori Launchpad and are also assigned to a Role. Depending on the general site settings users may personalize the tiles within a group. Refer to the lesson 'Describing SAP Fiori Launchpad Objects' for more details.

2. Which of the following UT technologies can be configured using Tiles besides the SAPUI5 applications?

Choose the correct answers.

☒ **A** SAP GUI for HTML

☒ **B** WebDynpro ABAP

☐ **C** WebDynpro Java

☒ **D** Business Server Page

You are correct! The SCP Portal Service allows you to create tiles for SAPUI5, SAP GUI for HTML, WebDynpro ABAP and Business Server Pages. Other UI technologies cannot be launched directly in the Fiori launchpad but may be integrated via URL tiles or through the use of an iFrame in a UI5 application. Refer to the lesson 'Exploring the App Types and Tile Types' for more details.

3. Subscribed content, e.g. Fiori Cloud for S4/HANA, is copied into your subaccount.

Determine whether this statement is true or false.

☐

True

☒

False

You are correct! Subscribed Content, such as Fiori Cloud for S4/HANA or Business suite, is hosted in a Provider Account. The sources stay within the provider account and are consumed during runtime in the subscriber account. Consumers do not need to deploy applications in their subaccounts, but simply subscribe to the provider application. Refer to the lesson 'Consuming Site Content' for more details.

UNIT 3

SAP Fiori Launchpad Site Development

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UNIT OBJECTIVES

Add and remove custom Items in the Options menu

Create apps to access external resources

Understand OData structure

Access API documentation

Unit 3

Lesson 1

Adding and Removing Custom Items in the Options Menu



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Add and remove custom Items in the Options menu

Addition and Removal of Custom Items in the Options Menu

SAPUI5 offers various APIs that can be used to extend the SAP Fiori launchpad UI and capabilities. For example, the SAP Fiori launchpad renderer (from the `sap.ushell.renderers.fiori2` namespace) can be used to extend the UI component in the launchpad sites. The methods of the `RendererExtensions` API from the renderer allow an item to be added to or removed from the Options menu.

To add/remove a button from the Options menu, use the following methods:



```
addOptionsActionButton(oButton[, sLaunchpadState1],[sLaunchpadState2],  
[sLaunchpadState3])
```

```
removeOptionsActionButton(oButton[, sLaunchpadState1],[sLaunchpadState2],  
[sLaunchpadState3])
```

The parameters are as follows:

`oButton`

This is the button to be added. The button should have an icon, text, tooltip, and a press callback.

`sLaunchpadState1/sLaunchpadState2/sLaunchpadState3`

This is optional. It refers to the launchpad state in which to add the button.

For example, to add a button to the Options menu, you need to implement a specific logic. You need to determine the launchpad state parameter values:

`Launchpadstate.App`

This is the launchpad state when running a SAP Fiori app.

`Launchpadstate.Catalog`

This is the launchpad state when the tile catalog is open.

`Launchpadstate.Home`:

This is the launchpad state when the home page is open.

If no launchpad state is passed, the button is added in all states

The following is sample code to add a button to the Options menu:

```
<SCRIPT language = "JavaScript">
...
var button1 = new sap.m.Button();
var rendererExt = sap.ushell.renderers.fiori2.RendererExtensions;
rendererExt.addOptionsActionSheetButton(button1,rendererExt.LaunchpadState.Home,
rendererExt.LaunchpadState.Catalog,rendererExt.LaunchpadState.App);
...
</SCRIPT>
```



LESSON SUMMARY

You should now be able to:

- Add and remove custom Items in the Options menu

Unit 3

Lesson 2

Creating Apps to Access External Resources



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Create apps to access external resources

Creation of Apps to Access External Resources

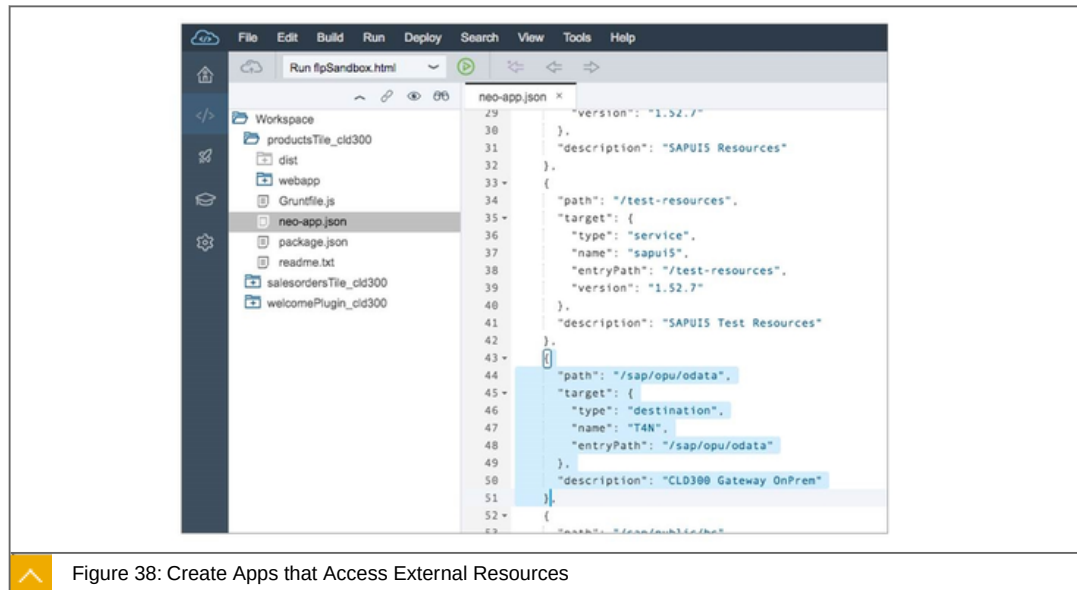


Figure 38: Create Apps that Access External Resources

The SAP Fiori launchpad on cloud provides an automatic routing mechanism for accessing external resources. It automatically intercepts all HTTP calls and determines the calls relevant for the application context and launchpad context. Developers only need to write the code once and deploy it, and no specific coding for accessing back-end systems or loading custom SAPUI5 libraries is required.

To implement this, the neo-app.json file is used. This is an application descriptor file, and you can use it to configure the behavior of your HTML5 application.

For example, to access a back end via a destination, you can use the following sample code:

```
...
{
  "path": "/sap/opu/odata",
  "target": {
    "type": "destination",
    "name": "T4N",
    "entryPath": "/sap/opu/odata"
  },
  "description": "CLD300 Gateway OnPrem"
},
}
```

**Note:**

The automatic routing mechanism assumes the SAP Fiori launchpad is the sole entry point for running business applications. If a different routing mechanism is in use to access the back-end systems or load custom SAPUI5 libraries, SAP recommends switching to the automatic mechanism. Otherwise, turn off the automatic routing mechanism by adding a header named `disableInterception` to the `headerWhiteList` section of your `neo-app.json` file.

**LESSON SUMMARY**

You should now be able to:

- Create apps to access external resources

Unit 3

Lesson 3

Understanding OData Structure



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Understand OData structure

OData Structure

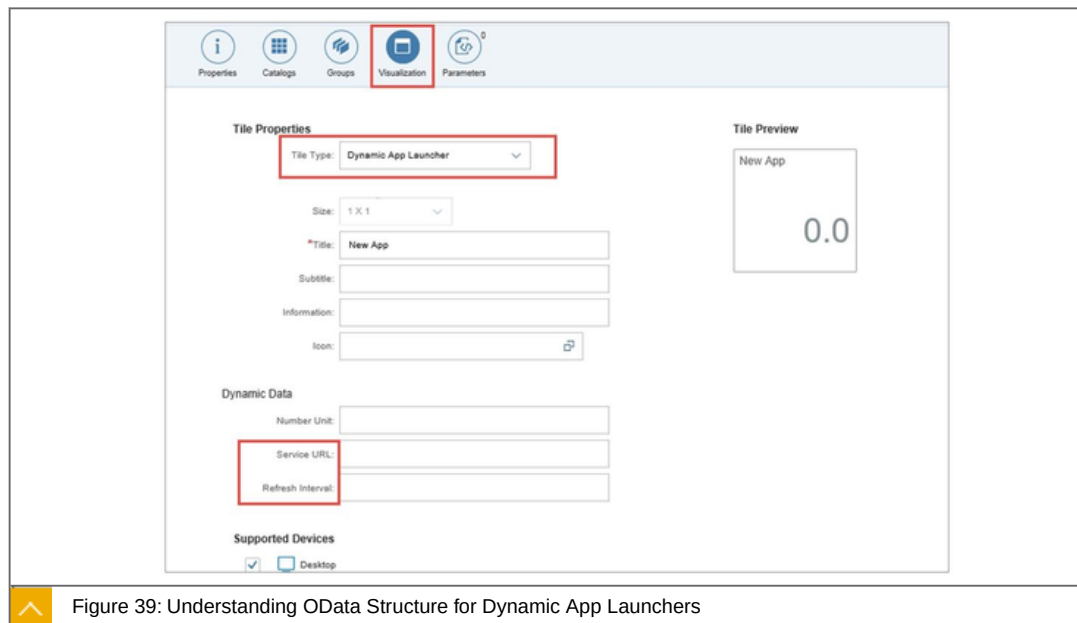


Figure 39: Understanding OData Structure for Dynamic App Launchers

The display of dynamic content is commonly required in the SAP Fiori launchpad. To support this use case, you need a dynamic tile (Dynamic App launcher) and an OData service that pulls information from a back-end system.

When configuring the dynamic tile in this case, several important visualization properties are required:

Tile type: Dynamic App Launcher

Service URL: OData service from which data should be read

Refresh interval

You also need to create an OData service that returns the configuration properties. The configuration properties configured in the OData service are as follows:

icon: icon name

info: text to be displayed at the bottom of the tile

infoState: the color of the tile is adapted according to the value of this property

number: number to be displayed in the top right corner of the tile.

numberDigits: number of digits to be displayed following the decimal separator

numberFactor: a factor for scaling numbers

numberState: the color of the number is adapted according to the value of this property

numberUnit: unit to be displayed below the number

stateArrow: displays an arrow indicating the trend

Subtitle

targetParams: list of key-value-pairs separated by ampersands. These parameters are passed to the application as business parameters or URL parameters.

Title



Note:

If the service returns an entity collection (rather than a single entity), all values from the number elements are accumulated.

The following is sample code:

```
{
  "d": {
    "icon": "sap-icon://travel-expense",
    "info": "Quarter Ends!",
    "infoState": "Critical",
    "number": 43.333,
    "numberDigits": 1,
    "numberFactor": "k",
    "numberState": "Positive",
    "numberUnit": "EUR",
    "stateArrow": "Up",
    "subtitle": "Quarterly overview",
    "title": "Travel Expenses",
  }
}
```



LESSON SUMMARY

You should now be able to:

Understand OData structure

Unit 3

Lesson 4

Accessing API Documentation



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Access API documentation

API Documentation

The SAP Cloud Platform Portal offers its own API catalog in the SAP API Business Hub. The SAP API Business Hub is a central catalog that brings together APIs from different SAP cloud products. Using the Hub, you can discover and consume digital content packages with APIs, pre-packaged integrations, business services, and sample apps from SAP and selected partners. You can access the SAP API Business Hub at <https://api.sap.com>.

The SAP Cloud Platform Portal offers several APIs for developers to use, for example, the SAP Fiori launchpad on cloud APIs. You can access the API documentation at <https://api.sap.com/shell/discover/contentpackage/SAPCLOUDPLATFORMPORTAL?section=ARTIFACTS>.

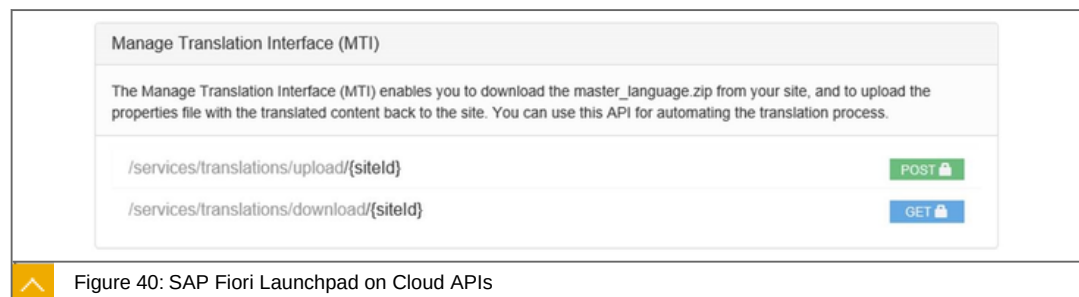


Figure 40: SAP Fiori Launchpad on Cloud APIs

You can easily create freestyle and SAP Fiori launchpad style business sites for employees, customers, and partners. The portal service offers a set of capabilities to design, build, manage, authorize, and analyse usage of your sites. To use the APIs, obtain an OAuth token at https://help.hana.ondemand.com/cloud_portal/frameset.htm?e655a4b76dd24a98ba53c9a6334d5e56.html. To obtain the portal service base URL, in the SAP Cloud Platform Cockpit, in your subaccount, click the portal service tile and copy the Go To Service link. On productive accounts, the portal service base URL should be in the following format: `https://fipnwc-{subaccountID}.dispatcher.{landscapeDomain}`

The following are some examples:

- Custom SAPUI5 account version
- Manages the SAPUI5 version for the account
- Translations API

Enables downloading of the master_language.zip from a site, and uploading of the properties file with the translated content back to a site. This API can be used for automating the translation process.

OAuth

OAuth defines an open protocol for allowing secure API authorization of desktop, mobile, and web applications through a simple and standard method. APIs that are protected by the OAuth service of SAP Cloud Platform can only be used by OAuth authenticated users. The authenticated users do not have to be recognized by any IDP, but they must possess the relevant OAuth tokens.

For each method of an API (also known as scope), there is a different token type. For example, in the Manage Translation Interface API, download master language (GET) requires a "read" token and upload translations (POST) requires a "write" token.

To access APIs that are protected by OAuth, OAuth configuration steps are required.

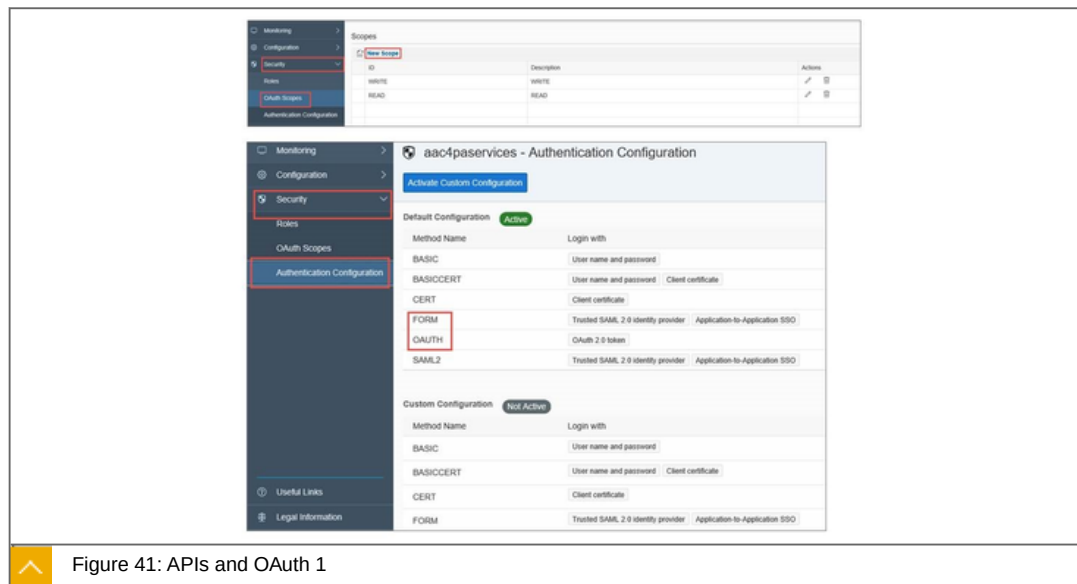


Figure 41: APIs and OAuth 1

For provider applications, complete the following configuration steps:

1. In the SAP Cloud Platform Cockpit, log on to the provider account.
2. Choose Applications Java Applications and select the provider application.
3. Choose Security OAuth Scopes and create the two scopes: read and write.
4. Choose Security Authentication Configuration .
5. Under OAUTH, select the OAuth 2.0 token option.
6. Under FORM, select the OAuth 2.0 token option.
7. Restart the application.

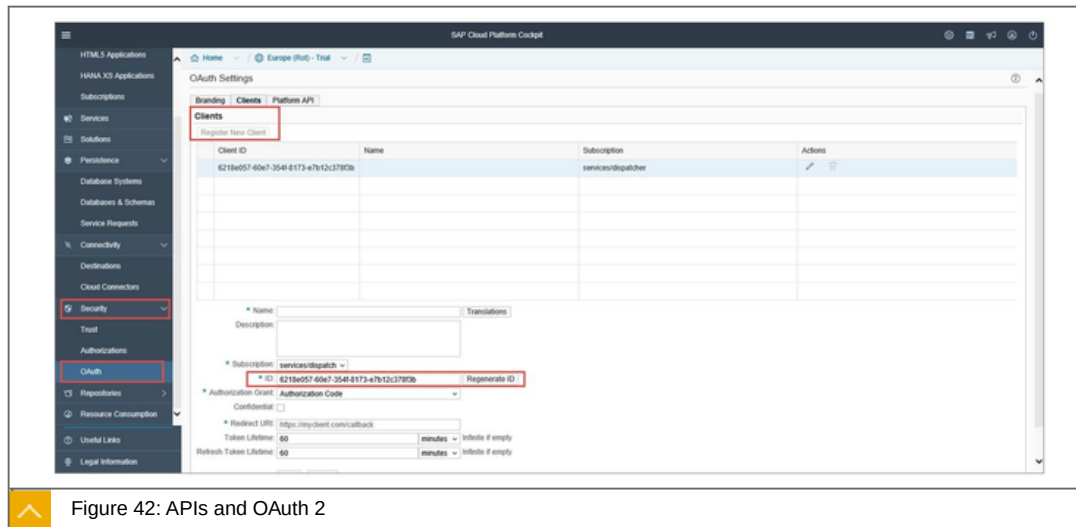


Figure 42: APIs and OAuth 2

For subscriber accounts, configuration steps should be completed by any customer who wants to use OAuth protected APIs as follows:

1. In the SAP Cloud Platform Cockpit, log on to your account.
2. Choose Security → OAuth → Clients and register a new client with a secret value. Encode the following key pair using a BASE64 encoding: <clientId>:<secret>.
3. Copy and save the encoded result you receive: <encoded_secret>.
4. For each scope you want to use, generate a Bearer token using the SAP Cloud Platform OAuth service. Send a POST to the REST API [https://oauthasservices-
<account>.hana.ondemand.com/oauth2/api/v1/token?
grant_type=client_credentials&scope=<scope>](https://oauthasservices-<account>.hana.ondemand.com/oauth2/api/v1/token?grant_type=client_credentials&scope=<scope>) (hana.ondemand.com - is for the productive landscape) with the header Authorization:Basic <encoded_secret>.
5. Copy and save the Bearer tokens you get: <read_token> and <write_token>.



LESSON SUMMARY

You should now be able to:

Access API documentation

Learning Assessment

1. What do you need to develop to add a custom button to the Fiori Launchpad Header?

Choose the correct answer.

- ☐ A Master Detail application
- ☐ B Overview Page
- ☐ C Fiori Launchpad Shell Plugin
- ☐ D Portal Widget

2. In which of the following file do you need to declare external resources in order to use them in an UI5 app?

Choose the correct answer.

- ☐ A component.js
- ☐ B neo-app.json
- ☐ C manifest.json
- ☐ D package.json

3. Where can you find information on the available APIs for SAP Cloud Platform Portal Service and the Fiori Launchpad?

Choose the correct answer.

- ☐ A api.sap.com
- ☐ B sapui5.hana.ondemand.com
- ☐ C help.sap.com
- ☐ D launchpad.support.sap.com

Learning Assessment - Answers

1. What do you need to develop to add a custom button to the Fiori Launchpad Header?

Choose the correct answer.

- ☐ A Master Detail application
- ☐ B Overview Page
- ☒ C Fiori Launchpad Shell Plugin
- ☐ D Portal Widget

You are correct! A SAP Fiori Launchpad shell plugin allows you to extend the shell of the SAP Fiori Launchpad. This way you can create custom options in the header, add a welcome screen or execute custom code under certain conditions. Refer to the lesson 'Adding and Removing Custom Items in the Options Menu' for more details.

2. In which of the following file do you need to declare external resources in order to use them in an UI5 app?

Choose the correct answer.

- ☐ A component.js
- ☒ B neo-app.json
- ☐ C manifest.json
- ☐ D package.json

You are correct! The neo-app.json file contains all project settings for SAP Web IDE and is created in the root folder of your project. It is a JSON format file consisting of multiple configuration keys. The most important setting for you to configure is the path where the SAPUI5 runtime is located when starting the app. Refer to the lesson 'Creating Apps to Access External Resources' for more details.

3. Where can you find information on the available APIs for SAP Cloud Platform Portal Service and the Fiori Launchpad?

Choose the correct answer.

- ☒ **A** api.sap.com
- ☐ **B** sapui5.hana.ondemand.com
- ☐ **C** help.sap.com
- ☐ **D** launchpad.support.sap.com

You are correct! The available APIs for any SAP product can be found at api.sap.com. Refer to the lesson 'Accessing API Documentation' for more details.

UNIT 4

Freestyle Site Development

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UNIT OBJECTIVES

- Get started with freestyle sites
- Use site templates
- Explain development flow
- Convert an application to a widget
- Describe project structure

Unit 4

Lesson 1

Getting Started with Freestyle Sites



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Get started with freestyle sites

Freestyle Sites Overview

Freestyle sites are very versatile. Unlike a SAP Fiori launchpad site, which offers a unified tile-based launcher experience, freestyle sites allow a fully customized experience in terms of the content and layout. For example, you can have a site with multiple pages, various page layout designs, and multiple applications and content on a page.

Freestyle sites are created based on templates. Templates can be from SAP, partners, or custom-made. When building a freestyle site, there are different design options to choose from as follows:

Create a site from one of the available site templates and use it as is with preconfigured pages, page templates, and built-in widgets.

Create a site from one of the available site templates, edit the predefined content, and extend it with your own page templates, pages, and apps.

Create a freestyle site with pages and apps and add a launchpad page.

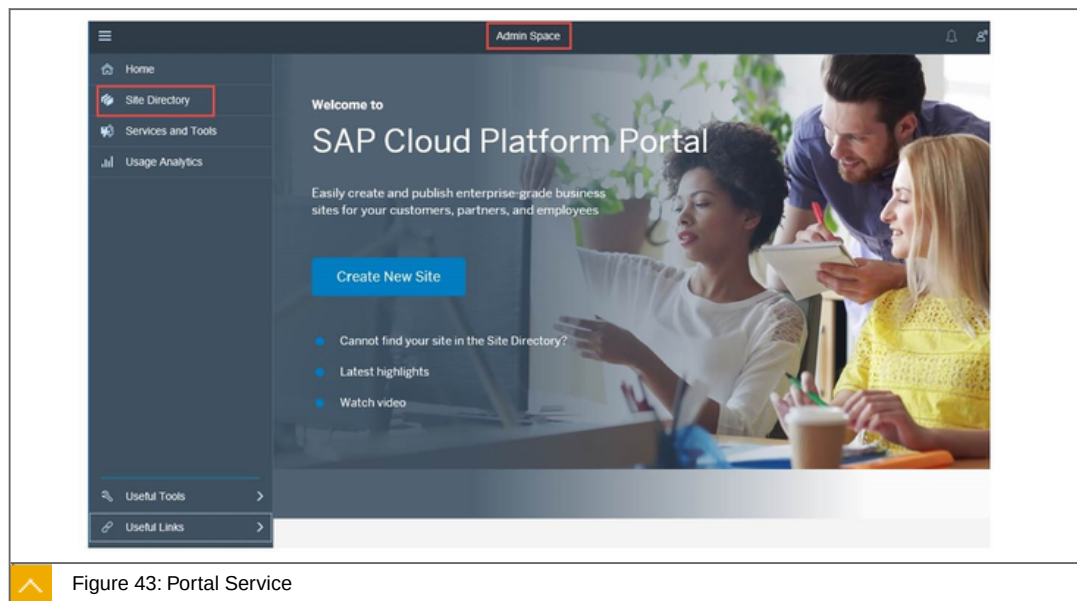


Figure 43: Portal Service

SAP Cloud Platform Portal provides various services and tools to help administrators manage their Admin Space (subaccount), which include sites, themes, analytics, and assets.

The following tools are available:

Site Directory

Using the Site Directory, you can create, edit, delete, duplicate, export, import, and publish your sites.

Site Designer

Using the Site Designer, you can create/maintain your site and site content (such as pages, apps, and page templates).

Site Building Blocks

The main site building blocks are as follows:

Site template

Page templates

Pages

Widgets

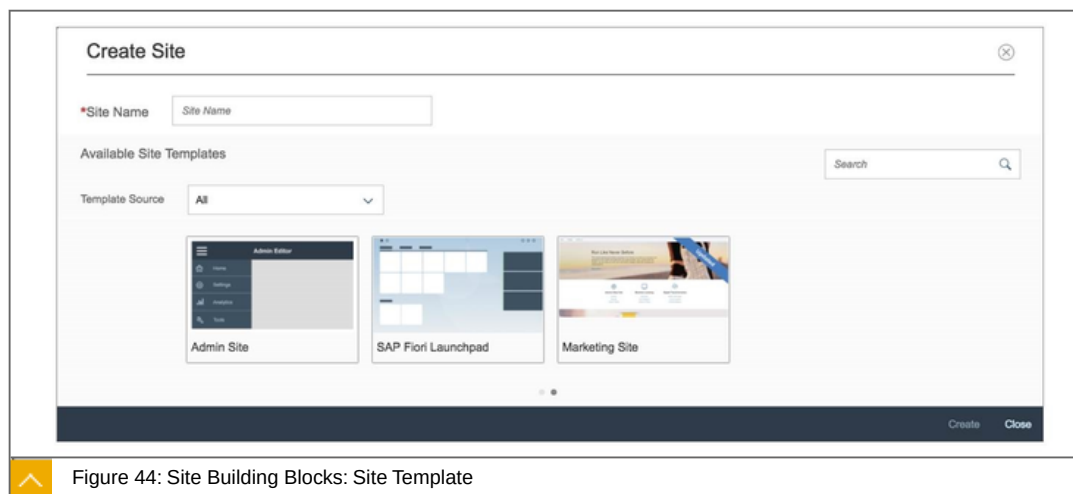


Figure 44: Site Building Blocks: Site Template

A site template contains page templates, pages, menus, and default content.

To create a site, you can use the following:

Out-of-the-box templates provided by SAP

Templates provided by SAP partners through subscription

Custom site templates created by developers using SAP Web IDE

A freestyle site instance is a copy of a site template. Once site templates have been deployed into the subaccount, administrators can create freestyle sites based on these using the Site Directory. Site content is maintained, customized, and configured using the Site Designer.

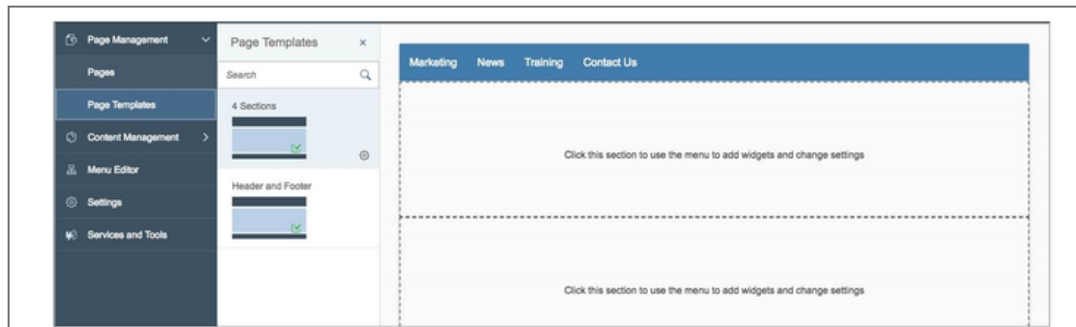


Figure 45: Site Building Blocks: Page Templates

A page template defines the layout of a page and is defined in the site template. The page layout is a predefined structure divided into content sections. Page templates are part of the site template and can come from SAP, a third-party provider, or developers who have created page templates using SAP Web IDE.

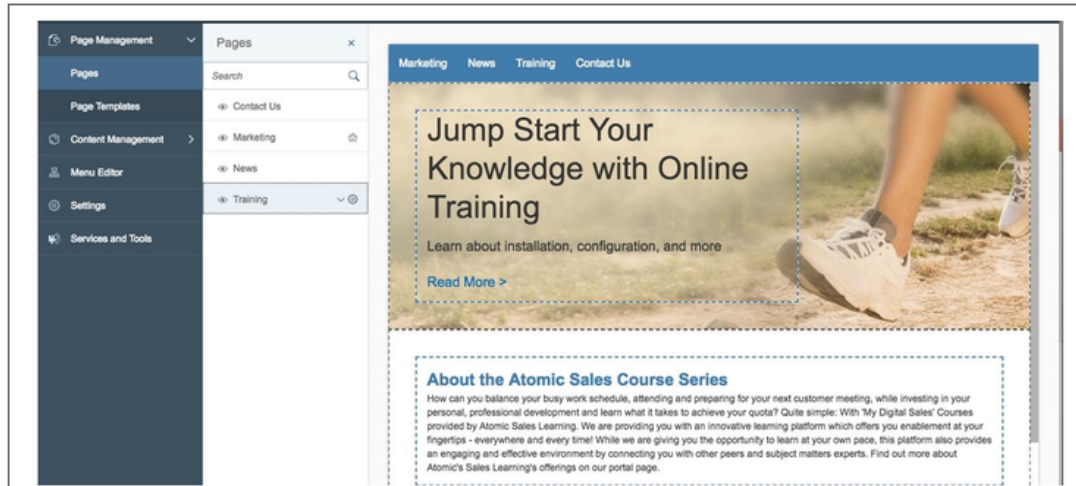


Figure 46: Site Building Blocks: Pages

A page is a copy of a page template (layout and predefined content). Multiple pages can be added to a freestyle site. Specific content and applications can be added into the content sections of the page. Page authorizations can be configured by administrators.

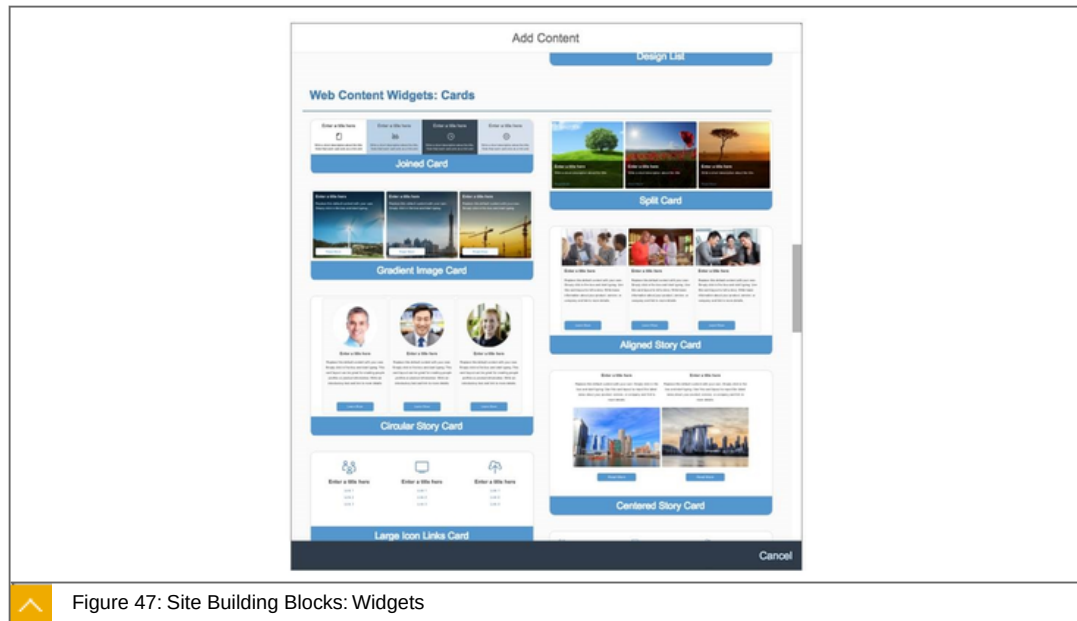


Figure 47: Site Building Blocks: Widgets

Widgets are web-based applications that can be added into page templates or pages. A widget could be an image, video, HTML, SAP Jam, or image carousel, for example. Various out-of-the-box widgets are provided by SAP, and you can also develop your own. Custom-developed SAPUI5 apps can also be converted into widgets by developers using SAP Web IDE.

Web content widgets are edited in a dedicated authoring tool called the Web Content Editor. You can do this editing, or it can be done by the site administrator or an authorized web content editor.



LESSON SUMMARY

You should now be able to:

- Get started with freestyle sites

Unit 4

Lesson 2

Using Site Templates



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Use site templates

Site Templates

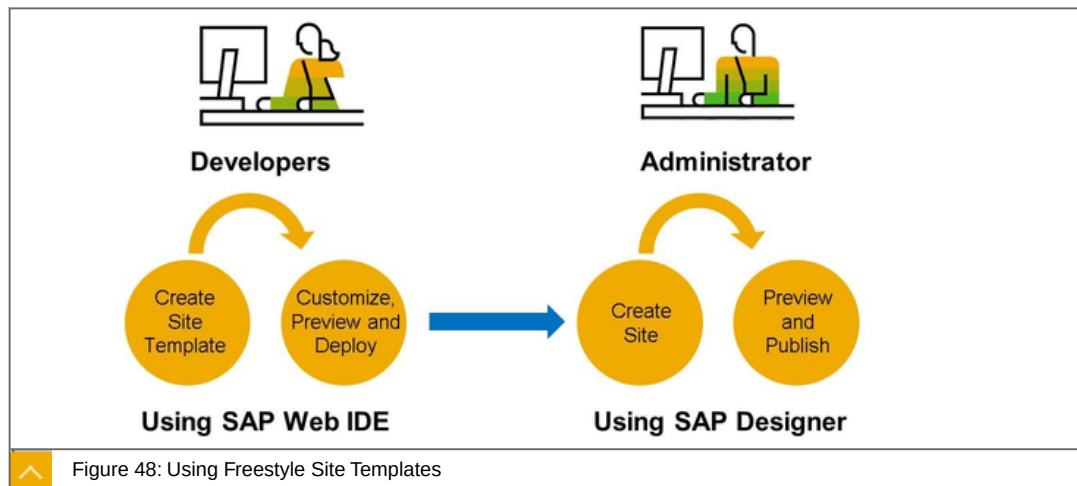


Figure 48: Using Freestyle Site Templates

Developers can take any template provided by SAP or other providers, customize the content, and create a new template from it using SAP Web IDE. The new template has to be deployed to the subaccount to make it available to site administrators in the Site Designer when they are creating a new site.



LESSON SUMMARY

You should now be able to:

Use site templates

Unit 4

Lesson 3

Explaining Development Flow



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Explain development flow

Development Flow

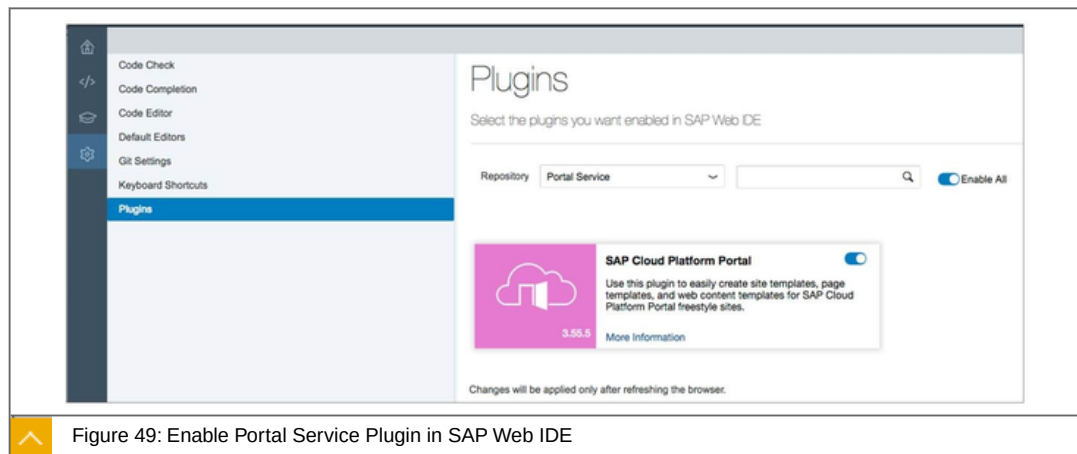


Figure 49: Enable Portal Service Plugin in SAP Web IDE

Developers use SAP Web IDE to create, customize, and deploy new site templates and make them available for portal administrators.

SAP Web IDE offers a custom plugin for developing freestyle site components in a structured manner. The plugin consists of a set of wizards, which support the following tasks:

- Creating site templates, page templates, pages, and widgets
- Converting a web application to a freestyle site widget
- Previewing a site in various form factors
- Deploying components (in a project package) to a subaccount on the cloud

To enable the portal plugin on SAP Web IDE:

1. Log on to the subaccount on SAP Cloud Platform.
2. Go to Services and under DevOps choose SAP Web IDE
3. Choose Plugins and in the Repository field, choose Portal Service.
4. Switch on the plugin and save.



Note:

For more details, see the Web IDE documentation at <https://help.hana.ondemand.com/webide/frameset.htm?0221845d73ad403ab2852142f3179177.html> under Plugins and Add-Ons.

The typical development flow is as follows:

1. Create a site template
2. Create a page template
3. Create pages
4. Create a widget template
5. Preview the site template
6. Deploy the site template to SAP Cloud Platform

Creating a Site Template

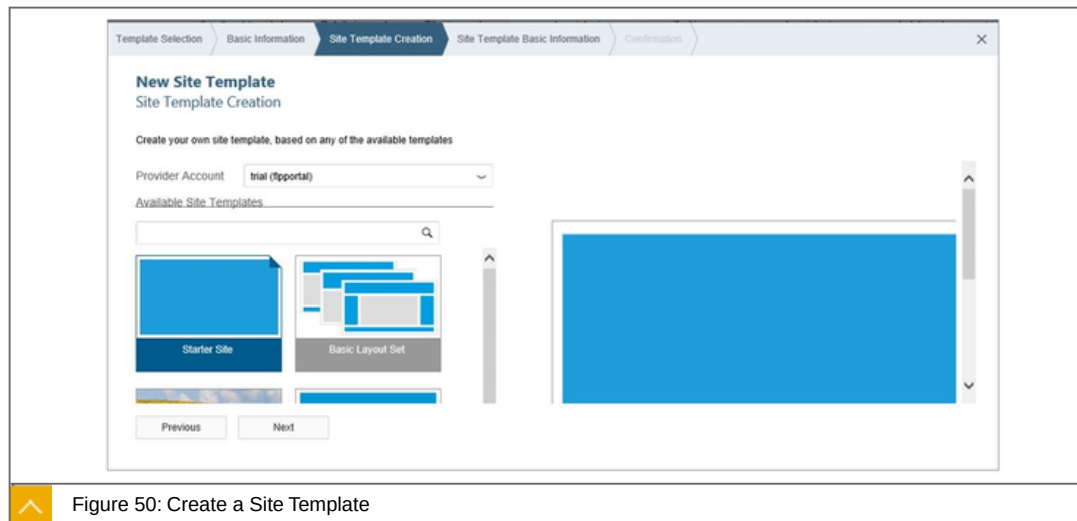


Figure 50: Create a Site Template

Custom site templates are required whenever there is a need to bring in a new site layout, new capabilities, or new content.

It is a prerequisite that developers enable the portal plugin on SAP Web IDE. Before creating a site template, developers must consider the kind of site template you want. The basic site template only defines the site layout, while the rich site template defines both layout and content for the site.

After deploying the site template, the site administrator can create site instances based on the deployed template and edit the site layout and content further as needed.

Site template sources can be as follows:

- Site templates provided by SAP

- Previously created site templates

- Site templates provided by other partners/providers through subscriptions

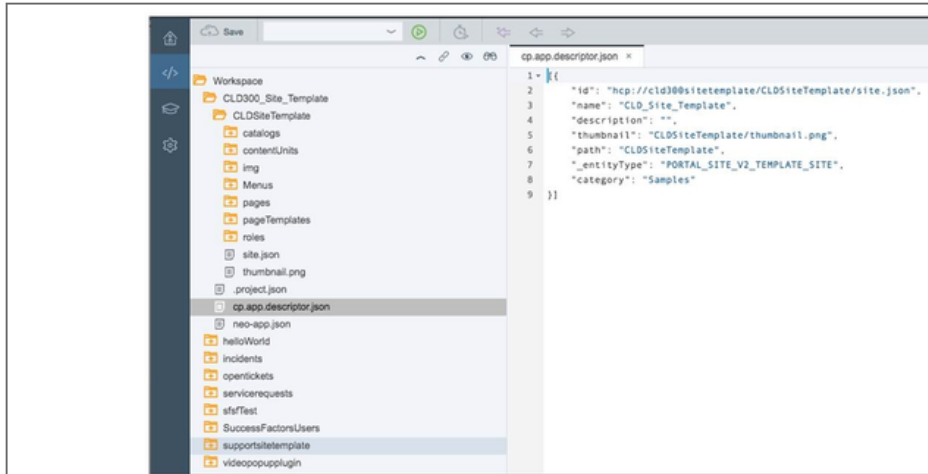


Figure 51: Create a Site Template

The SAP Web IDE wizard guides you through the main stages of creating a new site template. These stages are template selection, basic information, site template creation, and site template basic information.

To create a site template based on a template delivered by SAP:

1. Go to SAP Web IDE, create a new project, and choose Project from Template .
2. Choose Site Templates and provide the basic project information.
3. Select the relevant provider account to see all available site templates and choose the template you want.
4. Provide the site template name and confirm this.

Creating a Page Template

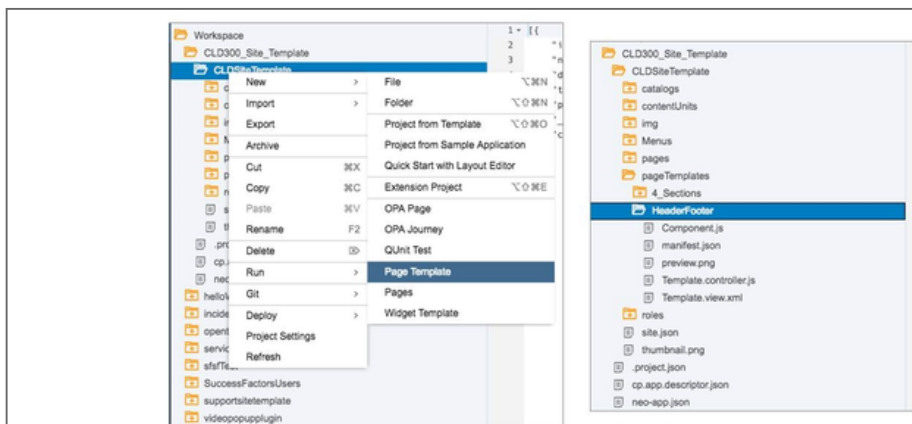


Figure 52: Create a Page Template

It is recommended to use a page template whenever specific layout or content needs to be repeated across several pages on a site. Instead of creating several similar pages, you can create a page template and the administrator can create pages based on it.

Before creating new page templates, check the pageTemplates folder of your project structure to see if there are any predefined page templates. Your site template inherits all the

page templates from its source site template. When creating a page template, it is recommended to use an existing page template as a source. For example, a new page template can be created based on the HeaderFooter page template provided by SAP.

To create a new page template:

1. Go to SAP Web IDE and create a new page template in the pageTemplates folder of your project structure (site template).
2. Select the relevant provider account to see all of the available page templates.
3. Choose a page template from the list (for example, HeaderFooter) and provide the new page template name.

When a new page template is created, a new folder structure is added to the pageTemplates folder.

Creating Pages

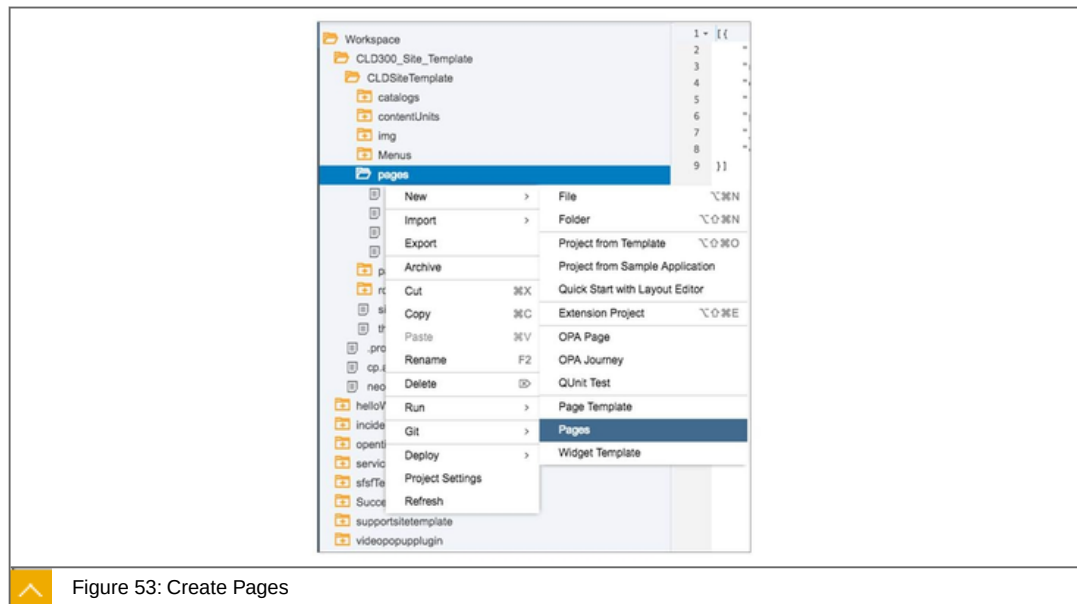


Figure 53: Create Pages

A page is an instance of a page template. Developers can add a page instance with predefined content to their site template so the page is always included when the administrator creates the site. Administrators can also modify that particular page if needed when they build their site.

To create a new page:

1. Go to SAP Web IDE and create a new page in the Pages folder of your project structure (site template).
2. Select the relevant provider account to see all of the available page templates.
3. Choose a page template from the list (for example, Starter Page) and provide the new page name.

When a new page is created, the page is added to the Pages folder.

Creating a Widget Template

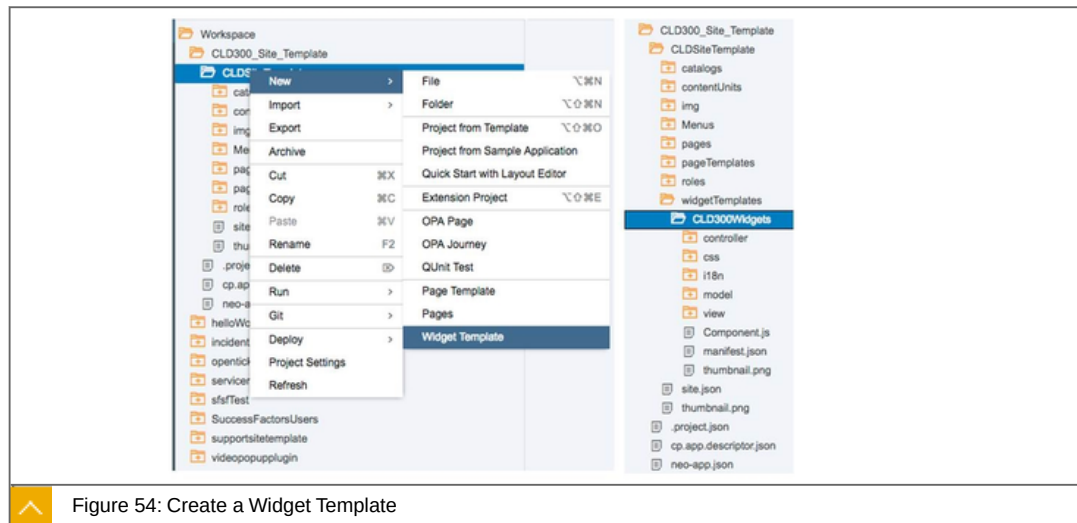


Figure 54: Create a Widget Template

Portal widgets are applications developed in SAPUI5, HTML, or JavaScript that are embedded into the pages of a site. Administrators can add them to the content section of a page. Developers can add widget templates to their site templates. These widgets become available to administrators, who can add them to any pages or page templates. The new widget template appears in the Site Designer catalog automatically when the site template is deployed into the subaccount.

Like other content, widgets can be created based on an existing widget template provided by SAP or other providers. You can also base these on custom lightweight SAPUI5 apps developed in your environment.

To create a widget template:

1. Go to SAP Web IDE and create a widget template in the site template folder by choosing **New** **Widget Template**.
2. Select the relevant provider account to see all of the available widget templates.
3. Choose a widget template from the list (for example, Starter) and provide the new widget template name and folder name.
4. Select an icon and save.

Deploying a Site Template to SAP Cloud Platform

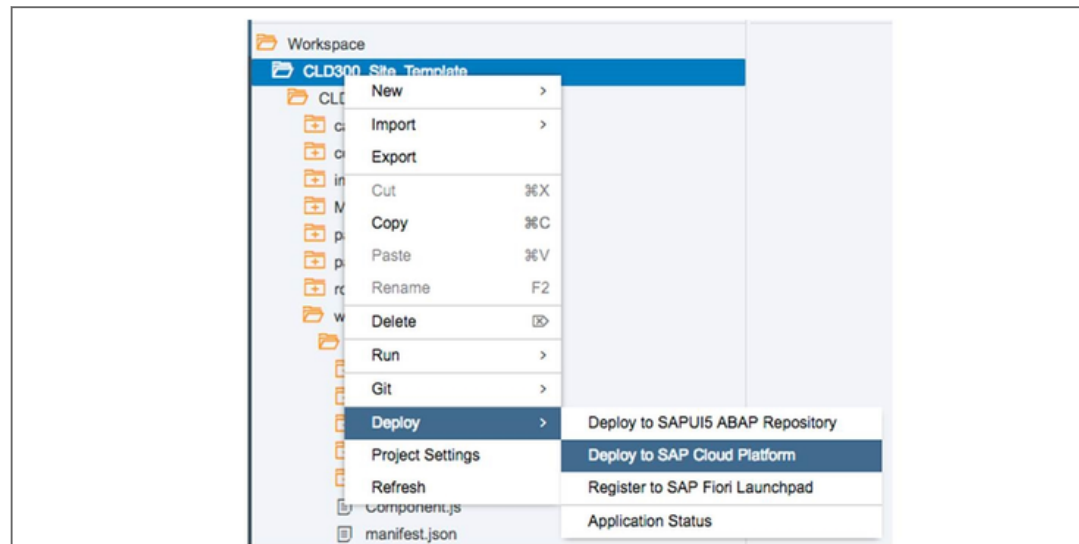


Figure 55: Deploy a Site Template to SAP Cloud Platform

When developers have completed a site template, the template must be deployed to SAP Cloud Platform to make it available in the admin tools for the portal administrators.

After the site template is deployed, changes made to the site template are not reflected in the admin tools. The updated template must be re-deployed, and a new site instance based on the updated template must be created. The updated template will overwrite the previously deployed version. If the administrator has already created a site using the original site template, that particular site instance will not be affected by the changes.

To deploy a site template:

1. Go to SAP Web IDE.
2. In the site template folder, choose **Deploy** **Deploy to SAP Cloud Platform**.

After the new site template is deployed, you can find it under **HTML5 Applications** in the SAP Cloud Platform Cockpit.



LESSON SUMMARY

You should now be able to:

- Explain development flow

Unit 4

Lesson 4

Converting an Application to a Widget



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Convert an application to a widget

Application to Widget Conversion

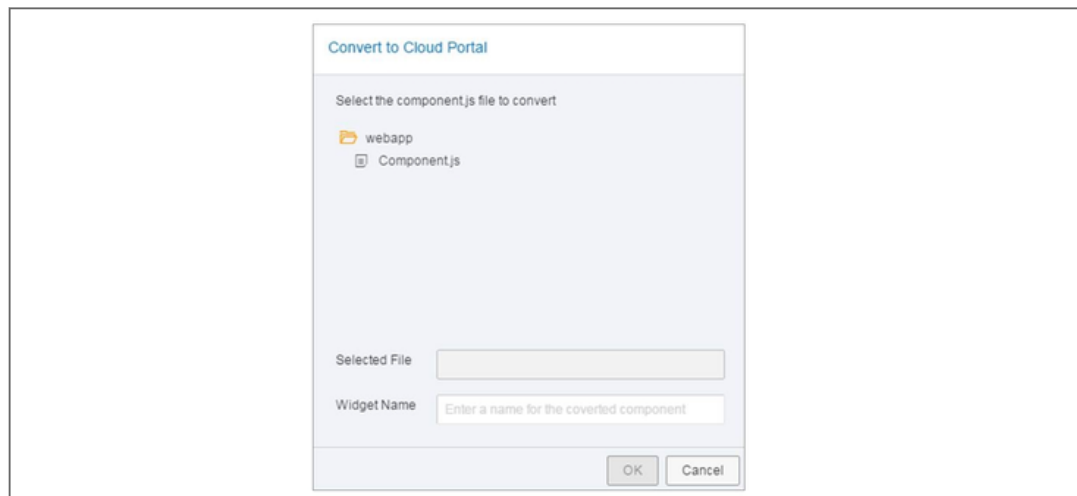


Figure 56: Convert an Application to a Widget

Developers can convert an existing HTML5 application to a portal widget provided the following constraints are met:

The HTML5 application has a Component.js file.

The HTML5 application does not have a cp.app.descriptor.json file. This file is essential for application discovery in the portal.

In the project.json file, the projectType does not include sap.site.template.

The conversion adds two new segments to the app descriptor and the application component.

To convert an HTML5 application to a widget:

1. Right-click the application and choose **Convert to Portal Service Component**.
2. Select the Component.js file that you want to convert and provide a name for the widget.

The application descriptor cp.app.descriptor.json and manifest.json are added to the root folder of your project during the conversion. After the conversion, the application must be deployed to SAP Cloud Platform to make the widget available for the admin tools. This means that administrators can add the widget to a site page section or page template section.



LESSON SUMMARY

You should now be able to:

- Convert an application to a widget

Unit 4

Lesson 5

Describing Project Structure



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Describe project structure

Project Structure

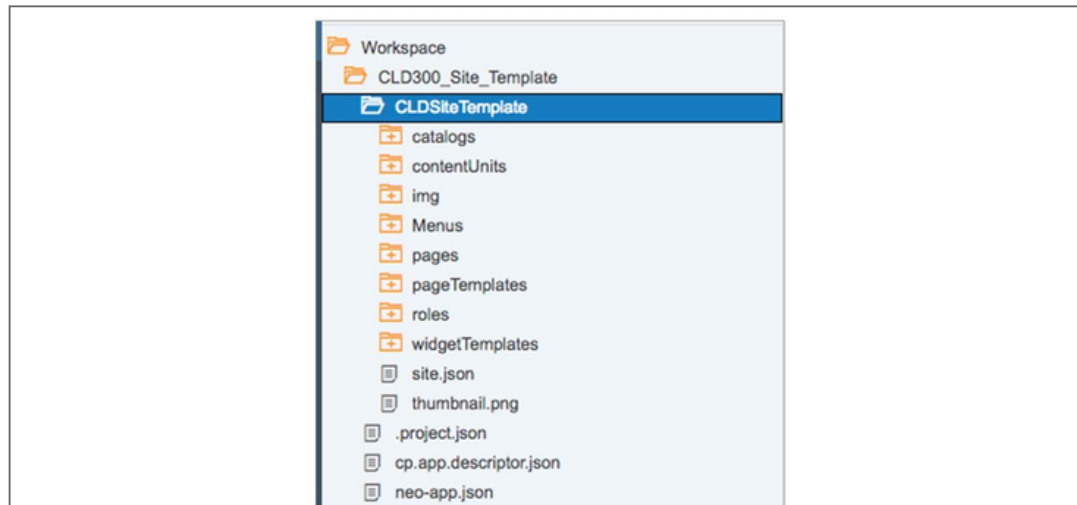


Figure 57: Project Structure

In a site template (which is a SAPUI5 project), you can find basic components like page templates, pages, and widget templates. There are also catalogs, menus, permission configurations, and other design elements. All of these project components are kept in folders under the project structure. Using SAP Web IDE, you can go to the development perspective to see the project structure, the components, and the source codes.

The application descriptor provides a central, machine-readable and easy-to-access location for storing metadata associated with an application or application component. The data is stored in json format in the manifest.json file with attributes in different namespaces. It contains, for example, the app ID, the version, and the data sources used, along with the required SAPUI5 components and SAPUI5 libraries.

In the SAP Cloud Platform Portal, you can find a manifest.json file for the following components: page template, widgets, and applications. Any instance of a component, for example, a page, does not have an associated manifest.json, and its json file includes only the custom metadata.

For a detailed description of a project and its components, see <https://help.sap.com/viewer/3ca6847da92847d79b27753d690ac5d5/Cloud/en-US/0828d5ec21c943e4a569a7940ec387a1.html>.



LESSON SUMMARY

You should now be able to:

Describe project structure

Learning Assessment

1. Which of the following elements can be part of a site template?

Choose the correct answers.

- ☐ **A** Page Templates
- ☐ **B** Widgets
- ☐ **C** Images
- ☐ **D** HTML Fragments

2. You can create your own site templates and use them with SAP Cloud Platform Portal Service.

Determine whether this statement is true or false.

- ☐ True
- ☐ False

Learning Assessment - Answers

1. Which of the following elements can be part of a site template?

Choose the correct answers.

- ☒ **A** Page Templates
- ☒ **B** Widgets
- ☒ **C** Images
- ☒ **D** HTML Fragments

You are correct. Portal freestyle sites are based on site templates. A site template contains all the necessary assets that make up a site, including pictures, widgets, text and HTML fragments, as well as the page templates for the individual pages. SAP Cloud Platform Portal Service comes with several Site Templates which can be used and modified. Refer to the lesson 'Getting Started with Freestyle Sites' for more details.

2. You can create your own site templates and use them with SAP Cloud Platform Portal Service.

Determine whether this statement is true or false.

- ☒ True
- ☐ False

You are correct! Developers can create site templates in SAP Web IDE, either from scratch or by modifying an existing one. Once they deploy the template project to your SAP Cloud Platform subaccount, you can then create site instances based on these site templates. Refer to the lesson 'Using Site Templates' for more details.

UNIT 5

Freestyle Site Administration

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UNIT OBJECTIVES

- Build a site
- Configure a site
- Describe how to customize a site with themes
- Manage site assets
- Work with the Web Content Editor
- Integrate SAP Fiori launchpad

Unit 5

Lesson 1

Building a Site



LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Build a site

Site Building

Building a site involves the following main tasks:

- Creating a site instance
- Working with page templates
- Adding pages to the site
- Adding content and apps to a page
- Configuring the menu structure
- Publishing the site

Through SAP Cloud Platform Portal, portal administrators can easily use the tools and services required to build their business sites. To create a site, use the Site Directory. You can then add content using the Site Designer.

Creating a Site Instance

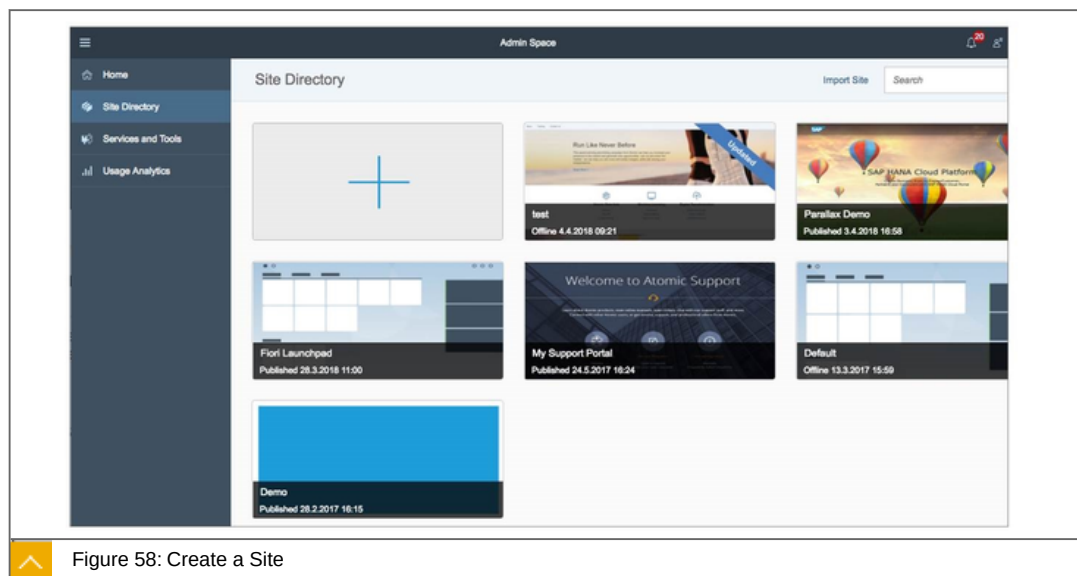


Figure 58: Create a Site

Sites are created in the Site Directory.

To create a site:

1. Log in to the Admin Space.
2. Select Site Directory and choose the option to create a site.
3. Provide a name for the site and select a site template from the available site templates.

You can filter by template source.

Working with Page Templates

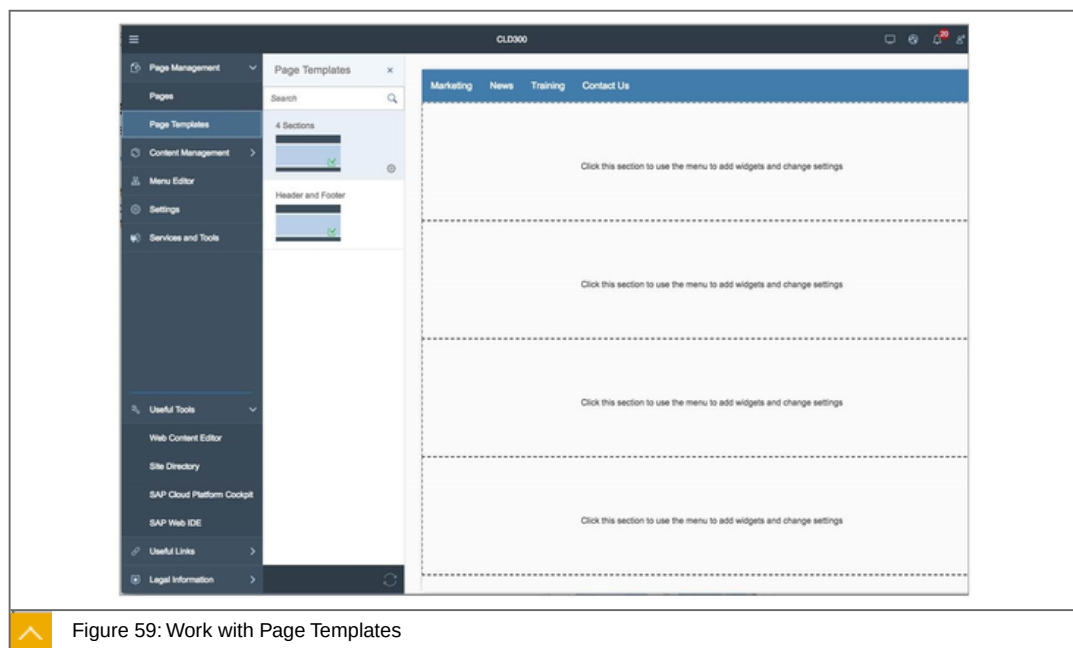


Figure 59: Work with Page Templates

Page templates provide a consistent look and feel across pages. The header and footer with fixed content (such as a company logo or contact information) is a good use case, because the fixed content appears in all the pages based on that page template. When a site is created based on a particular site template, you inherit all the page templates that were created by the developer in that source site template. The portal administrator can decide whether to implement those page templates when they create a new site.

To add fixed content to a template:

1. In the Site Designer, open the Page Template editor and select the relevant template from the list of available page templates.
2. To add content, click the section, choose the + button, and select the type of content to add.

For example, you can add an image widget.

The following figures shows the page template settings.

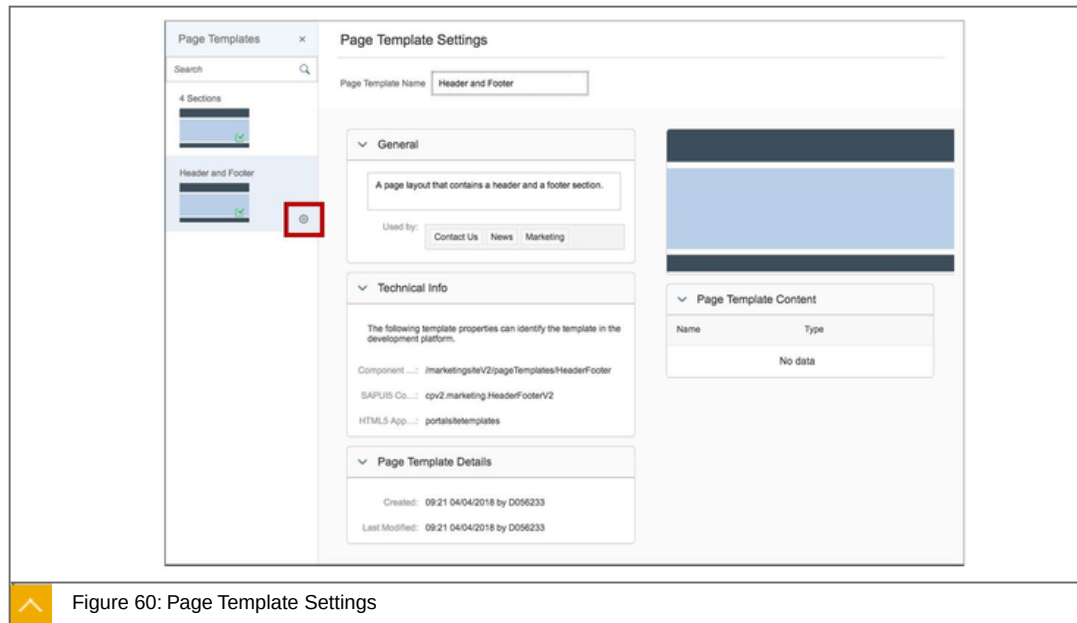


Figure 60: Page Template Settings

Adding Pages

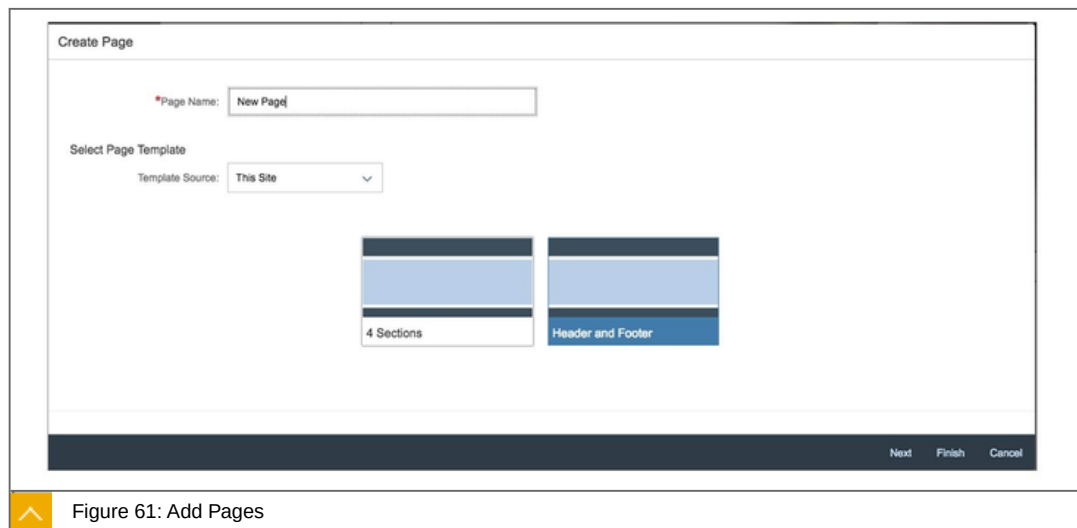


Figure 61: Add Pages

Pages can be added to a freestyle site. A page can contain content from one application, or a page can contain multiple content if the page is made up of multiple sections in which you can have different content in each. Pages are created from a page template. You can choose templates as follows:

- From the site template, meaning custom site templates

- From basic layouts, which are available for all freestyle sites, such as SAP Fiori launchpad or Anchor Navigation

New pages are added in hidden mode, and only become visible when you add them to the site menu.

There are different ways to add pages to your site (from Pages and Apps):

Add a new page.

Duplicate an existing page.

Add an app as a full page.

Adding Content to a Page

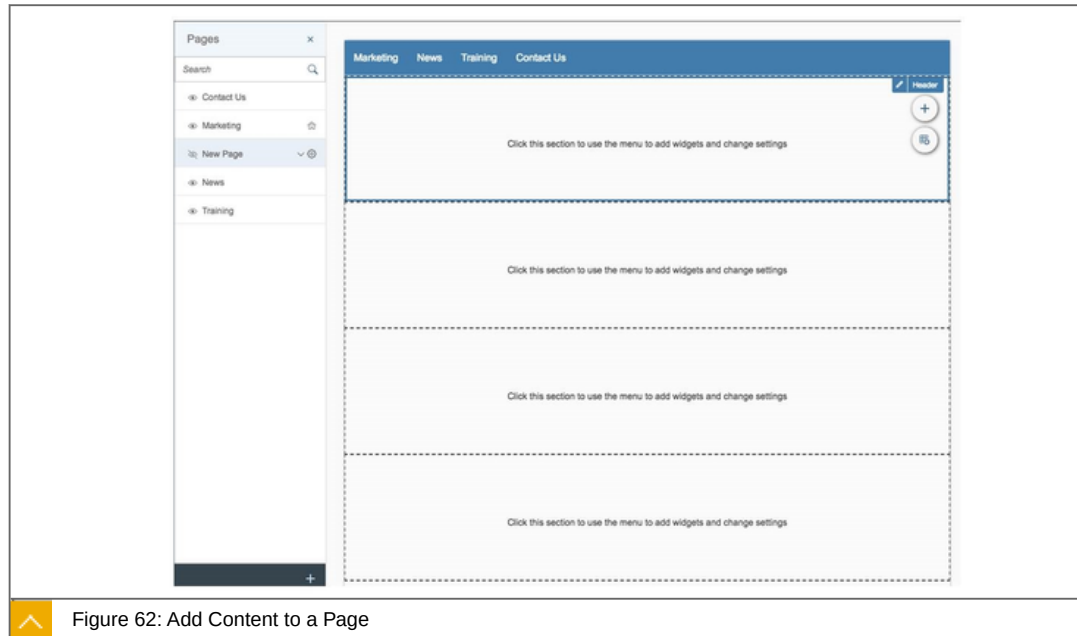


Figure 62: Add Content to a Page

When you create a page in a freestyle site, the page is based on a page template. The page is normally divided into sections, based on the page template you are using. Relevant content (widget) can then be loaded into the sections.

Administrators can decide the layout for a page section. The section layout usually depends on how many widgets are to be added and what device is to be used, for example, desktop, tablet, or mobile. Page section settings can be configured.

To add widgets to a section:

1. Open the section and add one or more widgets by choosing the + button.
2. Choose the widget type and then go to the settings to adjust the widget properties.
3. Edit the section settings (layout, background, and dimensions).

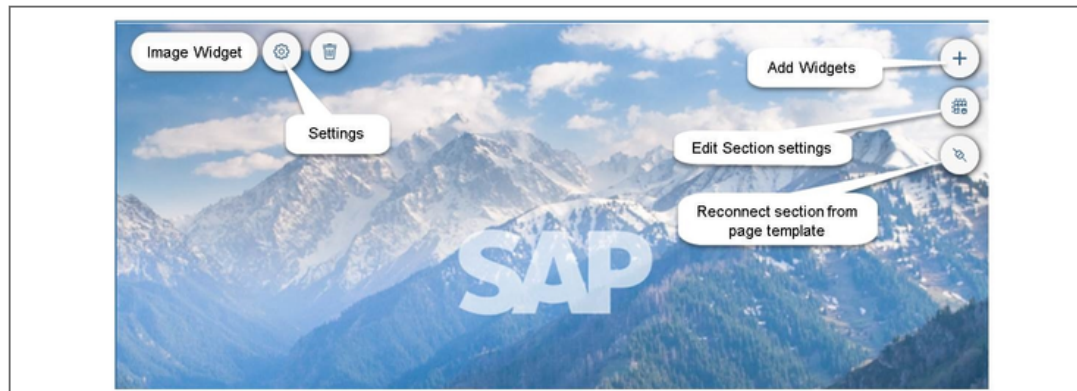


Figure 63: Add Content to a Section of a Page

You can add one or more widgets to any empty page sections. Widgets can be out-of-the-box widgets provided by SAP or SAP partners or portal widgets (converted from existing HTML5 applications).

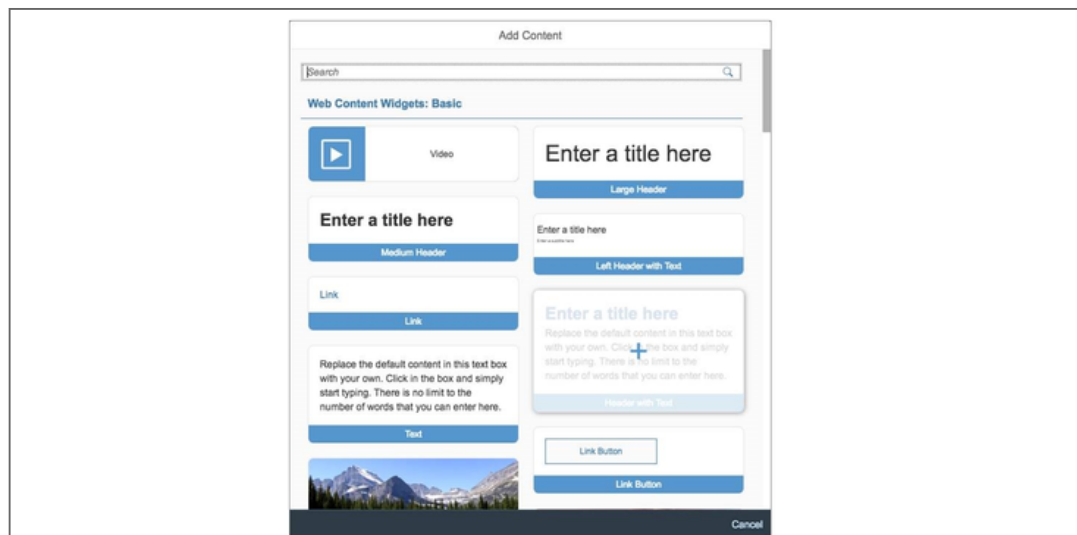


Figure 64: Widgets

A widget is the basic unit of content in a portal freestyle site. It's a simple, web-based application that you can embed in page sections of a site. Portal widgets are provided out-of-the-box by SAP or you can ask your developers to convert any SAPUI5 application into a widget (using SAP Web IDE). The widgets are available in a widget gallery in the Site Designer. You can access them directly from the page section in which you want to insert the widget.

The widget gallery includes all out-of-the-box widgets available from SAP as well as customized widgets created by your developers. These widgets are organized into two groups:

Web content widgets

Web content widgets are edited in a dedicated authoring tool called the Web Content Editor. You can edit these, or they can be edited by the site administrator or an authorized web content editor. In the widget gallery, they are organized into various categories:

- The Basic category includes widgets that consist of a single content item (such as an image, video, or link).
- The List, Cards, and Carousel categories include multiple content items that come with handy content snippets (small chunks of predefined content) that you can use as a starting point for adding content to your page.
- The Menu category contain content snippets that enable you to easily add predefined menus anywhere in a site page. You can link the menu items to any sections on the page or to other pages, apps, and external links.

Standard widgets

Standard widgets can only be edited in the Site Designer. It is the task of the administrator to edit this content. Standard widgets are organized into the following categories:

- Advanced

These are widgets with advanced settings that you can change. For example, if you select an image widget from this category, you can choose the settings icon and determine the image state, height, and alignment for both desktop and mobile devices. Some other examples of advanced widgets are HTML, Image, Rich Text Editor, List Builder, Tile Grid, Standard Carousel, and Login. Note that the Login widget is very useful for enabling your end users to log in and out of a public page that doesn't have a SAP Fiori header.

- Social

These include widgets for Facebook, Twitter, and other social networks, as well as widgets that you can add for users who want to connect to the SAP Jam social media platform to collaborate with colleagues.

Custom

These are widgets created by your developers and added to the widget gallery.

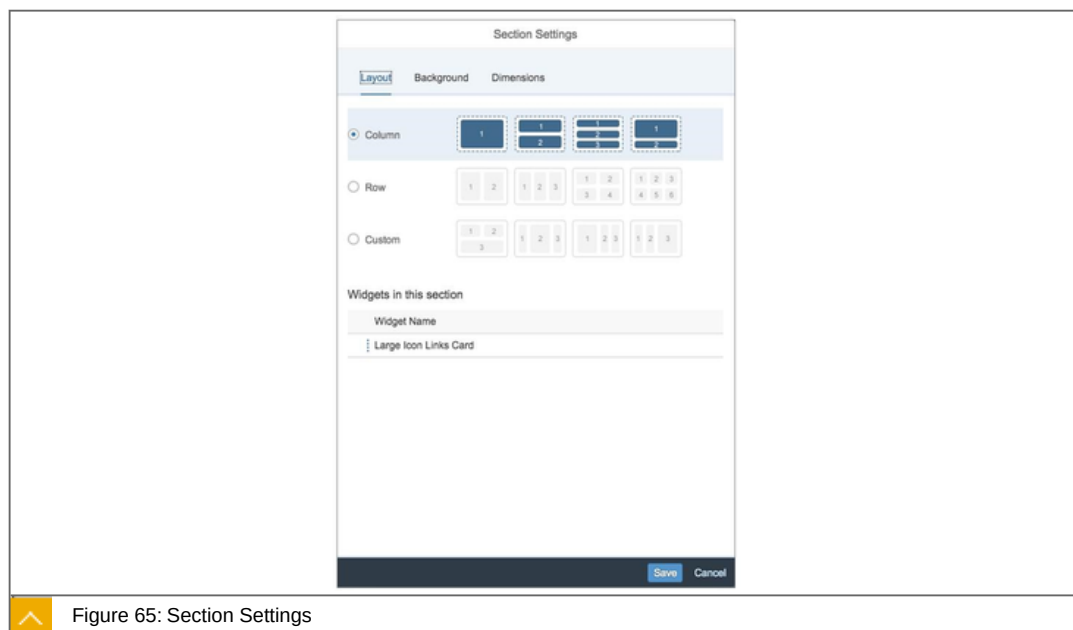


Figure 65: Section Settings

If you are going to add multiple widgets to a section, you can arrange them as follows:

By column: widgets are aligned top down

By row: widgets are aligned next to each other depending on the value you defined for the number of widgets in a row. If you have more widgets than the specified value, the widgets wrap to the next row.

Custom: widgets are displayed according to your device

In the section settings, the layout type can be selected as column, row, or custom. Using drag and drop, the widgets can be reordered. The background color or image can also be configured here as follows:

Image source: the image can be uploaded from your computer, from the Asset Repository, or a specific URL to the image file

Image state: this defines the behavior of the image when the image is not the same size as the section

The following dimensions can be configured:

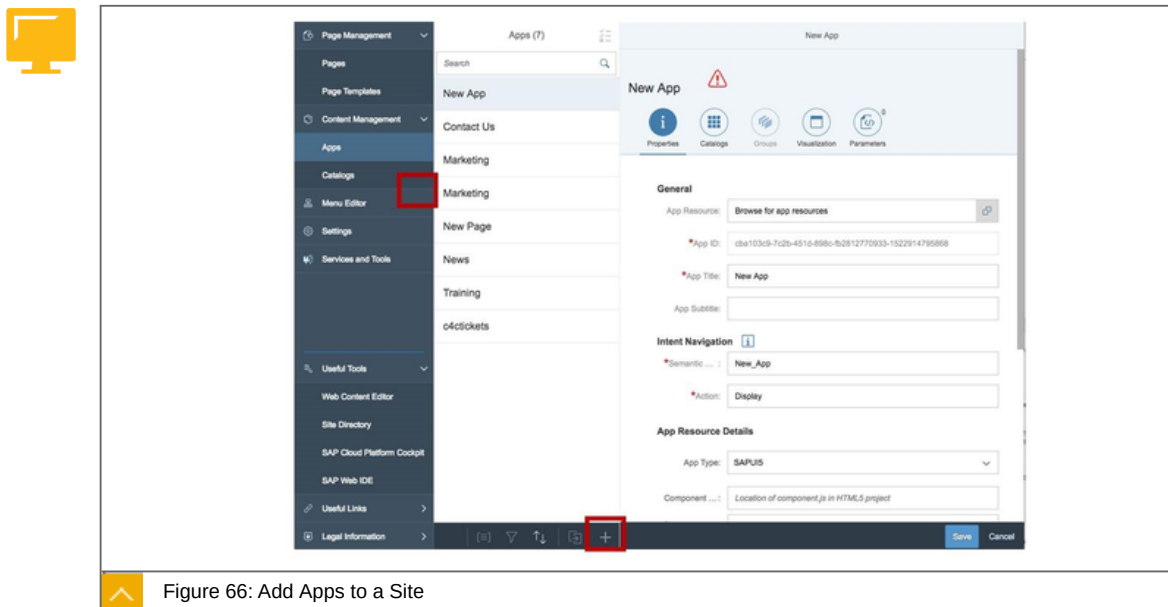
Margins: space between and around the widgets

Content width: amount of space the widgets use in a section

Minimum height: value that determines the minimum height of a section. The height of the section can be expanded if necessary but cannot be made smaller.

If there is predefined content in a section (from the page template that it inherited from), administrators have the option to replace it with other content. To replace the predefined content, the section needs to be disconnected from the page template. The widgets can then be added. You can reconnect to the page template if you want the predefined content.

Adding Apps to a Site



As well as using widgets to bring content into a freestyle site, you can add apps to the site. There are two ways to add apps using the Site Designer:

Add a full page app.

Add apps to an embedded launchpad page.

To add a full page app:

1. In the side navigation panel of the Site Designer, open the Menu Editor.
2. Choose the + button and choose Add Item .
3. Select app as your item type.
4. Search for and select your app to add it as a full page app in your site.
5. Save your changes.

If you preview the site, you can see that the app is part of the site menu and is displayed as a full page app in your site.

Administrators can integrate freestyles sites with SAP Fiori launchpad capabilities. One option is to use an embedded launchpad page. With this option, an app is added as a tile in the embedded launchpad page.

To add apps to an embedded launchpad page:

1. Under Pages and Apps, add a new page and choose the SAP Fiori launchpad page template for your launchpad page.
2. Under Content Management , add a new app tile. Select a tile type and provide the relevant visualization properties.
3. In the Catalogs section of the app, add the tile to a catalog.

This ensures that end users with a specific role can access the app.

4. Add the app to a group if needed.

Configuring the Menu Structure

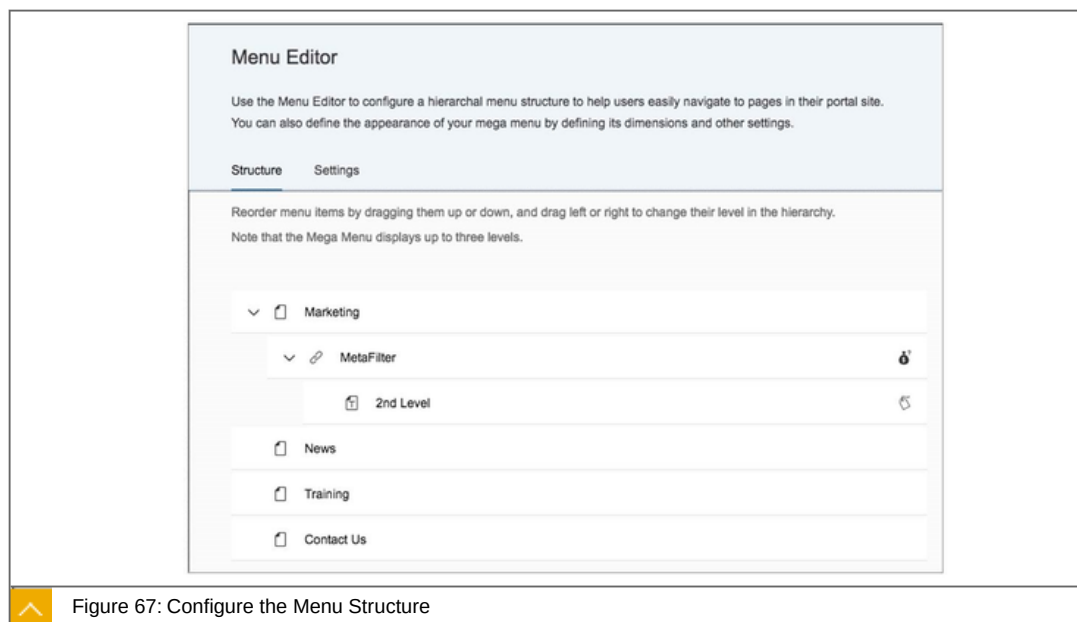


Figure 67: Configure the Menu Structure

Use the Menu Editor to configure the menu structure for your site. The menu structure is hierarchical and is made up of menu items. A three-level menu hierarchy is supported. A menu item can be an existing page or an app in your site. Menu titles can be used to group related pages or links together.

To assign a page or app to a menu, go to the Menu Editor and choose **Add to Menu**. By default, the item is added at the bottom of the menu structure. You can add an item at any level, allowing you to group and order a custom menu structure. An item in a menu could be one of the following:

Page: to navigate to a page within your site

App: to navigate to a full page app

Title: to group a number of links or apps together, making navigation easier for the end user

Link: to link to external sites

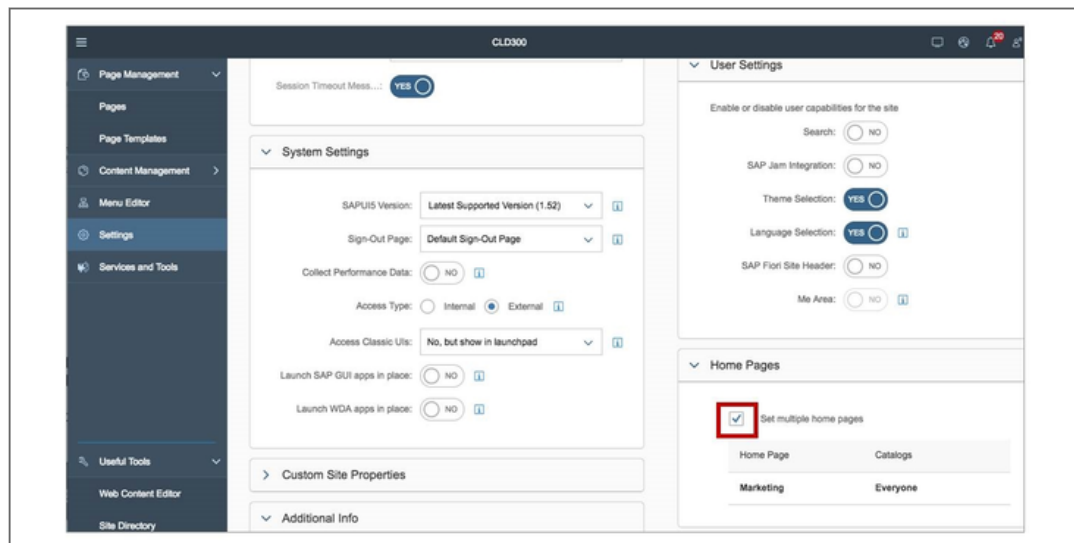


Figure 68: Set Multiple Home Pages

A freestyle site can have more than one home page so a different home page can be assigned to different roles. A role cannot be assigned to more than one home page.

To set up multiple home pages:

1. In the Site Designer settings, in the **Home Pages** section, select the **Set multiple home pages** checkbox.
2. Select the relevant page from **Pages and Apps** and choose **Select as Home Page**.

Publishing a Site

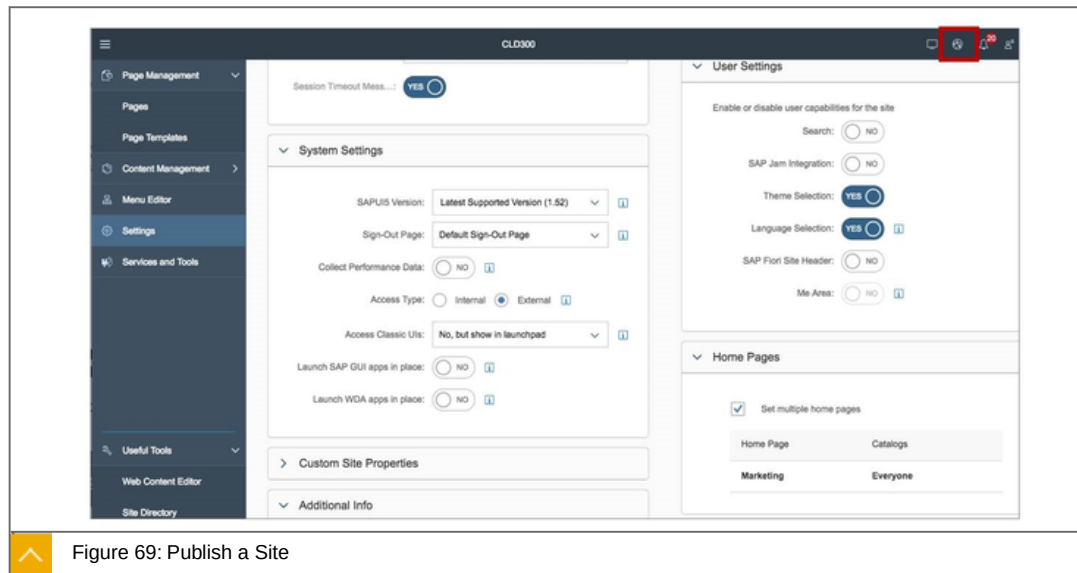


Figure 69: Publish a Site

When you are finished making changes to a site, you must publish it to make the latest changes available to end users. You can publish sites in the Site Directory or the Site Designer.



LESSON SUMMARY

You should now be able to:

- Build a site

Unit 5

Lesson 2

Configuring a Site



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Configure a site

Site Configuration

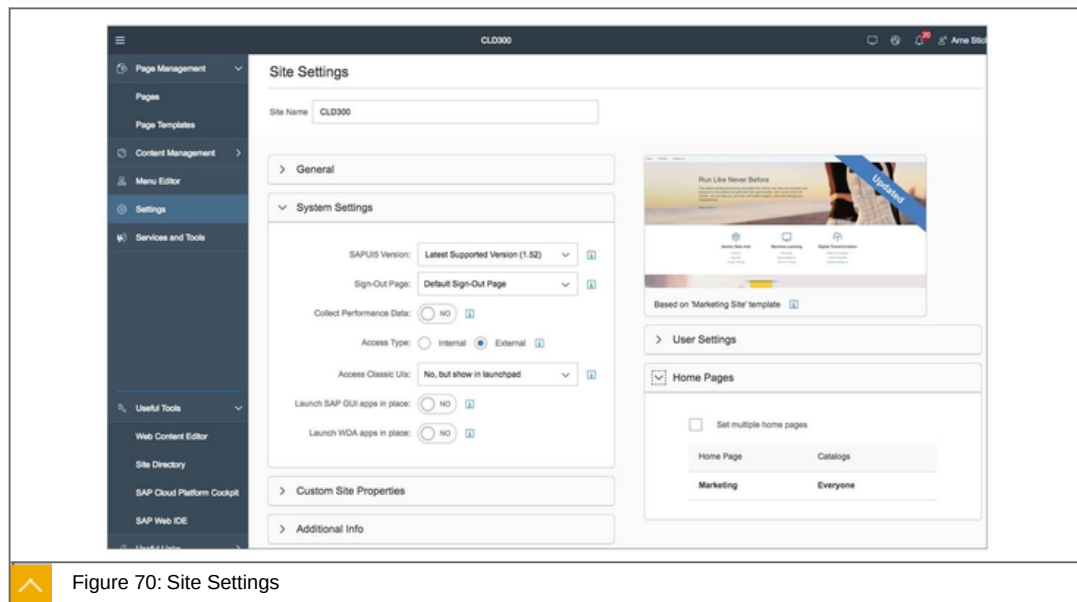


Figure 70: Site Settings

As shown in the figure, there are various site settings available as follows:

General

- Site status
- Site URL
- Site Alias

System Settings

- SAPUI5 Version

Choose between maintenance (latest stable version for production) or the latest supported version (latest version and features for testing).

- Sign-Out Page

Redirect users to a specific page after logout.

User Settings

- Search
- SAP Jam Integration
- Theme selection
- SAP Fiori site header
- Me Area

Custom Site Properties

Key-value pairs for customizing site implementations

Home Pages

- Set multiple home pages
- Assign different homepages to different business roles

Catalogs and Role-Based Access

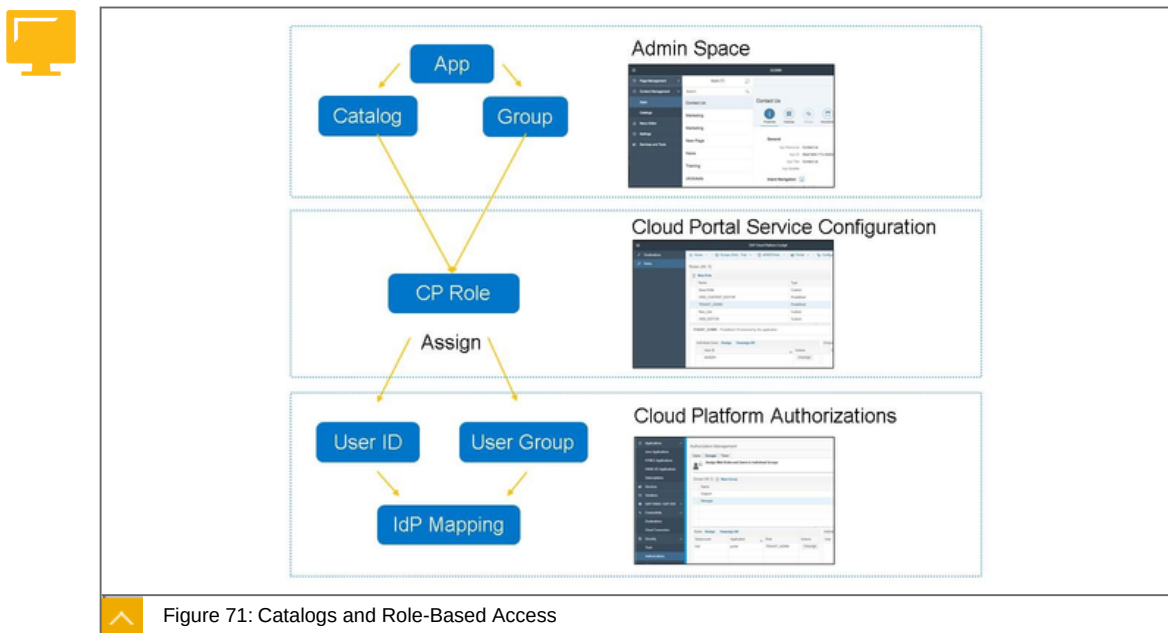


Figure 71: Catalogs and Role-Based Access

To control access to pages and apps, administrators can apply role-based access to the pages of freestyle sites using catalogs. In the role-based approach, roles are assigned to a catalog, and pages and apps are also assigned to a catalog. Users have access to all the pages and apps in the site that are assigned to the same catalog as their role. To implement more control, page authorizations can be set up to control access for each page.

To add and configure a catalog:

1. Add a catalog using the Content Management option in the Site Designer and provide a catalog name and description.
2. Select the apps to assign to this catalog using the Apps tab.

3. Select the roles to assign to this catalog using the Roles tab.

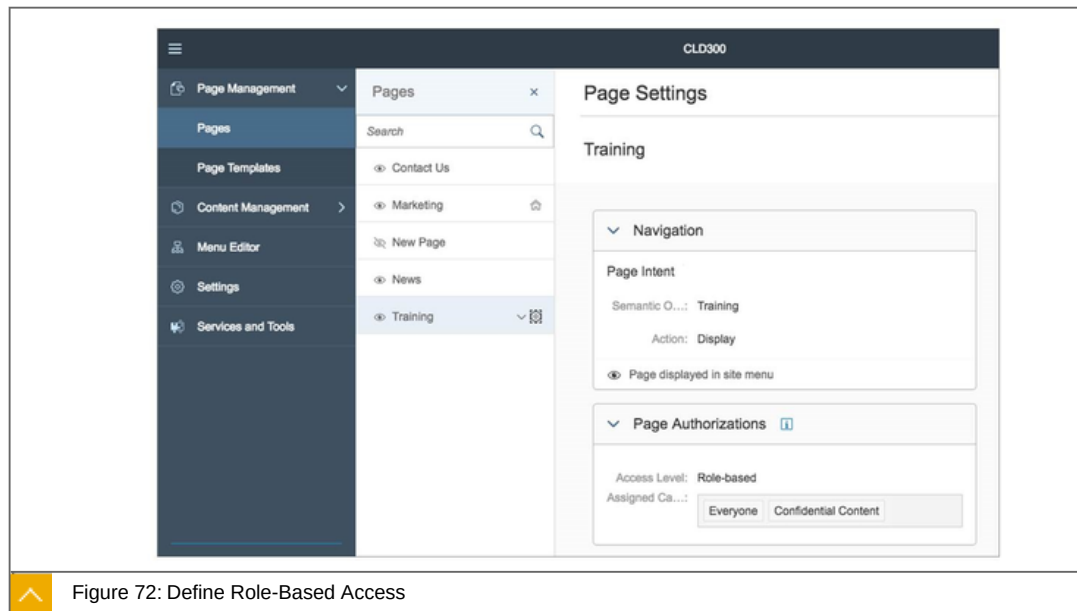


Figure 72: Define Role-Based Access

There are different access levels in SAP Cloud Platform Portal, and the recommended setup for portal sites is always role-based. Page authorizations can be set up to control access for each page as follows:

Guest: for unauthenticated users (cannot be modified) via Guests catalog

Public: both authenticated and unauthenticated users (cannot be modified) via Authenticated Users and Guests catalogs

Role-based: the administrator assigns the catalogs that contain the desired roles. Users with those roles can access the page.

To define role-based access for a page:

1. Go to the page settings for your page in the Site Designer.
2. In the Page Authorizations section, select Role-based as the access level
3. Add the catalogs.



LESSON SUMMARY

You should now be able to:

Configure a site

Unit 5

Lesson 3

Branding a Site



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Describe how to customize a site with themes

Site Branding

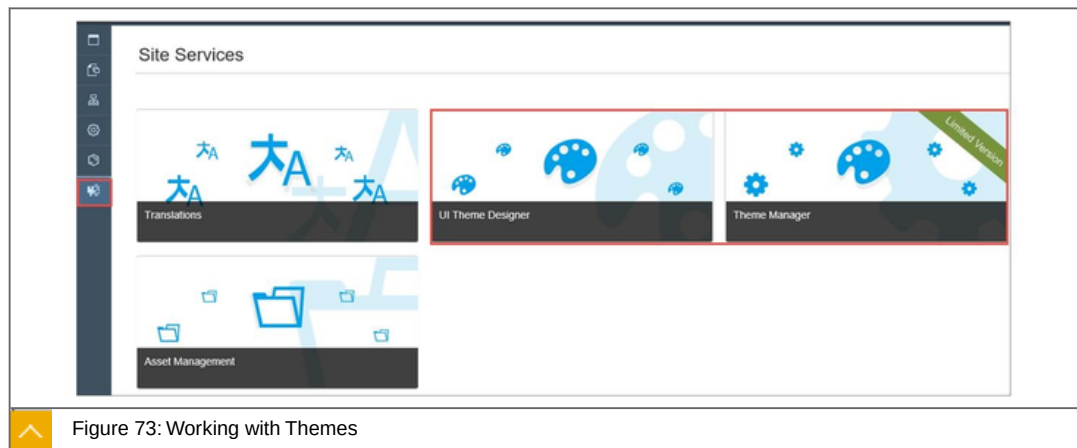


Figure 73: Working with Themes

To make your sites more interesting, you can customize the look and feel of your site using themes. For example, you can change the logo or the background images. The UI Theme Designer (SAP Cloud Platform Portal Service) is a browser-based tool for configuring the visual appearance of applications built with SAP UI technologies. You usually start theme customization at the site level in the Site Designer and subsequently assign it to sites using the Theme Manager.

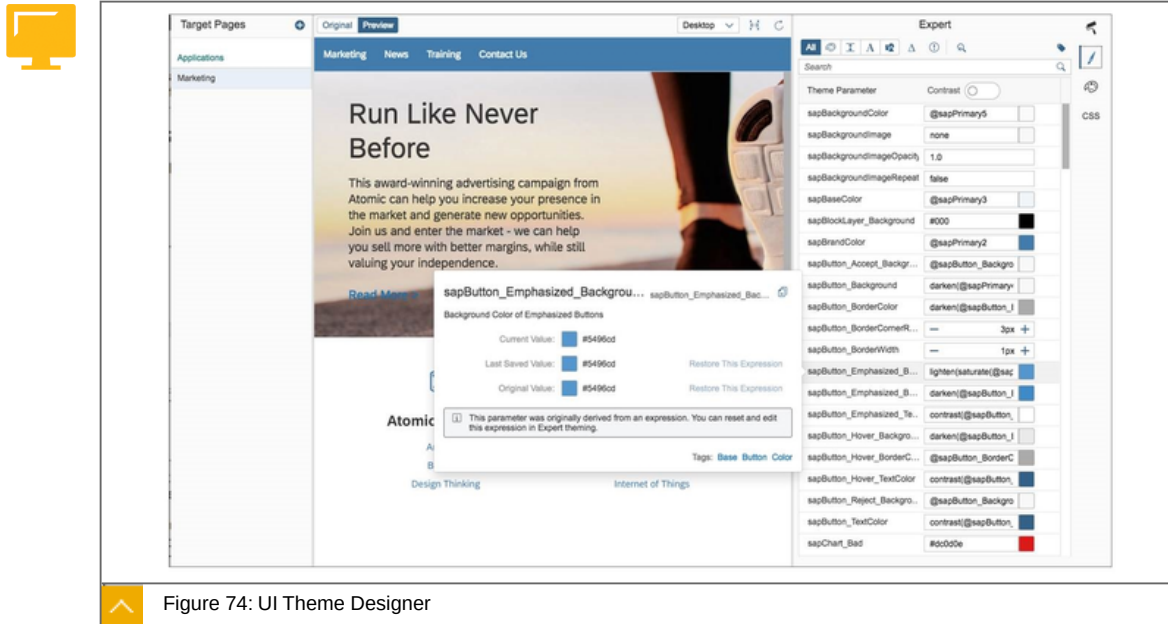


Figure 74: UI Theme Designer

To work with the UI Theme Designer:

1. In the UI Theme Designer, search for and select your theme.
2. On the Add Target Pages screen, enter the URL of the application you want or the available test suites to preview.
3. Choose the editing mode.

The following editing modes are available:

Quick: displays the basic properties that are common throughout different applications. You can change the logo and background images in Quick mode under Images.

Expert: displays all the parameters available for theming

Palette: allows you to add color swatches that will be readily available for selection in Expert mode and Quick mode

CSS: allows you to write your own custom CSS markup

4. Apply your changes.

For more details on how to work with the UI Theme Designer, see <https://eaexplorer.hana.ondemand.com/item.html?id=10599#!/facet/18>.

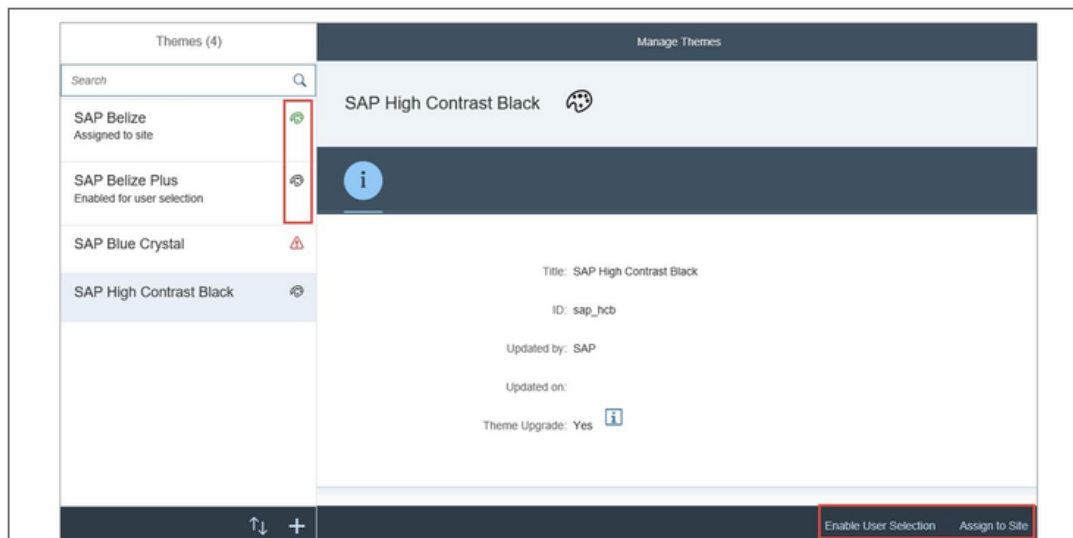


Figure 75: Configuring Themes

After themes are imported, administrators need to enable them and configure one of them as the default theme. Use the Theme Manager to enable and configure the themes.

A theme can be set up for end user selection at runtime using the **Enable User Selection** option. To assign a default theme to the site, select **Assign to Site**.

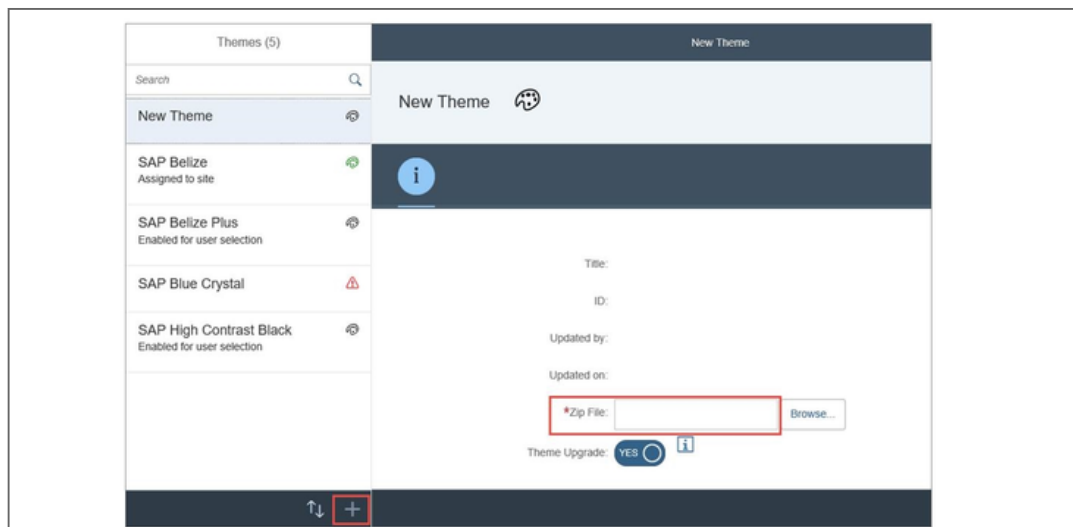


Figure 76: Transporting Themes

When you make changes to a theme, the theme needs to be exported from the UI Theme Designer and imported to a relevant portal account to be applied.

To transport a theme:

1. In the UI Theme Designer, export the theme.

A theme ZIP file is created in the **Downloads** folder during export.

2. In Theme Manager, create a new theme and upload the theme zip file from the export step.



LESSON SUMMARY

You should now be able to:

- Describe how to customize a site with themes

Unit 5

Lesson 4

Managing Site Assets

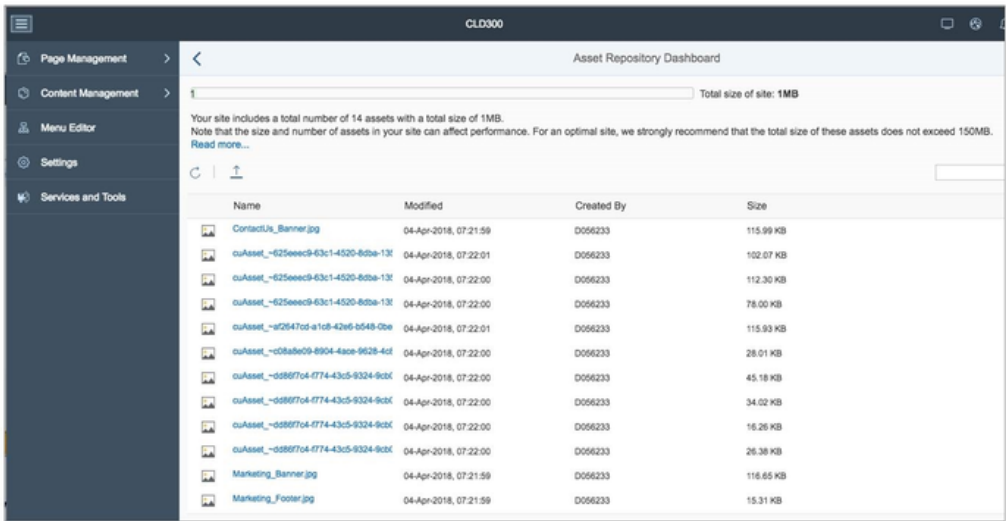


LESSON OBJECTIVES

After completing this lesson, you will be able to:

Manage site assets

Site Asset Management



The screenshot shows the SAP Asset Repository Dashboard. On the left is a navigation menu with options: Page Management, Content Management, Menu Editor, Settings, and Services and Tools. The main area displays the 'Asset Repository Dashboard' for user 'CLD000'. It shows a search bar, a filter for 'Total size of site: 1MB', and a message: 'Your site includes a total number of 14 assets with a total size of 1MB. Note that the size and number of assets in your site can affect performance. For an optimal site, we strongly recommend that the total size of these assets does not exceed 150MB. Read more...'. Below this is a table listing the assets.

Name	Modified	Created By	Size
ContactUs_Banner.jpg	04-Apr-2018, 07:21:59	D056233	115.99 KB
cuAsset_~625eeec9-63c1-4520-8dba-13f	04-Apr-2018, 07:22:01	D056233	102.67 KB
cuAsset_~625eeec9-63c1-4520-8dba-13f	04-Apr-2018, 07:22:00	D056233	112.30 KB
cuAsset_~625eeec9-63c1-4520-8dba-13f	04-Apr-2018, 07:22:00	D056233	78.00 KB
cuAsset_~a2647cd-a1cd-42a6-b548-0be	04-Apr-2018, 07:22:01	D056233	115.93 KB
cuAsset_~c08a8b09-8904-4a9c-9628-4cf	04-Apr-2018, 07:22:00	D056233	28.01 KB
cuAsset_~d58677c4-f774-43c5-9324-9c0f	04-Apr-2018, 07:22:00	D056233	45.18 KB
cuAsset_~d58677c4-f774-43c5-9324-9c0f	04-Apr-2018, 07:22:00	D056233	34.02 KB
cuAsset_~d58677c4-f774-43c5-9324-9c0f	04-Apr-2018, 07:22:00	D056233	16.26 KB
cuAsset_~d58677c4-f774-43c5-9324-9c0f	04-Apr-2018, 07:22:00	D056233	26.38 KB
Marketing_Banner.jpg	04-Apr-2018, 07:21:59	D056233	116.65 KB
Marketing_Footer.jpg	04-Apr-2018, 07:21:59	D056233	15.31 KB

Figure 77: Managing Site Assets

Every time an image or any other media file (such as video or HTML) is uploaded into a widget in a page section, it is automatically uploaded into the Asset Repository. Site assets in the repository are available for reuse in other pages and sections of the site. To reuse a site asset the next time a widget is created, select Asset Repository as the source and then select the file. The site assets can also be downloaded from the Asset Repository for editing or translation purposes. Administrators can view and manage their site assets using the Asset Management service.



LESSON SUMMARY

You should now be able to:

Manage site assets

Unit 5

Lesson 5

Working with the Web Content Editor



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Work with the Web Content Editor

Web Content Editor

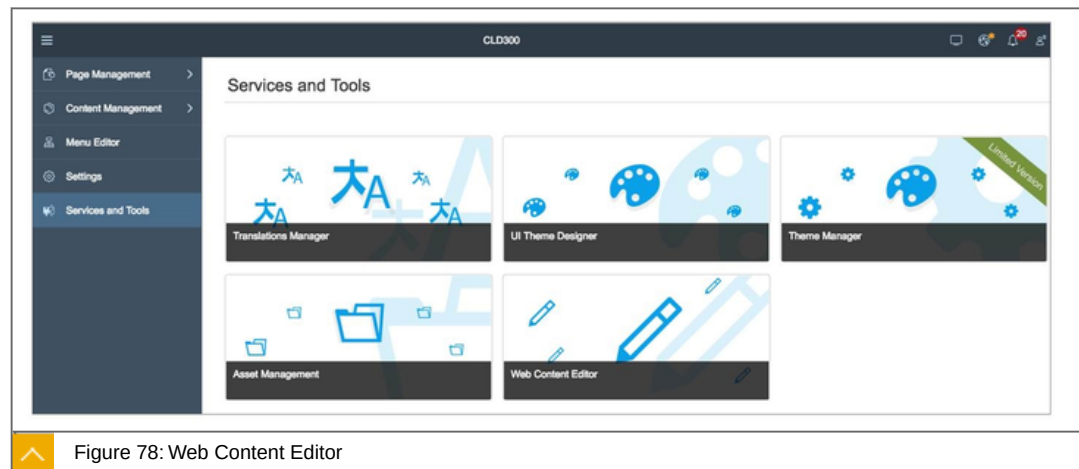


Figure 78: Web Content Editor

In SAP Cloud Platform Portal, web content is added to a site in the form of widgets that the site administrator selects from a widget gallery in the Site Designer. The actual content of a web content widget is created and maintained via the Web Content Editor, which can be accessed by users who have been assigned the `WEB_CONTENT_EDITOR` role.

A web content widget is made up of one or more content items, each with their own editable design elements (such as a title, text, images, and links).

Web content widgets are divided into categories:

Basic category

A basic web content widget is a single, standalone content item such as a title, link, or image. These widgets are nonrepeatable, meaning that you cannot add additional content items to these widgets.

List, Card, Carousel, and Menu categories

More complex web content widgets include multiple content items. These widgets are divided into the List, Card, Carousel, and Menu categories in the Site Designer widget gallery, depending on their layout and behavior. They are typically populated with handy content snippets (small chunks of predefined content). The content snippets include repeatable content items, each one with design elements such as an image, caption, or

link. Content snippets are repeatable, meaning that you can also add additional content items to these content snippets.

Using the Web Content Editor

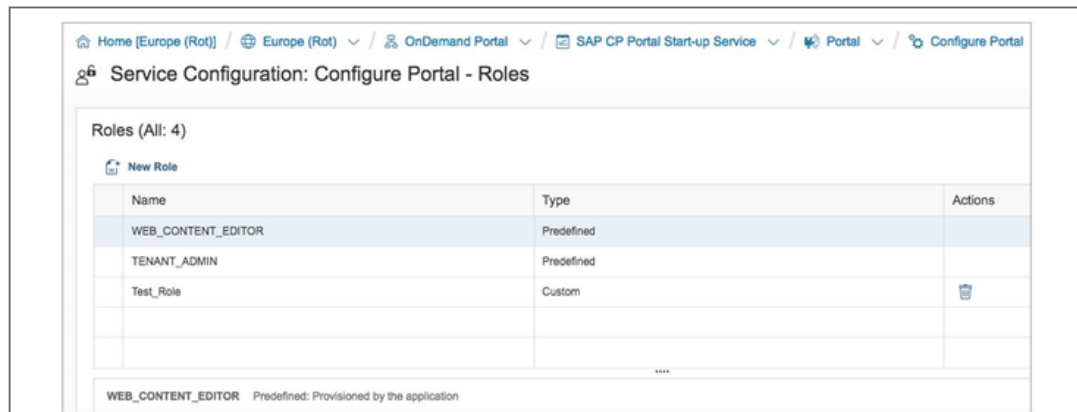


Figure 79: Accessing the Web Content Editor

To be able to access the Web Content Editor, users need to be assigned the predefined WEB_CONTENT_EDITOR role in the SAP Cloud Platform Cockpit. The site administrator then needs to share the link to the Web Content Editor. The link is available under Services and Tools Web Content Editor.

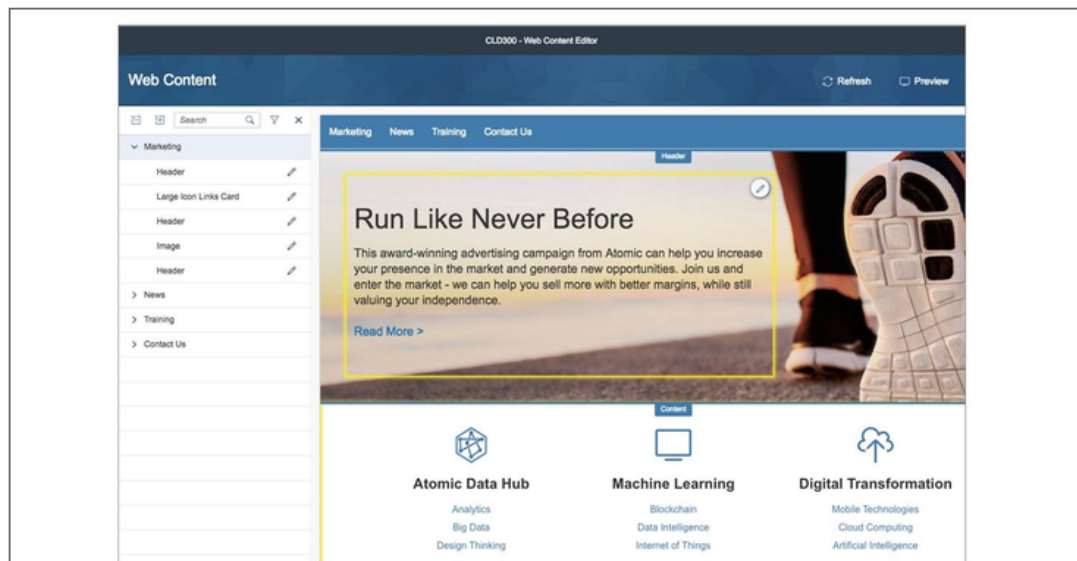
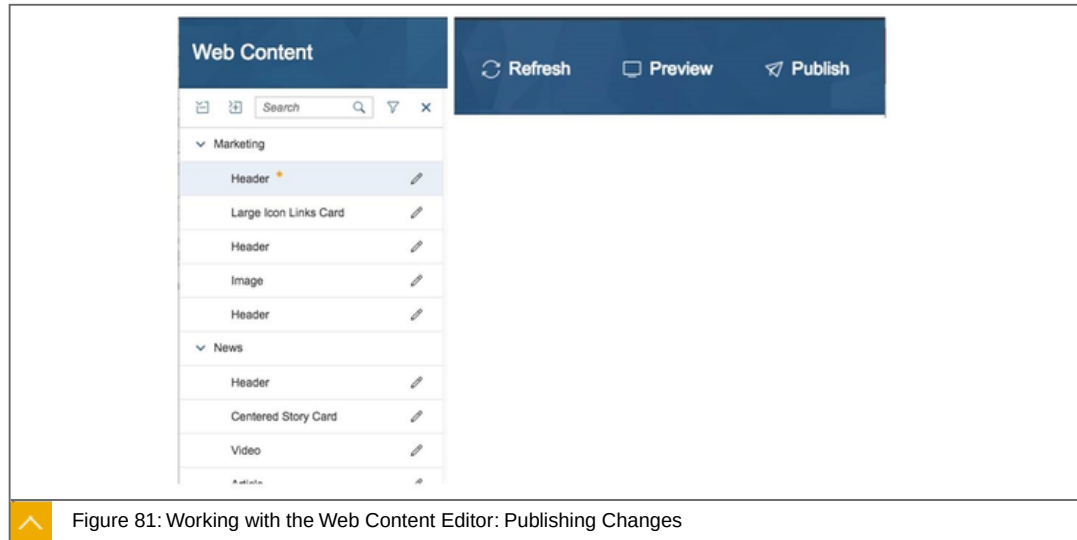


Figure 80: Working with the Web Content Editor

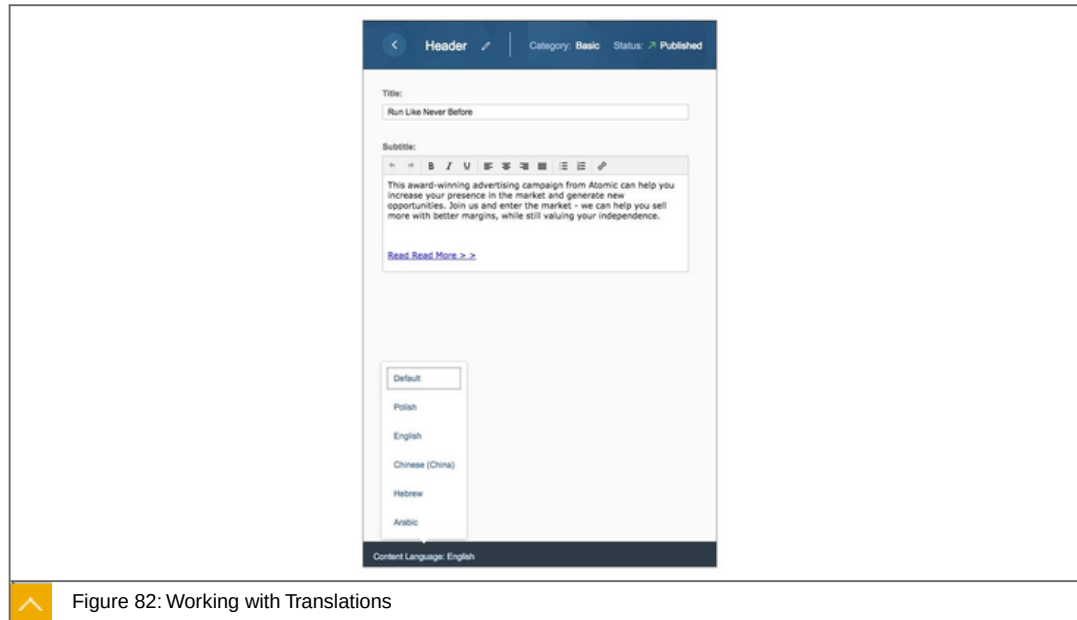
Content that can be edited is shown in the tree view on the left on each page, as well as on the main screen, highlighted in a yellow border. By selecting the widget or choosing the Edit button, you can add/modify the relevant widget content. On the left, you see a list of pages and page templates with their respective web content. On the right, you see a preview of the selected page or page template.

Edit web content as follows:

If the web content consists of multiple content items (such as a card widget that includes three content items, each with predefined content), the editor opens displaying each content item. Clicking on each content item displays its design elements on the right, and you can edit each of these.



Translations



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Site administrators then have the site translated into several other languages using the Translations service. To do this, a site administrator accesses the Translations service from the Site Designer, exports the site content for translation, and imports the translated content back to the site. The translated languages must then be activated to make them available to end users as additional language options.

You may want to translate web content directly from the Web Content Editor. When you open the Web Content Editor, the tool is displayed in the default language (the language used by the site administrator to build the site). As a web content editor, you simply select one of the activated languages from the dropdown list of content languages and start editing. In this way, you are assigning the language you select to the new content, without having to go through the entire translation process. The choice of languages for selection is determined by the languages that were enabled by the site administrator using the Translations service.



LESSON SUMMARY

You should now be able to:

- Work with the Web Content Editor

Unit 5

Lesson 6

Integrating SAP Fiori Launchpad



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Integrate SAP Fiori launchpad

SAP Fiori Launchpad Integration

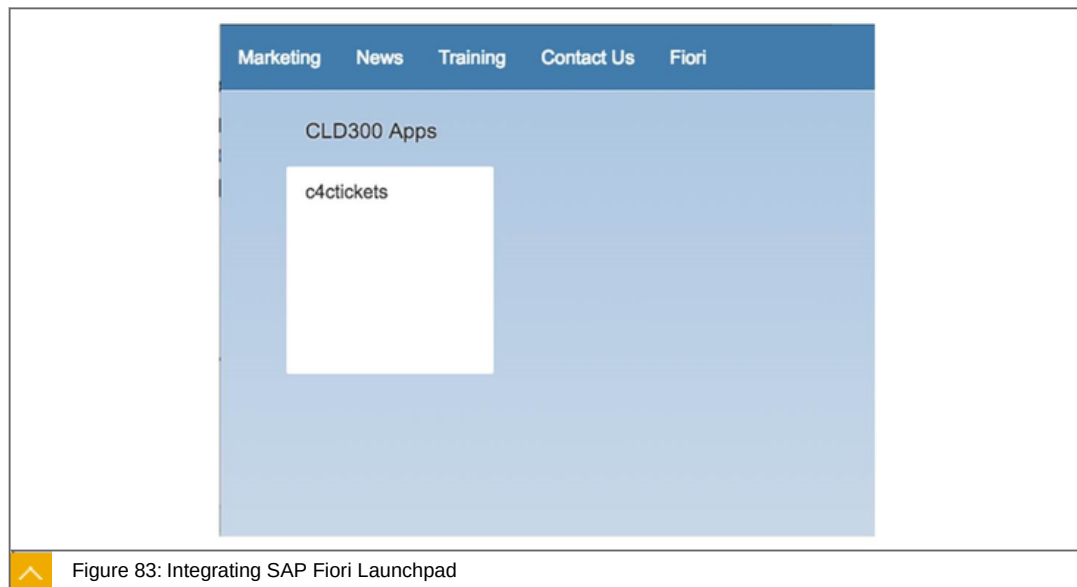


Figure 83: Integrating SAP Fiori Launchpad

SAP Cloud Platform Portal service is fully integrated with the SAP Fiori launchpad. Administrators can integrate the SAP Fiori launchpad when they create a freestyle site. This way, SAP Fiori becomes a part of the freestyle site, providing a custom site with a look and feel and content that leverages SAP Fiori, for example.

There are two ways to bring SAP Fiori launchpad capabilities to a freestyle site:

Create a freestyle site based entirely on the SAP Fiori launchpad.

Create a freestyle site with a SAP Fiori launchpad page.

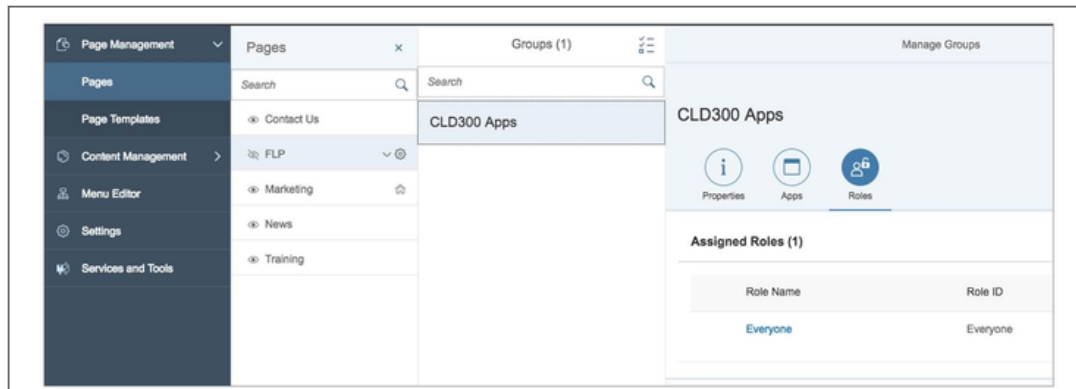


Figure 84: Creating a Launchpad Page

The following prerequisites must be met to support this implementation:

The site to which the page is added is based on a site template that contains the SAP Fiori launchpad page template.

The apps you want to include in the tile groups of the SAP Fiori launchpad page have been added to your site, either via the list of available apps in the Site Directory or via the application configuration in the Site Designer.

To add an SAP Fiori launchpad page, add a new page to the site and select the SAP Fiori launchpad as the source template.

To add tiles and groups to a SAP Fiori launchpad page, create a new group in the launchpad page, assign relevant roles to the group, as add apps to the group. The selected app is added a tile.



LESSON SUMMARY

You should now be able to:

Integrate SAP Fiori launchpad

Learning Assessment

1. Freestyle sites do not provide for role-based access like a Fiori Launchpad.

Determine whether this statement is true or false.

☐

True

☐

False

2. Which tools do you need to create and assign a theme for a SAP Cloud Platform Portal site?

Choose the correct answers.

☐

A Web IDE

☐

B Theme Manager

☐

C UI Theme Designer

☐

D Asset Management

3. Which service do you need when uploading pictures to a freestyle site?

Choose the correct answer.

☐

A UI Theme Manager

☐

B Web Content Editor

☐

C Page Templates

☐

D Asset Management

4. Which of the following element can be modified via the Web Content Editor?

Choose the correct answer.

☐

A Page Layout

☐

B Standard Widgets

☐

C Themes

☐

D Web Content Widgets

5. It is possible to integrate a SAP Fiori Launchpad into a Freestyle site.

Determine whether this statement is true or false.

☐

True

☐

False

Learning Assessment - Answers

1. Freestyle sites do not provide for role-based access like a Fiori Launchpad.

Determine whether this statement is true or false.

☐ True

☒ False

You are correct! It is possible to create content for freestyle sites that is role-based as well as content that does not require a user to log in or even a mix of the two. The required authorizations are set for each page in the Page Settings. Refer to the lesson 'Configuring a Site' for more details.

2. Which tools do you need to create and assign a theme for a SAP Cloud Platform Portal site?

Choose the correct answers.

☐ A Web IDE

☒ B Theme Manager

☒ C UI Theme Designer

☐ D Asset Management

You are correct! The SAP UI Theme Designer is used to create themes for UI5 applications, such as the SAP Cloud Platform Portal Site. Once a theme has been designed, it is deployed to a SAP Cloud Platform Account. It can then be assigned to a site using then Theme Manager in the site administration cockpit. Refer to the lesson 'Branding a Site' for more details.

3. Which service do you need when uploading pictures to a freestyle site?

Choose the correct answer.

- ☐ A UI Theme Manager
- ☐ B Web Content Editor
- ☐ C Page Templates
- ☒ D Asset Management

You are correct! Assets of a freestyle site, such as images, text files, HTML Fragments, etc. can be managed with the Asset Management service. Refer to the lesson 'Managing Site Assets' for more details.

4. Which of the following element can be modified via the Web Content Editor?

Choose the correct answer.

- ☐ A Page Layout
- ☒ B Standard Widgets
- ☐ C Themes
- ☐ D Web Content Widgets

You are correct! The Web Content Editor enables nontechnical users to easily create, edit, and maintain content of Web Content Widgets. Web Content editors can only modify the content of Web Content Widgets, so they are not able to modify themes, page or site layouts or standard widgets. Anyone assigned to the WEB_CONTENT_EDITOR role for a specific subaccount, can create and update the content. Refer to the lesson 'Working with the Web Content Editor' for more details.

5. It is possible to integrate a SAP Fiori Launchpad into a Freestyle site.

Determine whether this statement is true or false.

- ☒ True
- ☐ False

You are correct! By adding a page based on the SAP Fiori Launchpad page template, it is possible to add Fiori Launchpad functionality to any freestyle page. Refer to the lesson 'Integrating SAP Fiori Launchpad' for more details.

UNIT 6

Site Services

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UNIT OBJECTIVES

- Understand site services
- Manage translations
- Manage themes
- Transport content
- Describe site statistics

Unit 6

Lesson 1

Understanding Site Services



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Understand site services

Site Services

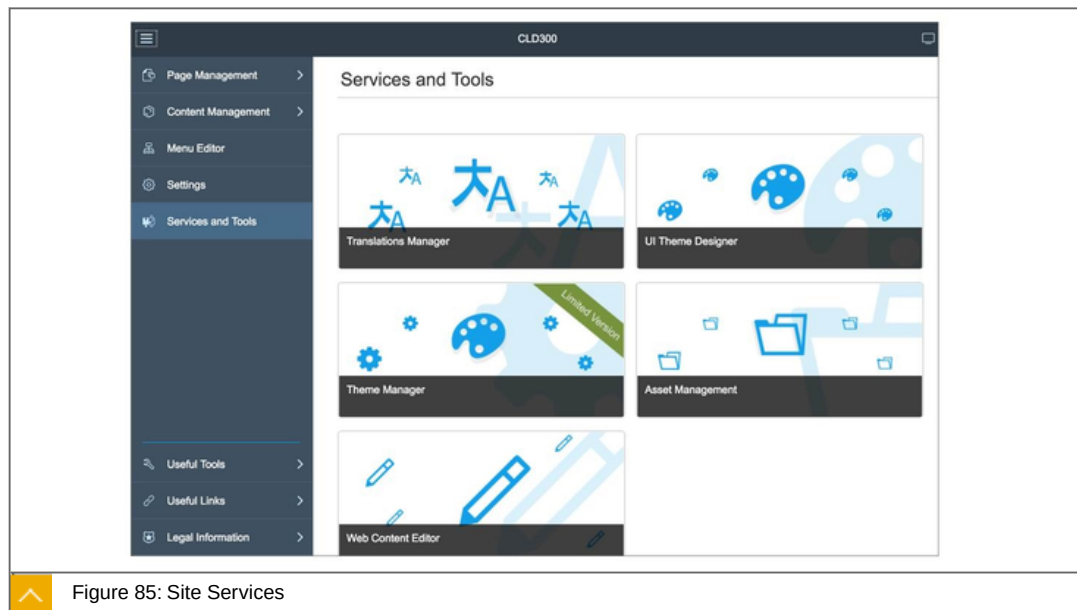


Figure 85: Site Services

Various services are provided to help site administrators manage their apps and portal sites. These include the following:

Translations service

Theming service (UI Theme Designer and Theme Manager)

Transport Manager

Web Content Editor

In this unit, we are going to look at these in detail.



LESSON SUMMARY

You should now be able to:

Understand site services

Unit 6

Lesson 2

Managing Translations



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Manage translations

Translation Management

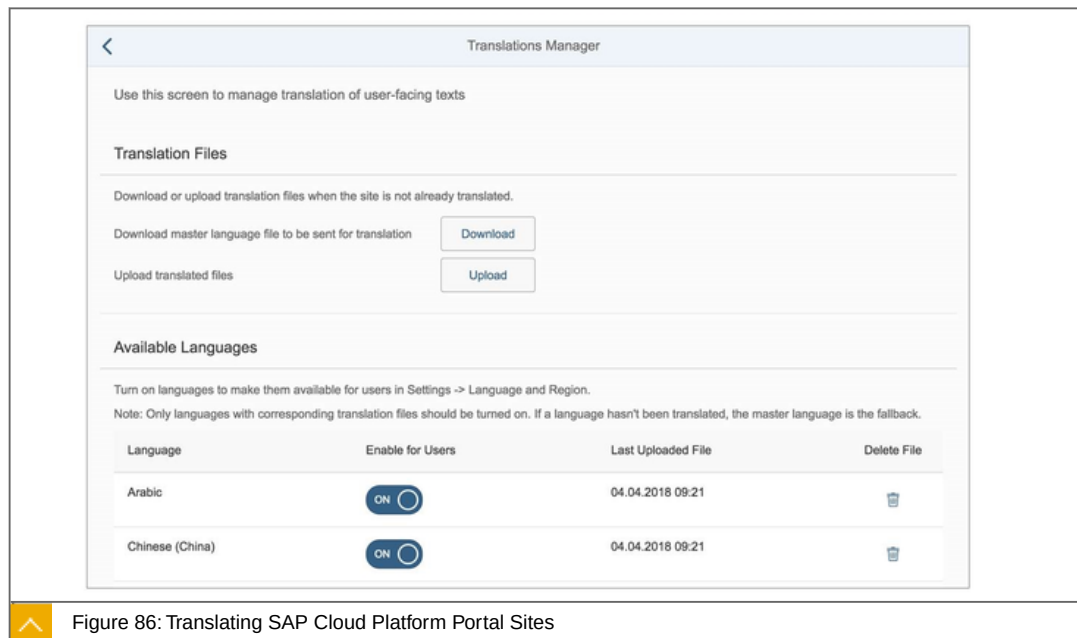


Figure 86: Translating SAP Cloud Platform Portal Sites

The Translations service allows administrators to manage the translations for their sites, whether these are freestyle sites or SAP Fiori launchpad sites. In the Translations Manager, you can do the following:

- Download the master language file.
- Upload new translation files.
- Maintain which languages are active in the site.

The main stages in the translation process occur in the following order:

1. Download master language file
2. Copy the master language file and translate the content
3. Upload the master language file in the Translation service

4. Activate the language so it becomes available to users

By default, SAP Cloud Platform Portal sites display in the browser language. In the case of the SAP Fiori launchpad, users can change the display language in the Me Area.

Master Language File



```
1 1...ldi.translation.uid=37cc70e6-251e-4060-a82c-afe116d52842
2 #Site ID:[37cc70e6-251e-4060-a82c-afe116d52842]
3 #Fri Apr 06 08:26:43 UTC 2018
4 8794ead4-94de-4b83-b29c-95977accf076_link1Tooltip-1=
5 d886f7c4-f774-43c5-9324-9cb85e9e6bd9_title-4=SAP Leonardo Design-Led Engagements Demystified
6 8794ead4-94de-4b83-b29c-95977accf076_link3DisplayText-3=Artificial Intelligence
7 d886f7c4-f774-43c5-9324-9cb85e9e6bd9_title-3=Atomic Design-Led Engagements
8 8794ead4-94de-4b83-b29c-95977accf076_link3DisplayText-2=Internet of Things
9 d886f7c4-f774-43c5-9324-9cb85e9e6bd9_title-2=Atomic Manufacturing Execution Custom Development
10 8794ead4-94de-4b83-b29c-95977accf076_link3DisplayText-1=Design Thinking
11 d886f7c4-f774-43c5-9324-9cb85e9e6bd9_title-1=Top Assets\ Atomic Analytics
12 c49199a8-7201-4071-80ea-32ba8de14f87_title=Buy Now, Pay Later
13 8794ead4-94de-4b83-b29c-95977accf076_link3Tooltip-3=
14 8794ead4-94de-4b83-b29c-95977accf076_link3Tooltip-2=
15 8794ead4-94de-4b83-b29c-95977accf076_link3Tooltip-1=
16 fdb7a5a6-b21a-46fc-b711-fa916882e88c_title=
17 625ecec9-63c1-4528-8dba-135729a9cb94_link0DisplayText-3=Read More
18 d886f7c4-f774-43c5-9324-9cb85e9e6bd9_link0DisplayText-4=REGISTER
19 625ecec9-63c1-4528-8dba-135729a9cb94_link0DisplayText-2=Read More
20 d886f7c4-f774-43c5-9324-9cb85e9e6bd9_link0DisplayText-3=REGISTER
21 625ecec9-63c1-4528-8dba-135729a9cb94_link0DisplayText-1=Read More
22 d886f7c4-f774-43c5-9324-9cb85e9e6bd9_link0DisplayText-2=REGISTER
23 d886f7c4-f774-43c5-9324-9cb85e9e6bd9_link0DisplayText-1=REGISTER
24 bfae9ff2-c2c6-4d00-a1ae-a7643f1b880_title=Recommended Training
25 8ef15306-03ec-44a9-8ee0-dff744fe5eeb_subtitle=<p>Learn about installation, configuration, and more</p>
26 77ca54a3-7508-4688-8a1a-49b33d89ee0d_link1DisplayText-3=Walldorf Dietmar-Hopp Alle
27 77ca54a3-7508-4688-8a1a-49b33d89ee0d_link1DisplayText-2=+1800-872-1727
28 77ca54a3-7508-4688-8a1a-49b33d89ee0d_link1DisplayText-1=marketing@sap.com
29 77ca54a3-7508-4688-8a1a-49b33d89ee0d_link1Tooltip-3=
30 77ca54a3-7508-4688-8a1a-49b33d89ee0d_link1Tooltip-2=
31 77ca54a3-7508-4688-8a1a-49b33d89ee0d_link1Tooltip-1=
32 625ecec9-63c1-4528-8dba-135729a9cb94_linkTooltip-3=
33 625ecec9-63c1-4528-8dba-135729a9cb94_linkTooltip-2=
```

Figure 87: Master Language File

To download text in the master language:

1. In your site, launch the Translation service from the SAP Fiori launchpad configuration cockpit. Choose **Services and Tools** → **Translations and Download**.

A zip file called `master_language_YYYY_MM_DD.zip` is created. The zip file contains the master language texts and a `readme.txt` file that describes the translation process.

2. Extract the zip file and look for the source file `<SiteID>.properties`, which contains the master language texts

The content of the source file is in the following format: `<key> = <value>`. Note that `<key>` is the property key (do not change this) and `<value>` is the string to translate.

The master language is the language that the site was initially created in, so make sure to be consistent when creating your content.

Creating and Maintaining Translations



```
WCE_37cc70e6-251e-4060-a82c-afe116d52842_de.properties
WCE_37cc70e6-251e-4060-a82c-afe116d52842_se.properties
WCE_37cc70e6-251e-4060-a82c-afe116d52842.properties
```

Figure 88: Create a Translation

After you have finished translating the strings for a locale (language-country), rename the properties file using one of the following naming conventions:

```
<SiteId>_<Language>-<Country>.properties / WCE_<SiteId>_<Language>-<Country>.properties

<SiteId>_<Language>.properties / WCE_<SiteId>_<Language>.properties
```

Note that <Language> should be according to the ISO 639 convention, and <Country> should be according to the ISO 3166 convention. The following are examples:

```
4c736e0c-a094-46f1-9ae5-a602eb23b2b9_en-US.properties
```

```
4c736e0c-a094-46f1-9ae5-a602eb23b2b9_en.properties
```

You can create your translations manually with any text editor. The upload and download of the language files can be done via the SAP Cloud Portal API, allowing for further automation and extension of the translation process. For an example, see the blog at <https://blogs.sap.com/2016/11/02/automatic-translation-process-for-cp-using-jenkins-and-github/>.

SAP Cloud Platform also offers the Translation service, which can be used for creating and maintaining machine-assisted translations.



Figure 89: Maintaining Available Translations

Administrators can manage all available language translations using the Translations service. Language translations can be activated or deactivated for end users.

To maintain a language, go to the Translations service under Services and Tools in the Site Designer or the SAP Fiori launchpad configuration cockpit. All the languages are displayed in the Available Languages table.

To activate a translation, set the status to On. To deactivate a translation, set the status to Off. To delete a translation from the account, click the delete icon of the language entry in the table.

By default, the language displayed for the user is determined by the following:

URL parameter sap-language=<locale>

Browser locale settings (if there is no URL parameter)

The translation uses the following order of precedence:

1. <SiteId>_<Language>-<Country>.properties
2. <SiteId>_<Language>.properties

3. <SiteId>_en.properties



Note:

As a best practice, SAP recommends having the English translation as a fallback, instead of using the source texts of the site.

The end user can also select their desired language in the portal site by choosing **User Preferences** **Language**.



LESSON SUMMARY

You should now be able to:

- Manage translations

Unit 6

Lesson 3

Managing Themes



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Manage themes

Theme Management

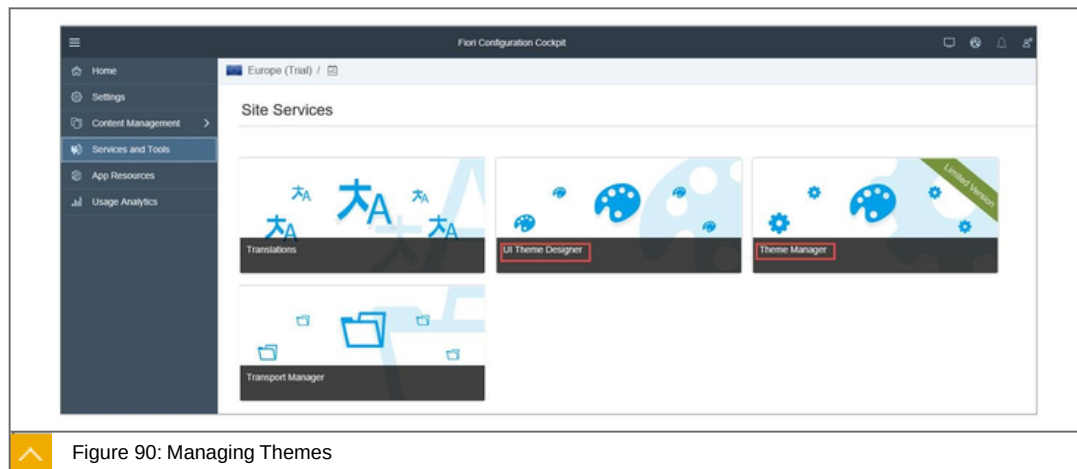


Figure 90: Managing Themes

To make a portal site more interesting, a custom theme is often used instead of the default theme provided by SAP. This custom theme is created using the UI Theme Designer, and administrators can use the Theme Manager to assign the theme and make it available to the portal site.

It is important to maintain compatibility between the site theme and the SAPUI5 version. SAPUI5 is automatically upgraded once a new version is available in SAP Cloud Platform. This means that all custom SAPUI5 themes are broken, because the themes only work for the SAPUI5 version for which they were built and exported from the UI Theme Designer. To make sure the portal sites and applications are displaying properly, the custom themes need to be upgraded whenever a newer SAPUI5 version becomes available. This upgrade is done using the Theme Manager.

To upgrade a custom theme:

1. Go to the Theme Manager using the SAP Fiori launchpad configuration cockpit. Alternatively, in the Admin Space, choose **Services and Tools**.
2. Set the Theme Upgrade option to **On**.



Note:

When you activate the Theme Upgrade option, the theme is checked for compatibility. If the theme is not compatible, the Theme Upgrade option is turned off automatically. Before you can activate this option again, you need to export the theme from the UI Theme Designer and re-import it into the Theme Manager.

For more details, refer to SAP Note 2140280 at <https://launchpad.support.sap.com/#/notes/2140280> and SAP Note 2119552 at <https://launchpad.support.sap.com/#/notes/2119552>.



LESSON SUMMARY

You should now be able to:

Manage themes

Unit 6

Lesson 4

Transporting Content



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Transport content

Transport of Content

The following table provided information about transporting launchpad content:



Table 7: Transporting Launchpad Content

Transport Content	What Is Included?	Name of Transport File
Content package	Site content: HTML5 applications Themes SAP Cloud Platform roles Site configuration: Site settings App configurations Catalogs Groups Roles Translations	SAP_exported_solution_<siteID>.mtar Note that all the items in a content package are assembled into a single .mtar file.
Site	Site configuration: Site settings App configurations Catalogs Groups Roles Translations	SAP_exported_site_<site ID>.zip

Transport Content	What Is Included?	Name of Transport File
Roles	One or more roles	SAP_exported_roles_<site ID>.zip
Catalogs	One or more catalogs	SAP_exported_cata-logs_<site ID>.zip

Launchpad site content can be transported between subaccounts, landscapes, and regions. Administrators can perform these tasks using the export and import options in the Site Directory or the Transport Manager service in the SAP Fiori launchpad configuration cockpit.

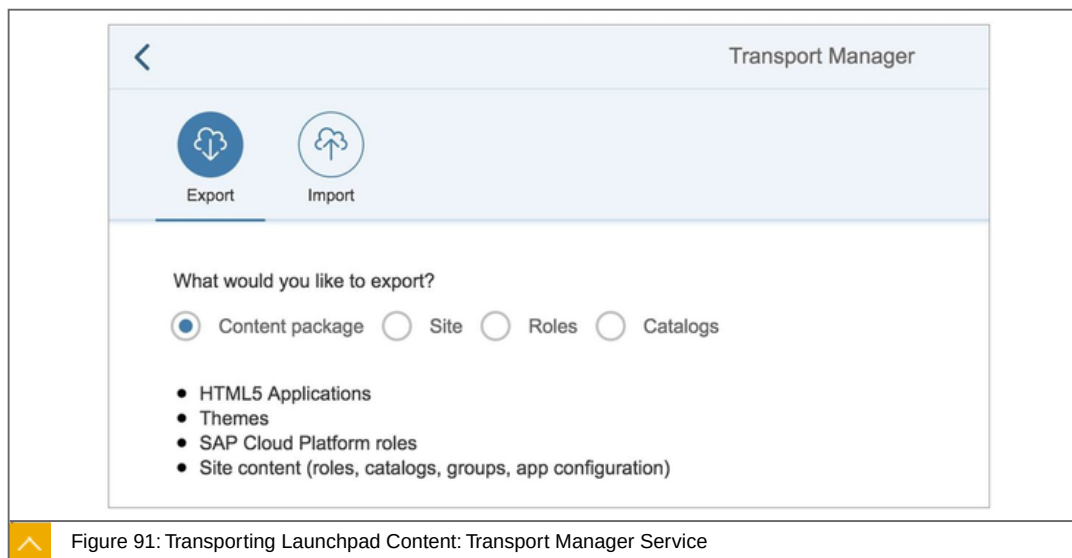


Figure 91: Transporting Launchpad Content: Transport Manager Service

To export content, use the Transport Manager (under Services and Tools) in the configuration cockpit of the source account or landscape. Select the content to export: content package (site content and configuration), site (site configuration only), roles, or catalogs. Save the exported file.

Note the following:

For roles, one or more roles can be selected.

For catalogs, one or more roles can be selected.

To import content, use the Transport Manager (under Services and Tools) in the configuration cockpit of the source account or landscape. Import the exported file, which must be a .zip or .mtar file. Use the Add and Override option to add new content and override existing content. Use the Replace All option to replace all the existing content with the imported content.

The following table provides information about transporting freestyle site content:



Table 8: Transporting Freestyle Site Content

Transport Content	What Is Included?	Name of Transport File
Content package (site content and configuration)	Site content: Site template, pages, apps, and widgets Site assets (such as images and documents) Themes SAP Cloud Platform roles Site configuration Site settings and page settings Launchpad configuration (if any), including app configurations, catalogs, groups, and roles Translations	SAP_exported_solution_<siteID>.mtar Note that all the items in a content package are assembled into a single .mtar file.
Site	Site configuration: Site settings and page settings Launchpad content (if any), including app configurations, catalogs, groups, and roles Translations	SAP_exported_site_<siteID>.zip

Freestyle site content can be transported between subaccounts, landscapes, and regions. Administrators can perform these tasks using the export and import options in the Site Directory.

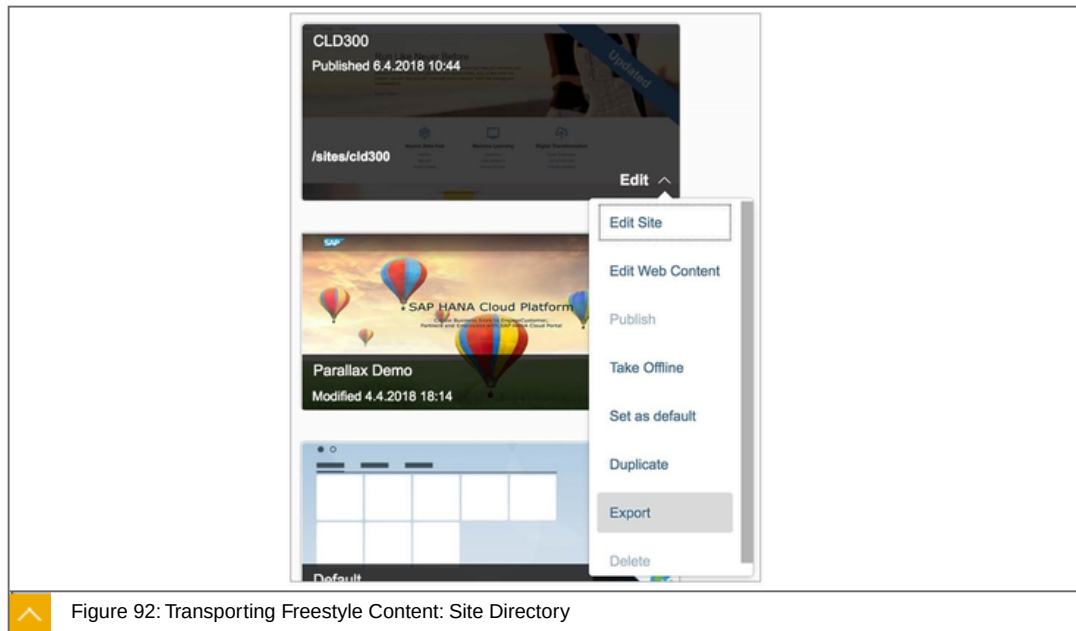


Figure 92: Transporting Freestyle Content: Site Directory

To export content, go to the Site Directory of the source account or landscape and open the freestyle site to export. Select the content to export the content: content package (containing site content and configuration) or site (containing configuration only). Save the exported file.

To import content, go to the Site Directory of the source account or landscape and choose the Import Site button at the top right of the header bar. Import the exported file, which must be a .zip or .mtar file.



LESSON SUMMARY

You should now be able to:

- Transport content

Unit 6

Lesson 5

Describing Site Statistics



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Describe site statistics

Site Statistics

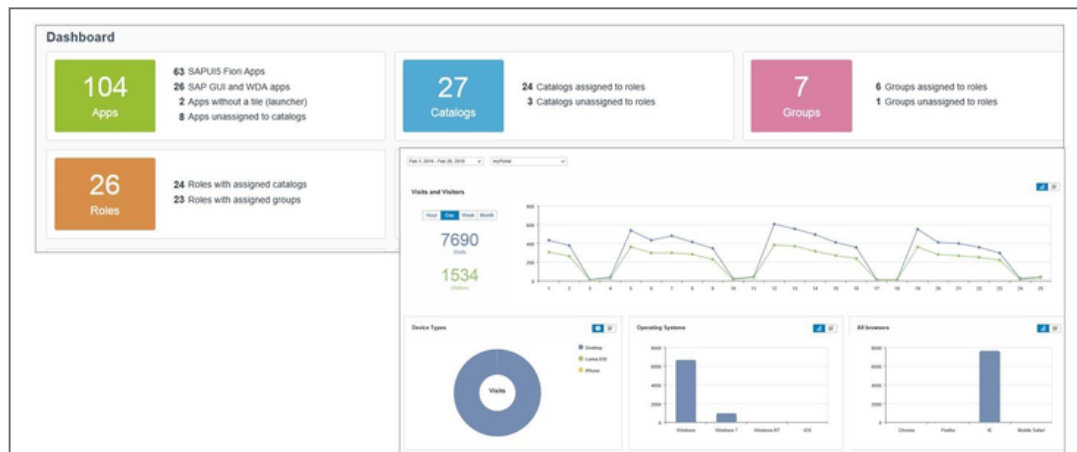


Figure 93: Site Statistics

The Usage Analytics service can be used to view site statistical information. Site statistics (SAP Fiori launchpad site or freestyle site) can be displayed as charts, graphs, or tables in the Usage Analytics dashboard.

Site statistical information includes the following:

Visits and visitors

Measures web traffic by hour, day, week, or month, during a selected period of time. The number of visits indicates the total number of visits, whereas the number of visitors indicates visits from a unique IP. Anonymous users are counted as a single unique user for the number of visitors, but they are all added to the total number of visits.

Device types

Displays the device types from which the site was accessed

Operating systems

Displays the operating systems from which the site was accessed

Browsers

Displays the browsers from which the site was accessed



LESSON SUMMARY

You should now be able to:

Describe site statistics

Learning Assessment

1. The Translation Manager will create translations for you site.

Determine whether this statement is true or false.

☐

True

☐

False

2. Which elements of a site are included when exporting a Site Configuration for either Freestyle or Fiori Launchpad pages?

Choose the correct answers.

☐

A Site Settings

☐

B Themes

☐

C Translations

☐

D SAP Cloud Platform Roles

Learning Assessment - Answers

1. The Translation Manager will create translations for you site.

Determine whether this statement is true or false.

☐ True

☒ False

You are correct! The Translation Manager allows for the creation of a Master language file which can then be used to create translations of a site. Each translation is then uploaded with the Translation Manager and can then be made available to users. Refer to the lesson 'Managing Translations' for more details.

2. Which elements of a site are included when exporting a Site Configuration for either Freestyle or Fiori Launchpad pages?

Choose the correct answers.

☒ **A** Site Settings

☐ **B** Themes

☒ **C** Translations

☐ **D** SAP Cloud Platform Roles

You are correct! When exporting a site configuration, the following elements are included: site settings, app configurations, catalogs, groups, roles, and translations. Refer to the lesson 'Transporting Content' for more details.

UNIT 7

Integration with Other Services

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UNIT OBJECTIVES

- Explain integration scenarios
- Collaborate using SAP Jam
- Describe a SAP Smart Business KPI tile
- Integrate with SAP Enterprise Portal

Unit 7

Lesson 1

Explaining Integration Scenarios



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Explain integration scenarios

Integration Scenarios

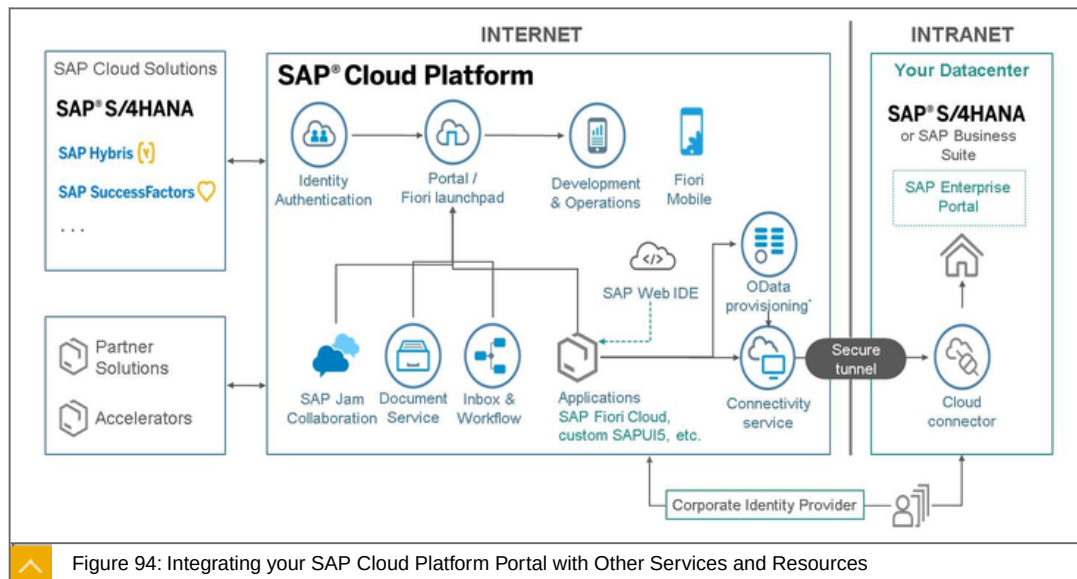


Figure 94: Integrating your SAP Cloud Platform Portal with Other Services and Resources

With SAP Cloud Platform Portal you can integrate other services and resources into your portal experience, whether these are from SAP solutions in the cloud, on-premise solutions, or partner solutions.

For example, you can do the following:

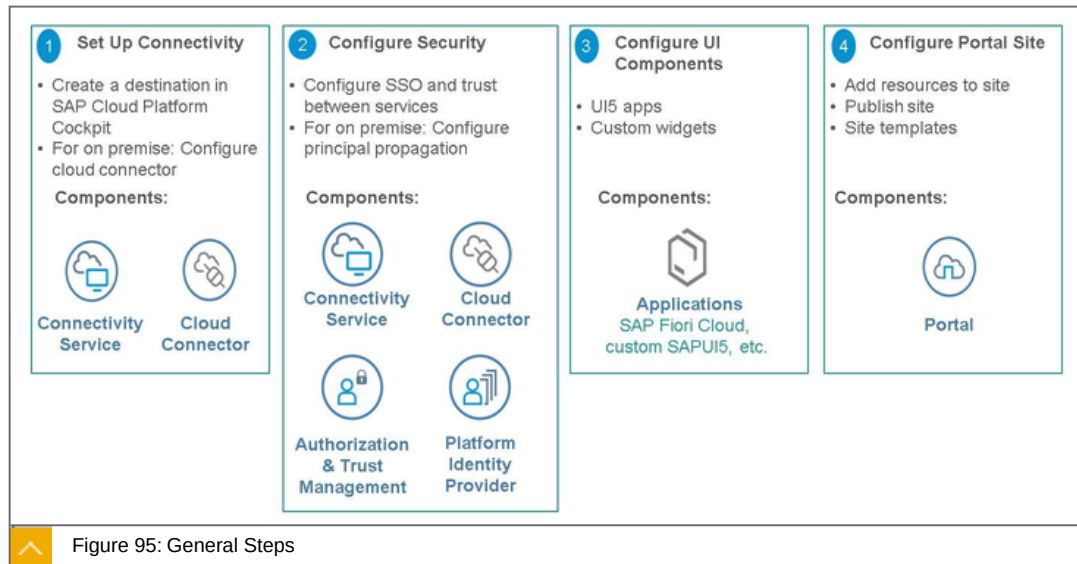
- Integrate SAP JAM Collaboration for sharing documents and discussions.

- Use the document service to store and retrieve unstructured data.

- Utilize OData services from SAP S/4HANA Cloud or SAP SuccessFactors in your SAP Fiori launchpad and custom applications.

- Mobilize your SAP Fiori launchpad with SAP mobile development and operations.

- Integrate content from your on-premise SAP Enterprise Portal, such as Web Dynpro , SAP GUI, BSP, and other applications.



The following is a general overview of the configuration involved in the integration of other services and resources:

1. Set up connectivity.

This involves creating a destination to the service or resource. It is for connecting to an internal resource, for example, SAP Enterprise Portal, in your intranet.

2. Configure security for SSO between the portal service and the required service.

If the service also resides on SAP Cloud Platform, this step is not needed (unless you are using multiple identity providers).

3. Deploy UI components.

If required, deploy the front-end components to your SAP Cloud Platform account, for example, UI5 apps, widgets, or site templates.

4. Configure the portal site.

You can now add the content to your portal site, using the site administration tools to configure your integrated components.

The following are integration scenarios:



1. URL link – configure URL or use iFrame
2. Template integration, for example, classic UIs and Bex – configure in portal services
3. On-premise custom applications – via EP integration
4. Custom UI5 app – deploy in SAP Cloud Platform and use in site
5. Third party, for example, Google Maps, Twitter, and CFL – create widget (freestyle only)
6. Other cloud services with UI – If SAP Cloud Platform and UI5, use UI in site. If other, use iFrame/redevelop in UI5
7. Other cloud services with web service/Odata – build custom UI



LESSON SUMMARY

You should now be able to:

- Explain integration scenarios

Unit 7

Lesson 2

Collaborating Using SAP Jam



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Collaborate using SAP Jam

SAP Jam Collaboration

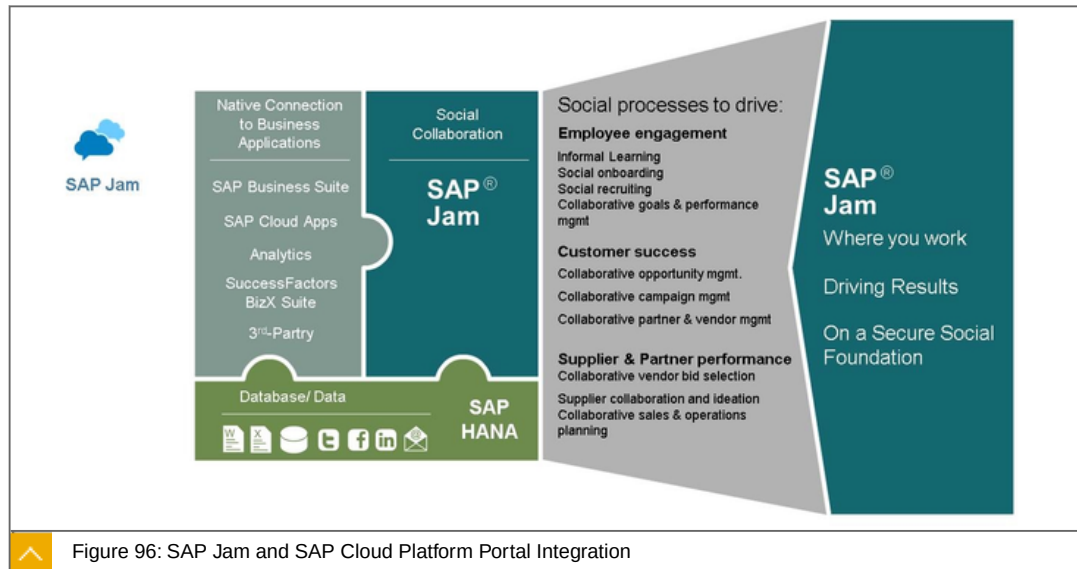


Figure 96: SAP Jam and SAP Cloud Platform Portal Integration

As a standard collaboration tool in SAP, SAP Jam brings people, processes, content, and business data together on a secure social foundation. For SAP Cloud Platform Portal sites, SAP Jam can be integrated with either the SAP Fiori launchpad or freestyle sites to enhance the portal experience.

The following are prerequisites to set up integration between SAP Jam and portal sites:

From SAP Cloud Platform:

- Enable the SAP Jam service on SAP Cloud Platform.
- Set up a trust with SAP Jam.
- Configure a destination to SAP Jam.

From SAP Jam, the end user needs to have an SAP Jam account (that is, an invitation to a group).

There are various SAP Jam collaboration capabilities in the SAP Fiori launchpad:

SAP Jam tiles configuration

Two different types of tiles can be found in a designated SAP Jam catalog in the launchpad, and they can be added to any tile group. These are viewing notifications and viewing group information. Configuration examples include the following:

- Single tile: for viewing latest notifications received in the SAP Jam account
- Multiple tiles: one for each SAP Jam group to which users belong, where users can see when their last group activity occurred and navigate directly to the SAP Jam group

Photo

End users can view their photo next to their name. The end user photo defined in the landscape is used automatically without any further configurations.

Share and Discuss menu controls configuration

Share and Discuss menu controls can be added to an app by using API services. End users can communicate with a particular SAP Jam group to share or discuss relevant information displayed.

To enable the share and discuss feature, you must do the following:

1. Create a destination pointing to the back-end ABAP application using the SAP Cloud Platform Cockpit. Choose **Destinations** **New Destination** and configure the destination to SAP Jam.
2. Configure a destination path in the neo-app.json file of the SAP Fiori app. Use the following code:

```
/sap/bc/ui2/smi/rest_tunnel/Jam/"target: {type: "destination" name: "jam"}, description: "jam destination"}
```

3. Configure ABAP Social Media Integration (ABAP SMI) to allow the SAP Fiori launchpad and apps to use the ABAP SMI functions developed for SAP Fiori.



Note:

For the detailed steps, see the help documentation on configuring share and discuss features for a launchpad site at <https://help.sap.com/viewer/3ca6847da92847d79b27753d690ac5d5/Cloud/en-US/9b529041d0fe470d9c7c795eb4038e7a.html>.

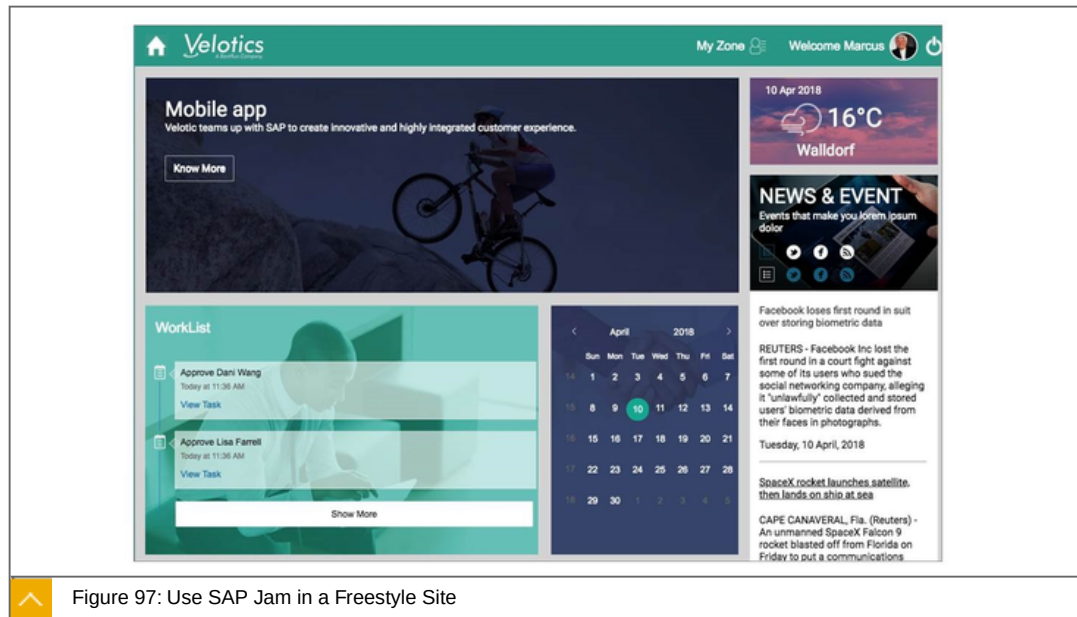


Figure 97: Use SAP Jam in a Freestyle Site

There are several SAP Jam widgets provided by SAP as follows:

SAP Jam Content

Displays content from a selected SAP Jam group. The end user can then navigate through the content and download files directly from the widget.

SAP Jam Group Feed

Displays the feed from a selected SAP Jam group in the site. The end user sees the group feed and can post to the group.

SAP Jam Search

Displays the results of a search query that you have defined in the widget. This enables the end user to search for content.



LESSON SUMMARY

You should now be able to:

- Collaborate using SAP Jam

Unit 7

Lesson 3

Describing a SAP Smart Business KPI Tile



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Describe a SAP Smart Business KPI tile

SAP Smart Business KPI Tile

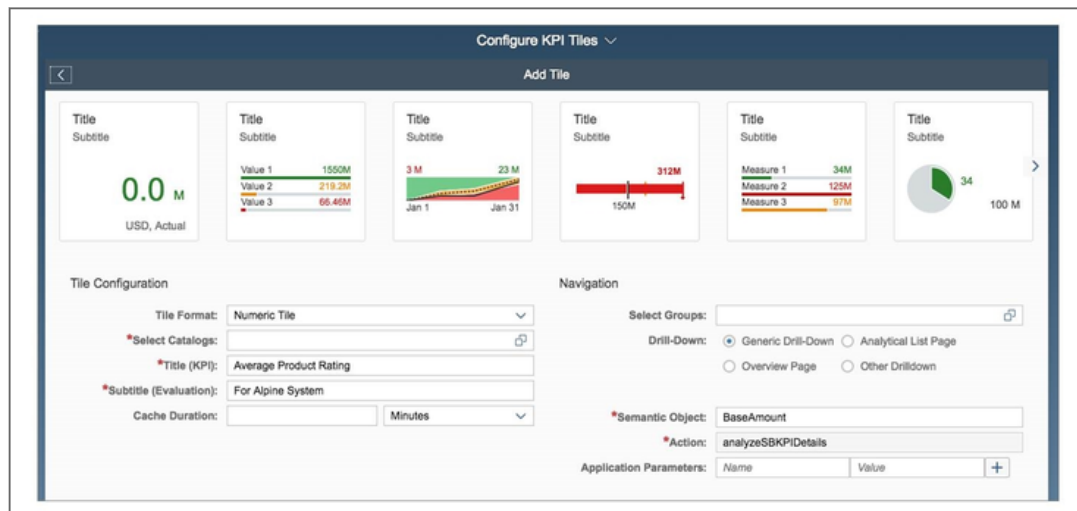


Figure 98: Creating an SAP Smart Business KPI Tile

SAP Smart Business is a framework for exposing key performance indicators (KPIs) and operational performance indicators (OPIs) as SAP Fiori applications without the need to write any code. SAP Smart Business is a business service in SAP Cloud Platform. It is now possible to bring together the analytic content from multiple business systems (on premise or on demand) into one single SAP Fiori launchpad.

To get started, it is a prerequisite to have the SAP Predictive Maintenance and Service solution. With this solution, you have the SAP Smart Business Modeler in the SAP Fiori launchpad configuration cockpit. The SAP Smart Business modeler apps are a set of SAP Fiori apps that you use to create and manage SAP Smart Business entities.

To create KPI tiles:

1. In the SAP Fiori launchpad configuration cockpit, use the SAP Smart Business Modeler editors to configure a KPI tile.

An app for the tile is added to the list of apps in the configuration cockpit.

2. Choose Content Management Apps.

3. Select the app and, under Properties, configure the assigned catalogs and the assigned groups.

**Note:**

For more information, see the following:

Documentation at http://help.sap.com/saphelp_uiaddon10/helpdata/en/28/bb30fccab54b4a965c00d02a6ae343/content.htm?frameset=/en/28/bb30fccab54b4a965c00d02a6ae343/frameset.htm

Blog describing how to use SAP Smart Business apps for SAP Fiori Cloud at <https://blogs.sap.com/2017/12/04/how-to-enable-your-sap-fiori-cloud-smart-business-apps/>

Blog describing how to set up the SAP Smart Business service at <https://blogs.sap.com/2016/04/27/how-to-subscribe-to-sap-smart-business-service/>

**LESSON SUMMARY**

You should now be able to:

Describe a SAP Smart Business KPI tile

Unit 7

Lesson 4

Integrating with SAP Enterprise Portal



LESSON OBJECTIVES

After completing this lesson, you will be able to:

Integrate with SAP Enterprise Portal

Integration with SAP Enterprise Portal

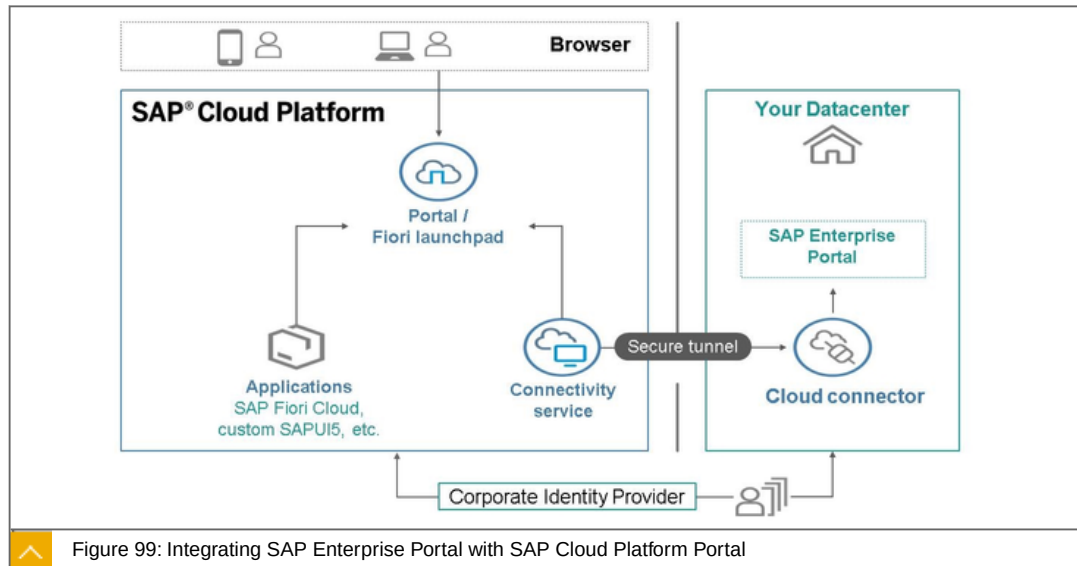


Figure 99: Integrating SAP Enterprise Portal with SAP Cloud Platform Portal

It is possible to use the on-premise SAP Enterprise Portal with the SAP Cloud Platform Portal. This allows the existing configuration and runtime of many classic UIs such as Web Dynpro ABAP, SAP GUI, BSP, and other applications to be used. Customers who want to use the SAP Cloud Platform Portal for their new applications and still want to access their on-premise apps can do so by making the SAP Enterprise Portal a content provider for the SAP Cloud Platform Portal. The on-premise apps are exposed and can then be added to a SAP Fiori launchpad site and/or a SAP Fiori launchpad page within a freestyle site.

The implementation steps are as follows:

1. Configure the cloud connector to allow access to the SAP Enterprise Portal.
2. Create a destination to the SAP Enterprise Portal in SAP Cloud Platform.
3. Configure your SAP EP to allow login via SAML2.0 and trust the corporate identity provider, for example, SAP Identity Authentication Service.
4. Configure the SAP Easy Access menu and Shell Plugin in the administration cockpit.



Note:

For a detailed description of the steps involved see the documentation at <https://help.sap.com/viewer/3ca6847da92847d79b27753d690ac5d5/Cloud/en-US/122bf2fe86034241a41d41cbd0d96b31.html>.

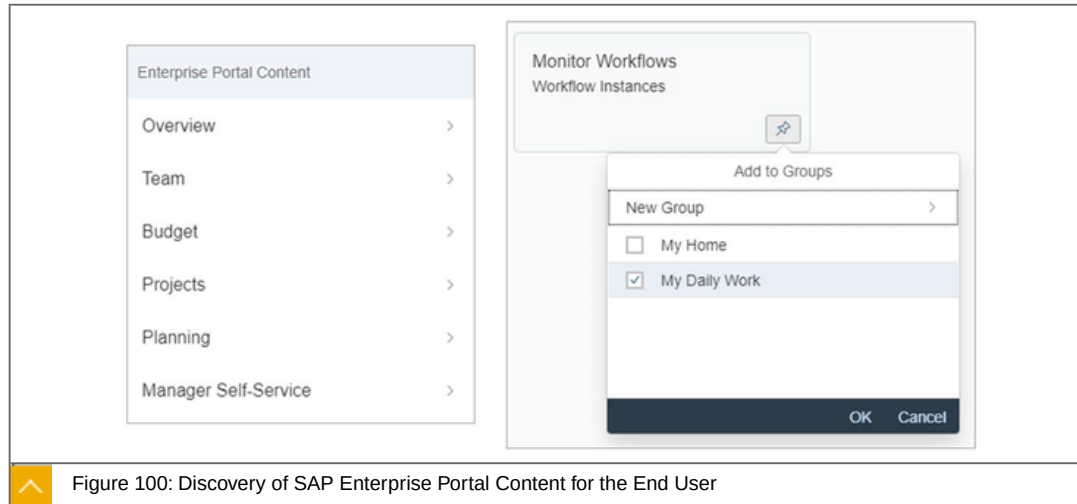


Figure 100: Discovery of SAP Enterprise Portal Content for the End User

Once the configuration is complete, end users are able to search for and add their on-premise apps to their SAP Fiori launchpad site and/or to a SAP Fiori launchpad page within a freestyle site. To do this, they perform the following steps:

1. Open the Me Area in the SAP Fiori launchpad.
2. Choose App Finder and open the User Menu tab.

The content structure from the SAP Enterprise Portal is available here.

3. Select the required apps by choosing the pin icon and selecting the groups in which you want the app tile to be displayed.



LESSON SUMMARY

You should now be able to:

Integrate with SAP Enterprise Portal

Learning Assessment

1. It is not possible to connect the SAP Enterprise Portal to the SAP Cloud Platform Portal Service.

Determine whether this statement is true or false.

- ☐ True
- ☐ False

Learning Assessment - Answers

1. It is not possible to connect the SAP Enterprise Portal to the SAP Cloud Platform Portal Service.

Determine whether this statement is true or false.

☐ True

☒ False

You are correct! It is possible to use the on-premise SAP Enterprise Portal with the SAP Cloud Platform Portal. This way the existing configuration and runtime of many classic UIs such as Web Dynpro ABAP, SAP GUI, BSP, and other applications can be used. Customers who want to use SAP Cloud Platform Portal for their new applications and still want to access their on-premise apps can do so by making SAP Enterprise Portal a content provider for SAP Cloud Platform Portal. The on-premise apps are exposed and can then be added to a launchpad site and/or to a launchpad page within a freestyle site. Refer to the lesson 'Integrating with SAP Enterprise Portal' for more details.